

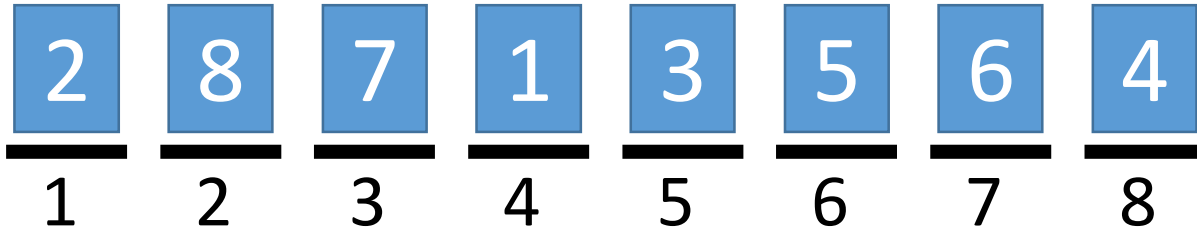


Quick Sort

Quicksort

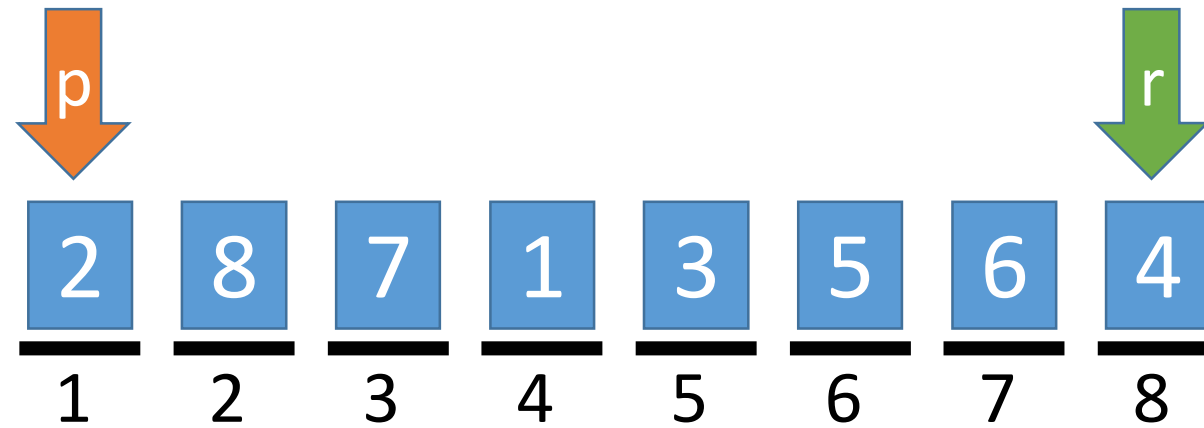
- Partition Function

Partition Function (A, p, r)

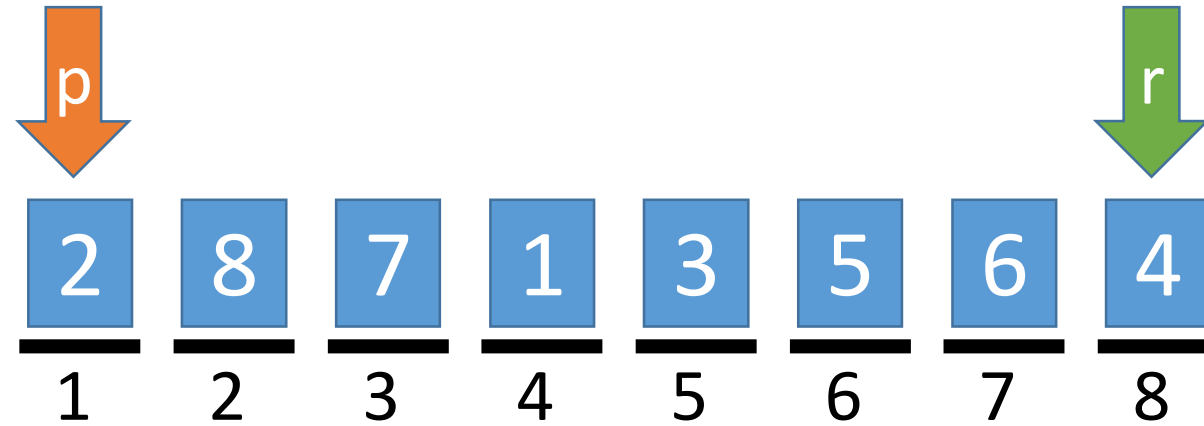


2	8	7	1	3	5	6	4
1	2	3	4	5	6	7	8

Partition Function (A, p, r)

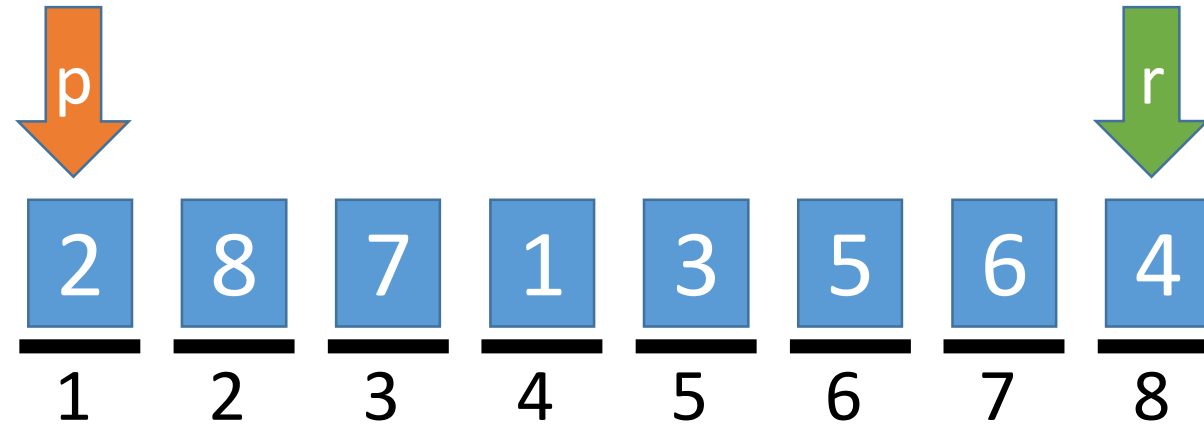


Partition Function (A, p, r)



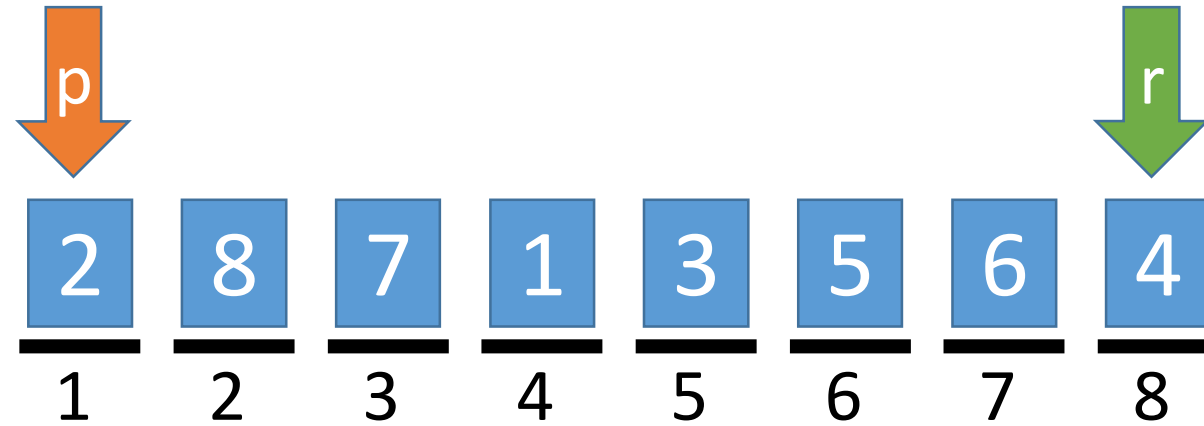
Pivot x

Partition Function (A, p, r)



Pivot x = 4

Partition Function (A, p, r)

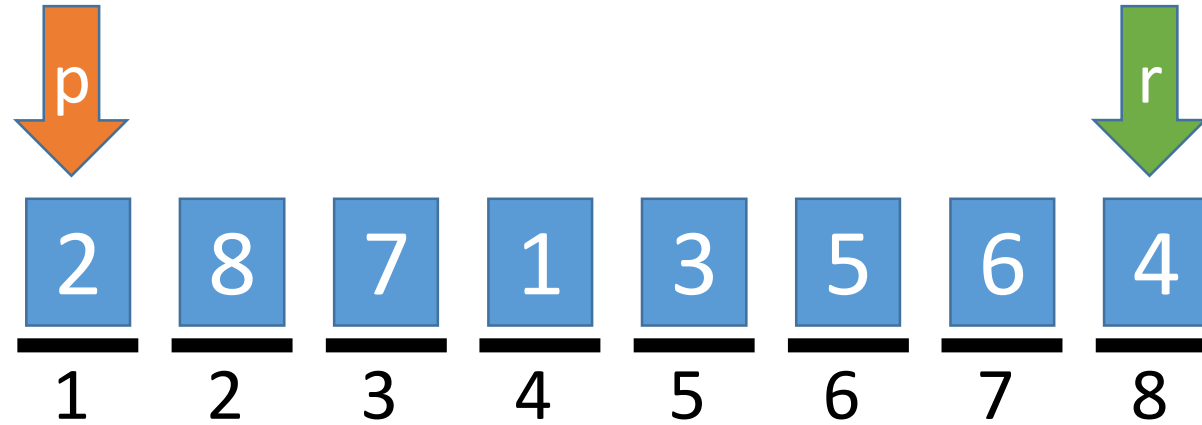


Pivot x = 4

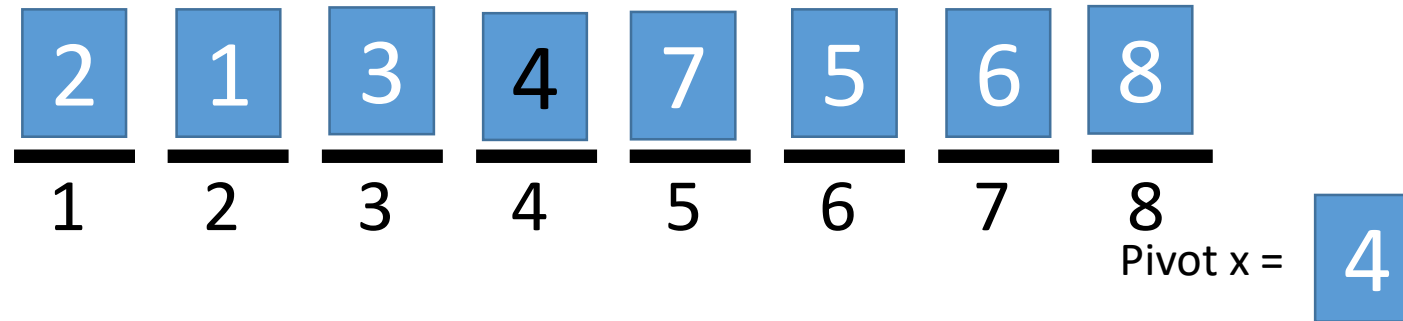
WHAT IS THE SIGNIFICANCE OF A PIVOT?

Partition Function (A, p, r)

Before partition
function:



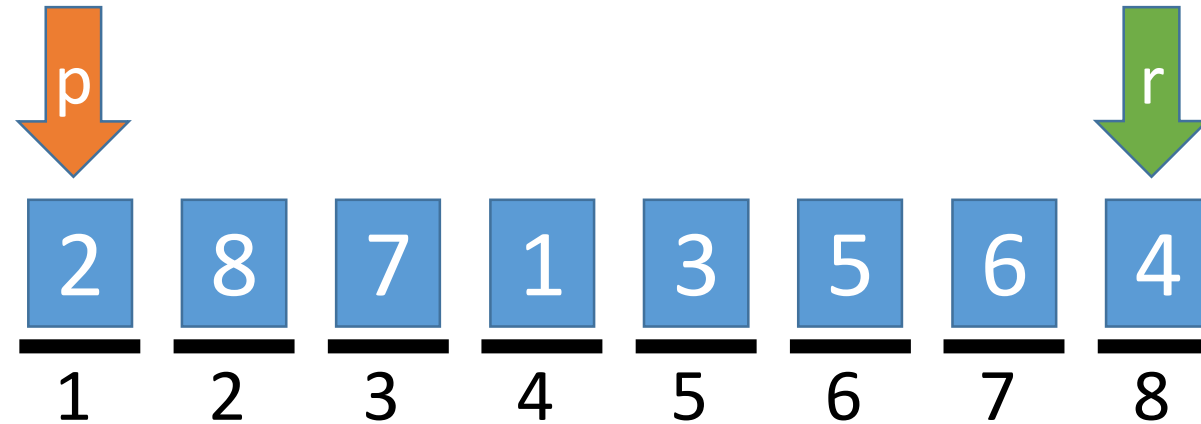
After partition
function:



WHAT IS THE SIGNIFICANCE OF A PIVOT?

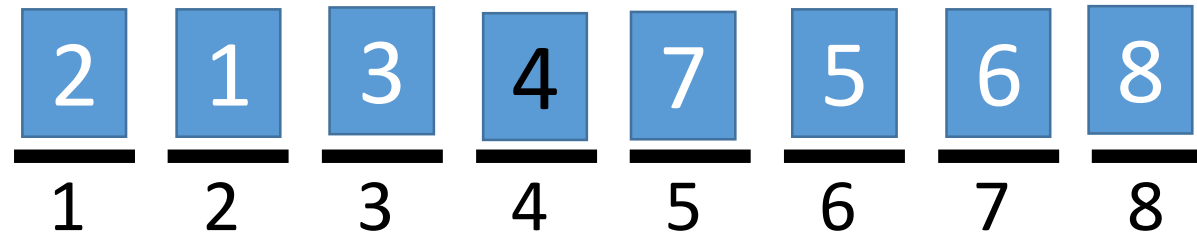
Partition Function (A, p, r)

Before partition
function:



NOTE:
Pivot is NOT
the center element

After partition
function:

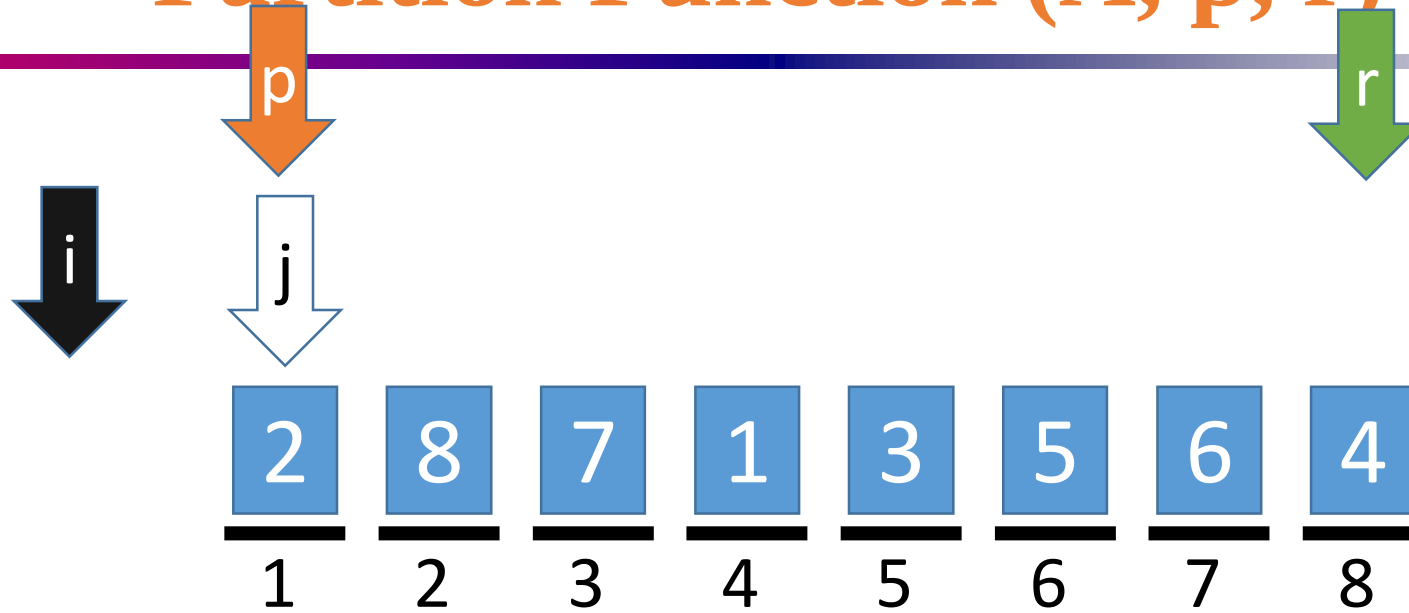


Pivot x =

4

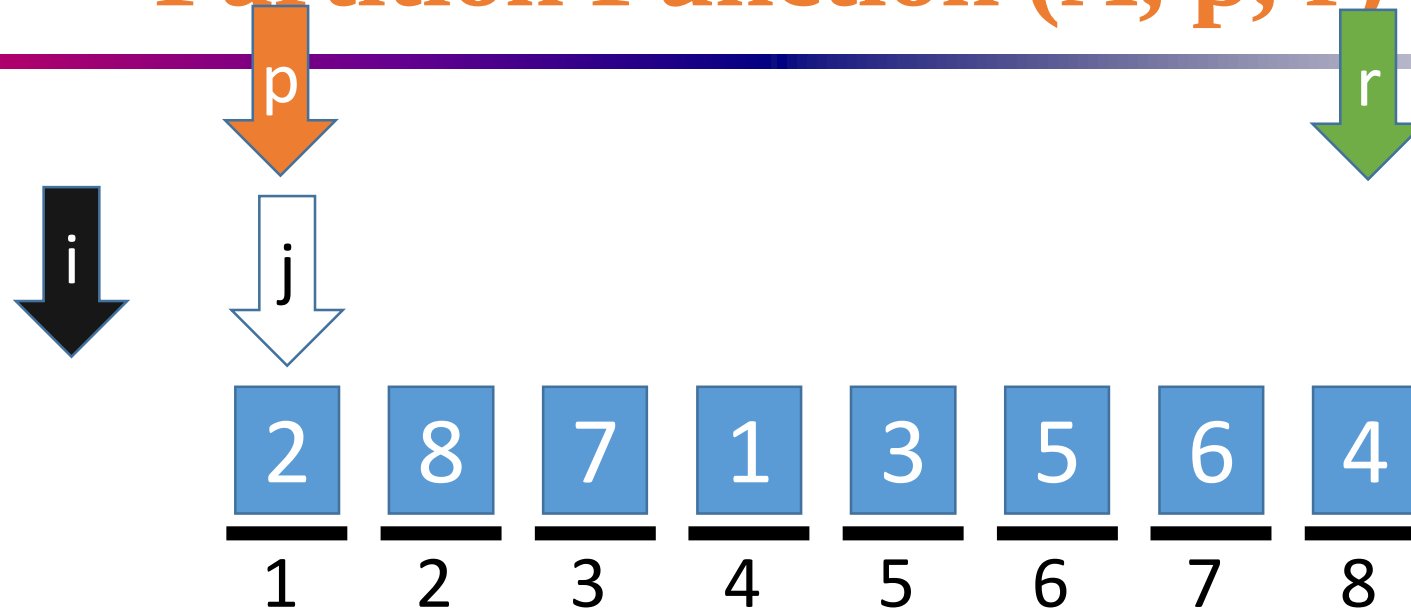
WHAT IS THE SIGNIFICANCE OF A PIVOT?

Partition Function (A, p, r)



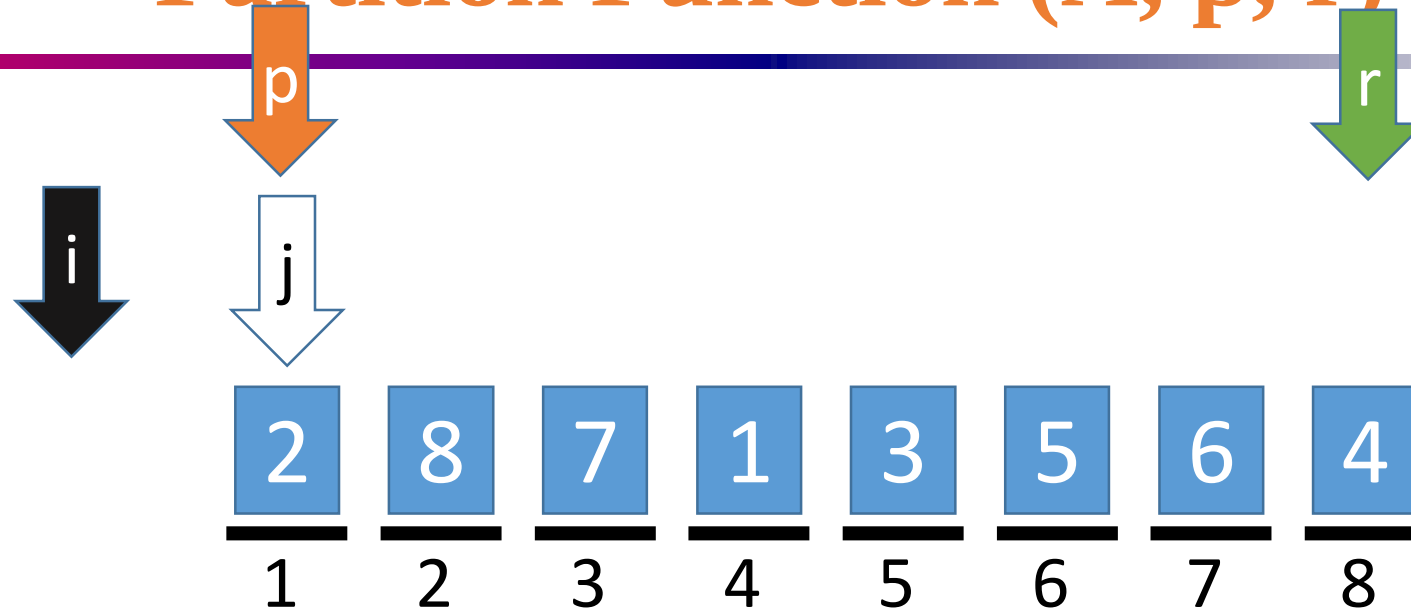
Pivot x = 4

Partition Function (A, p, r)



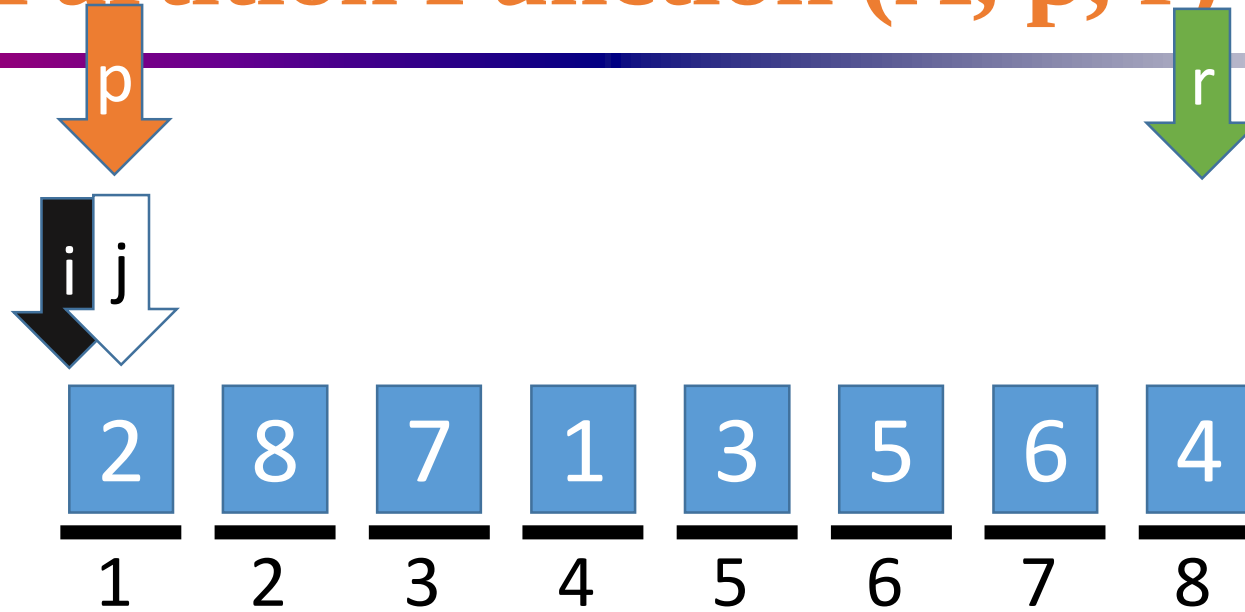
Pivot x = 4

Partition Function (A, p, r)



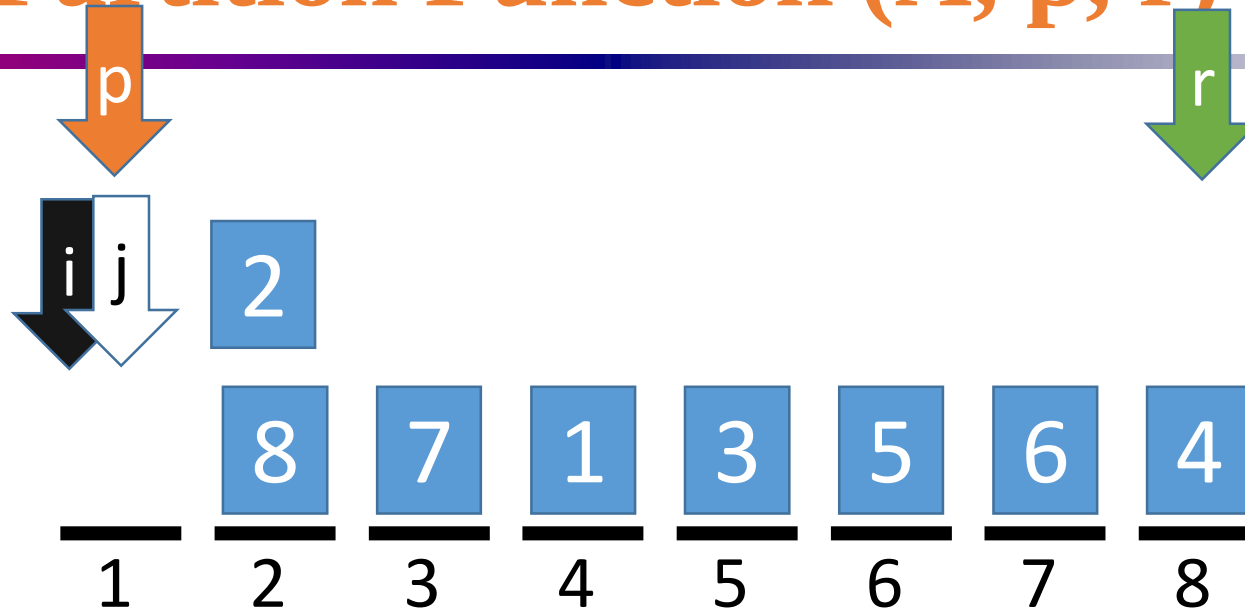
Pivot x = 4

Partition Function (A, p, r)



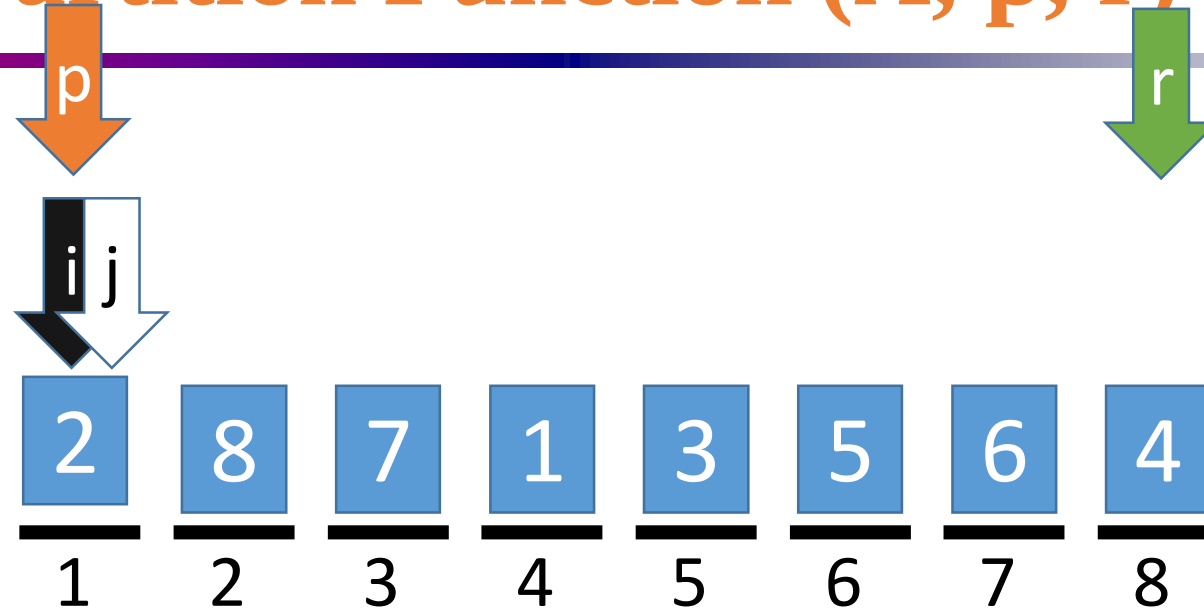
Pivot x = 4

Partition Function (A, p, r)



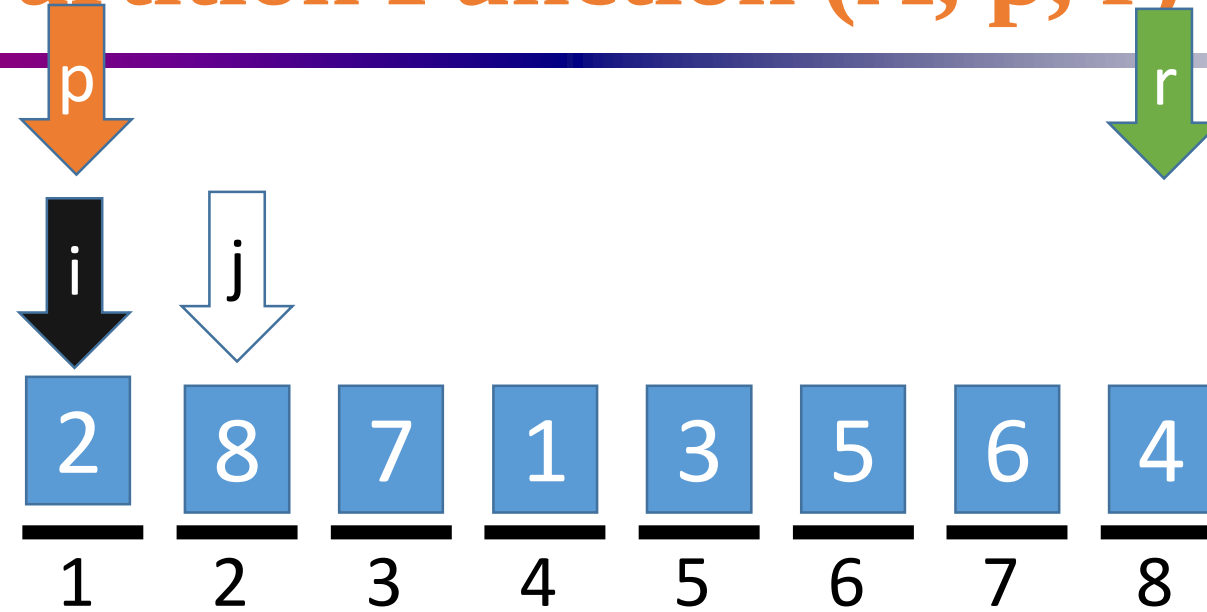
Pivot x = 4

Partition Function (A, p, r)



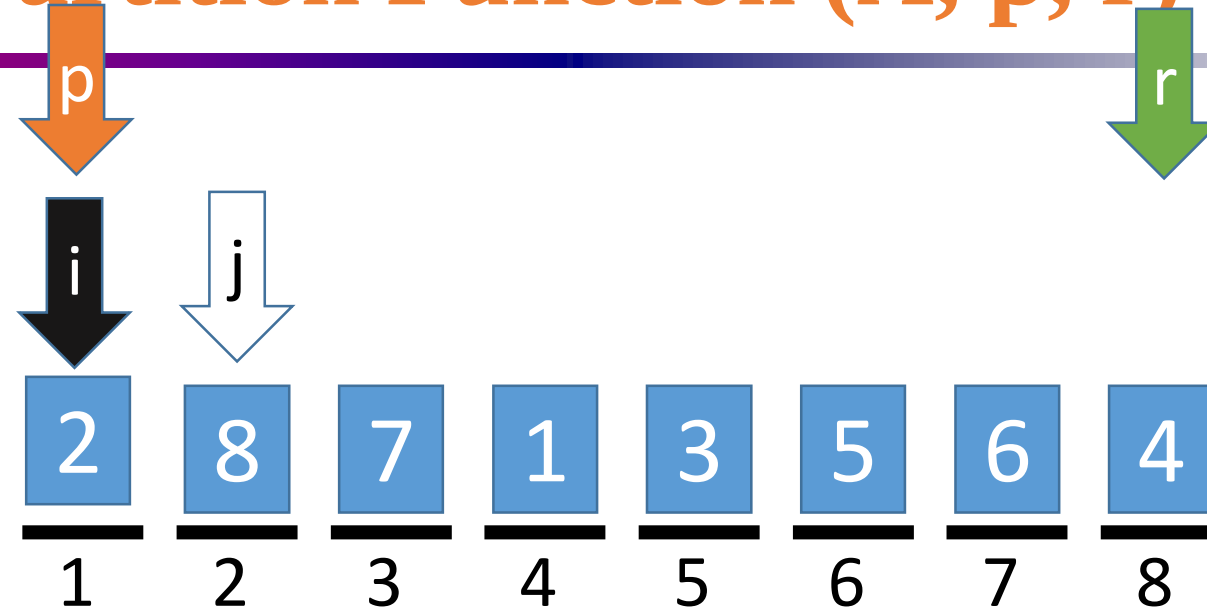
Pivot x = 4

Partition Function (A, p, r)



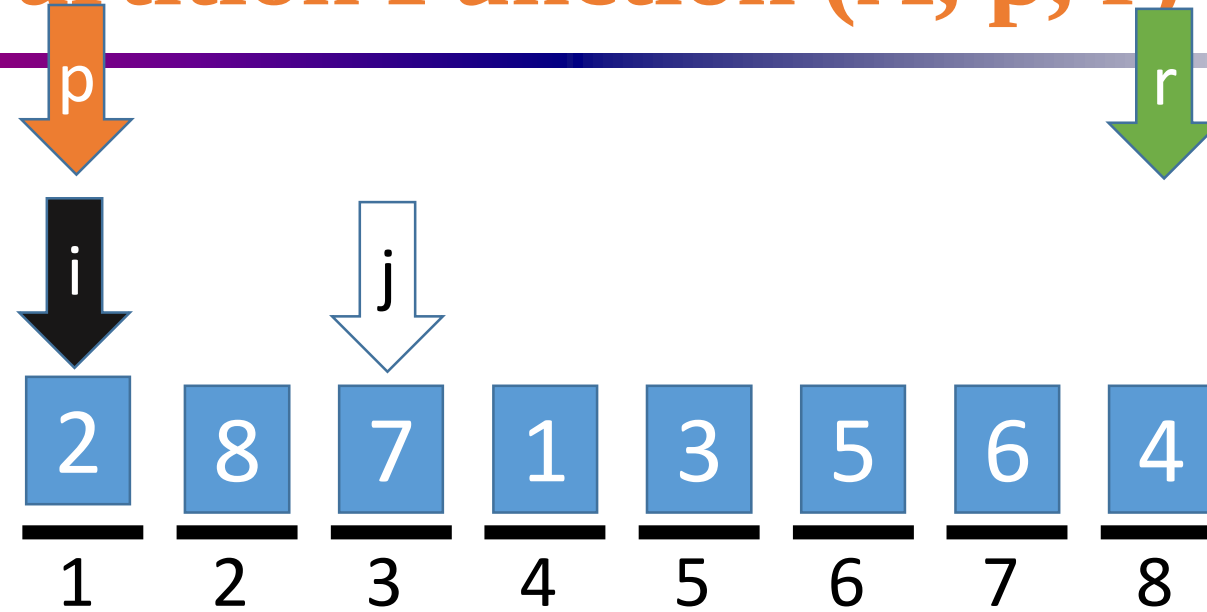
Pivot x = 4

Partition Function (A, p, r)



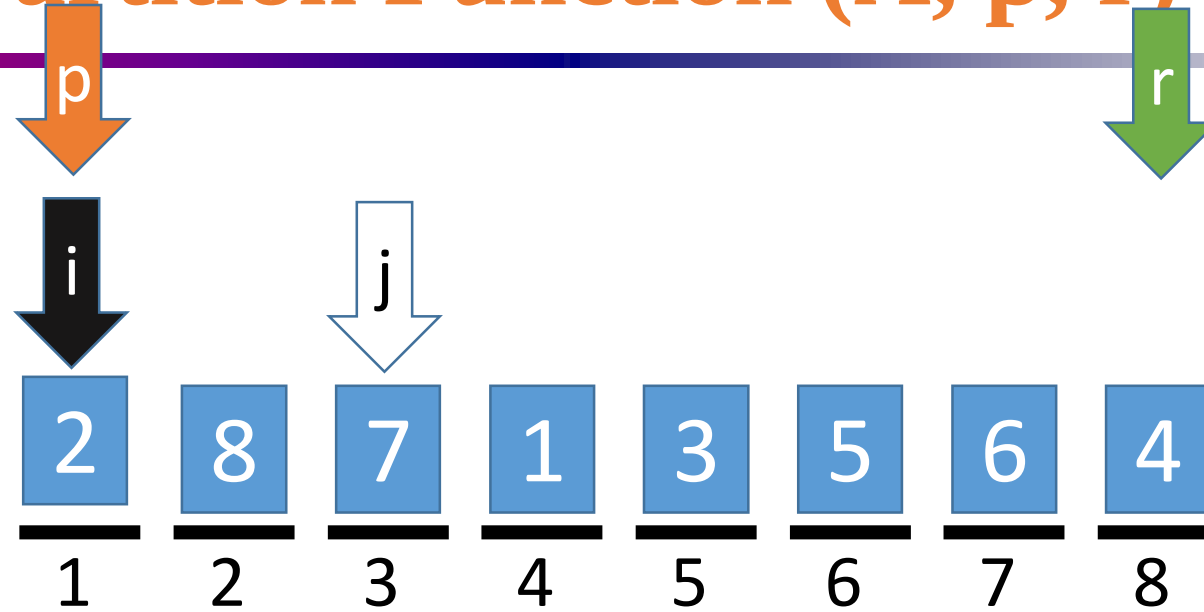
Pivot x = 4

Partition Function (A, p, r)



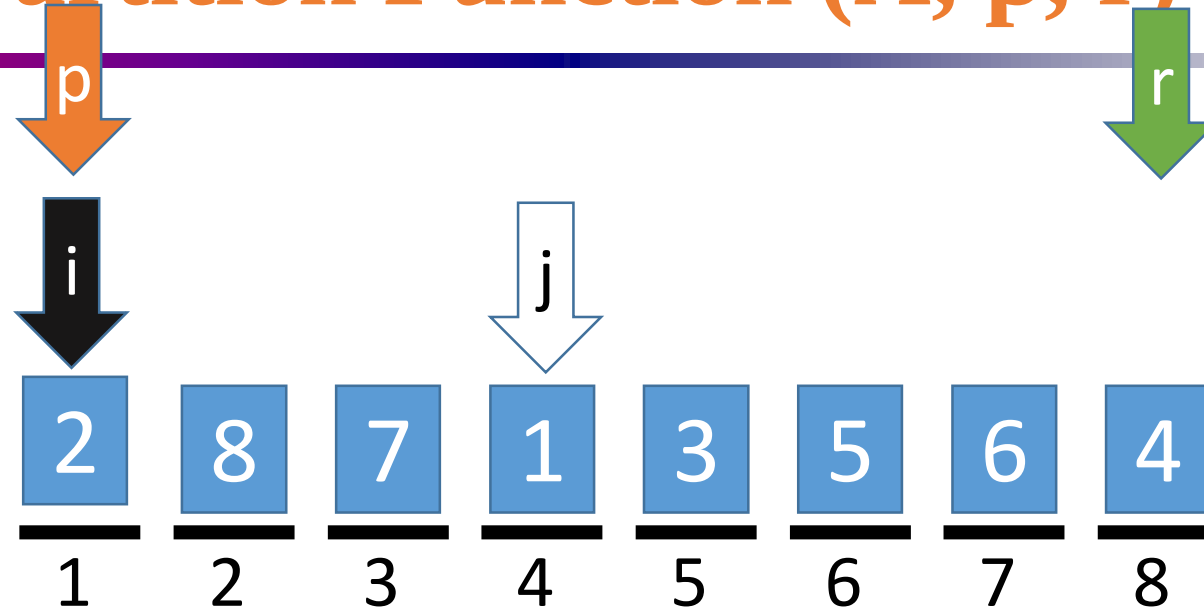
Pivot x = 4

Partition Function (A, p, r)



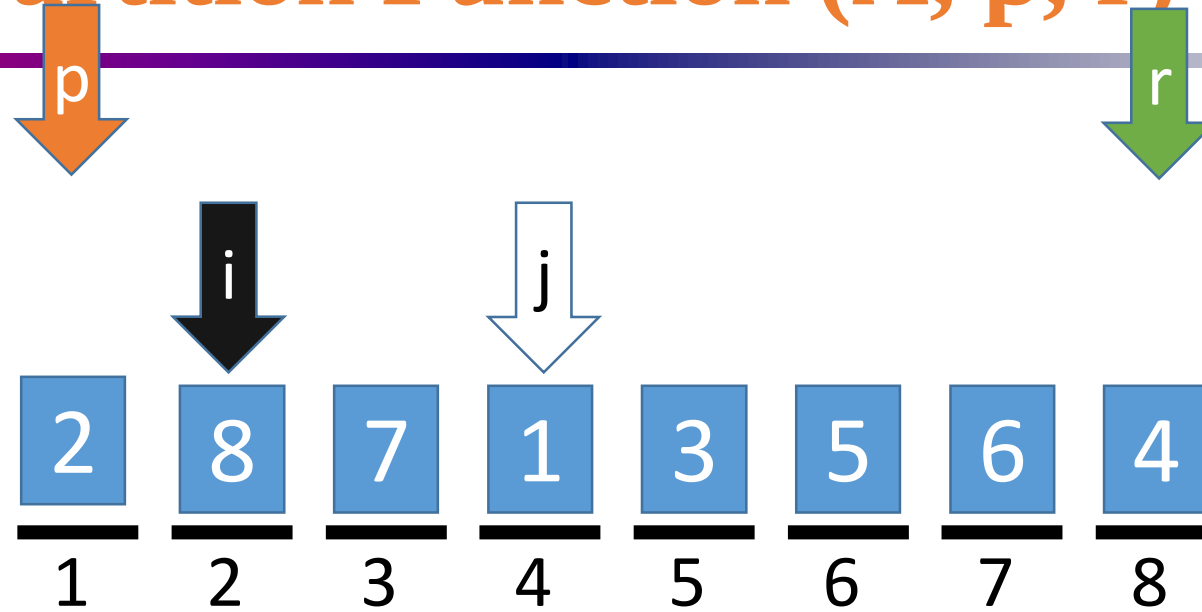
Pivot x = 4

Partition Function (A, p, r)



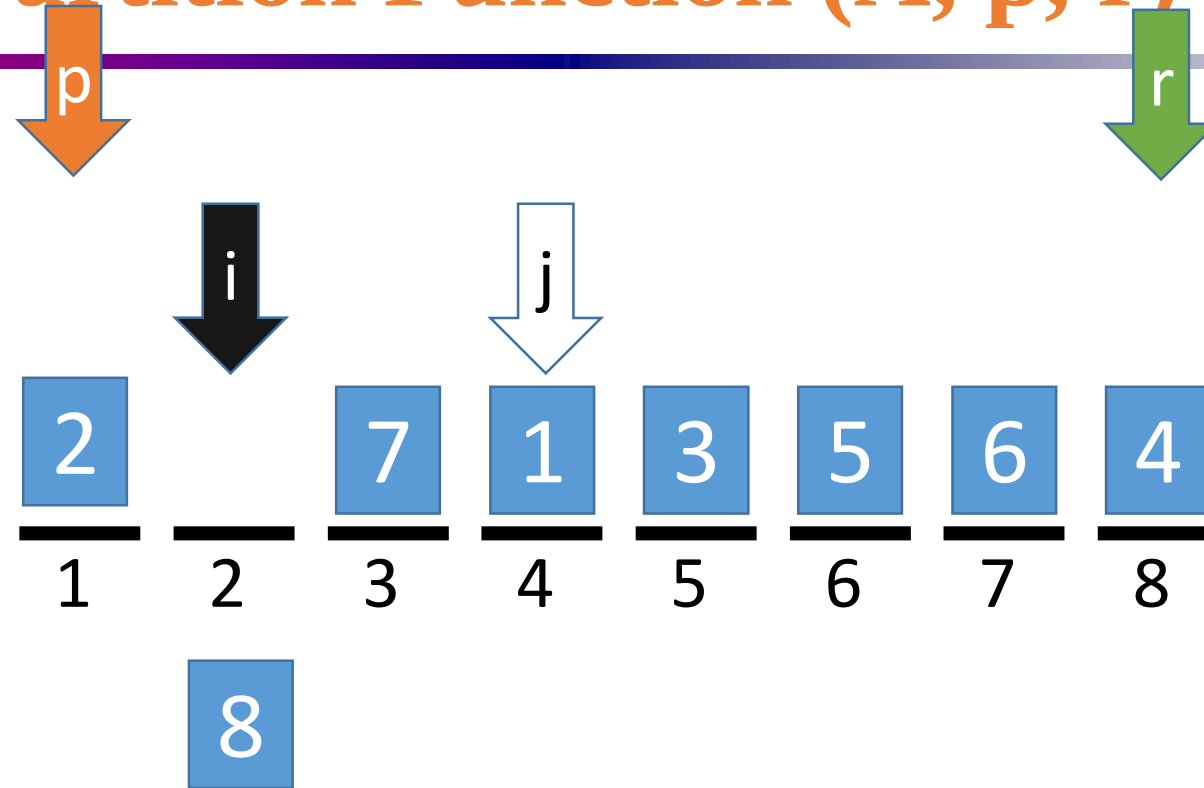
Pivot x = 4

Partition Function (A, p, r)



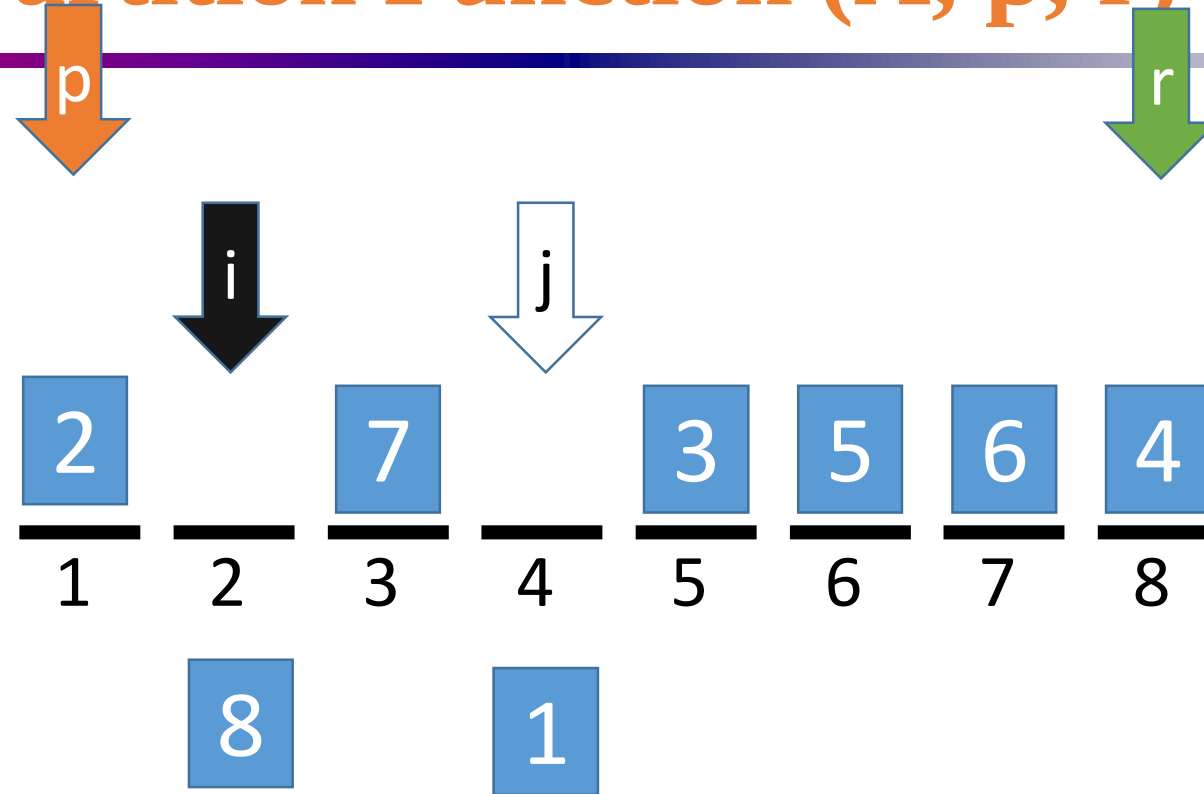
Pivot x = 4

Partition Function (A, p, r)



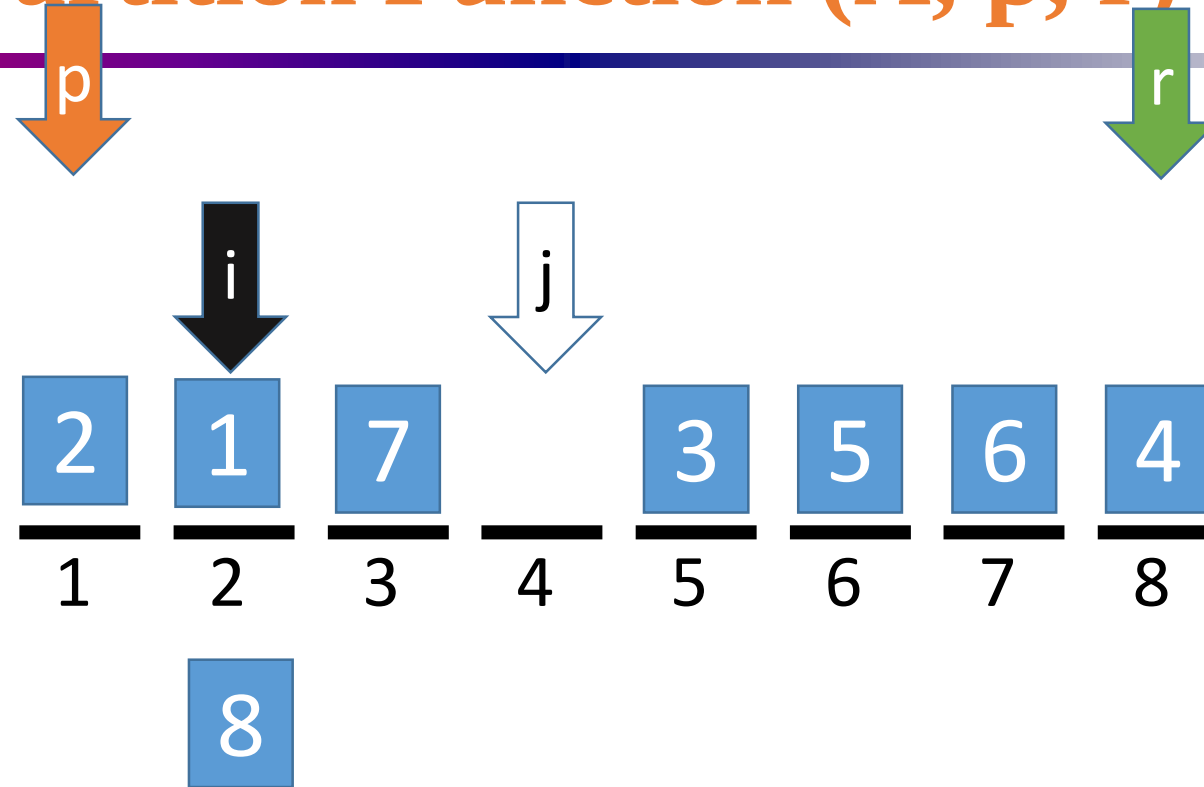
Pivot x = 4

Partition Function (A, p, r)



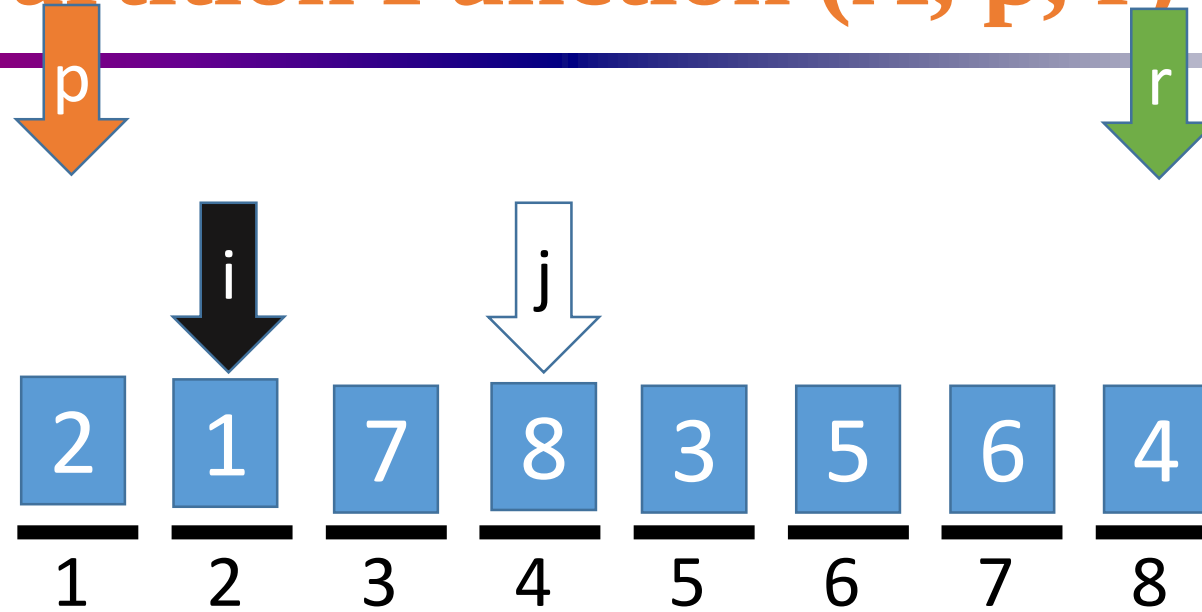
Pivot x = 4

Partition Function (A, p, r)



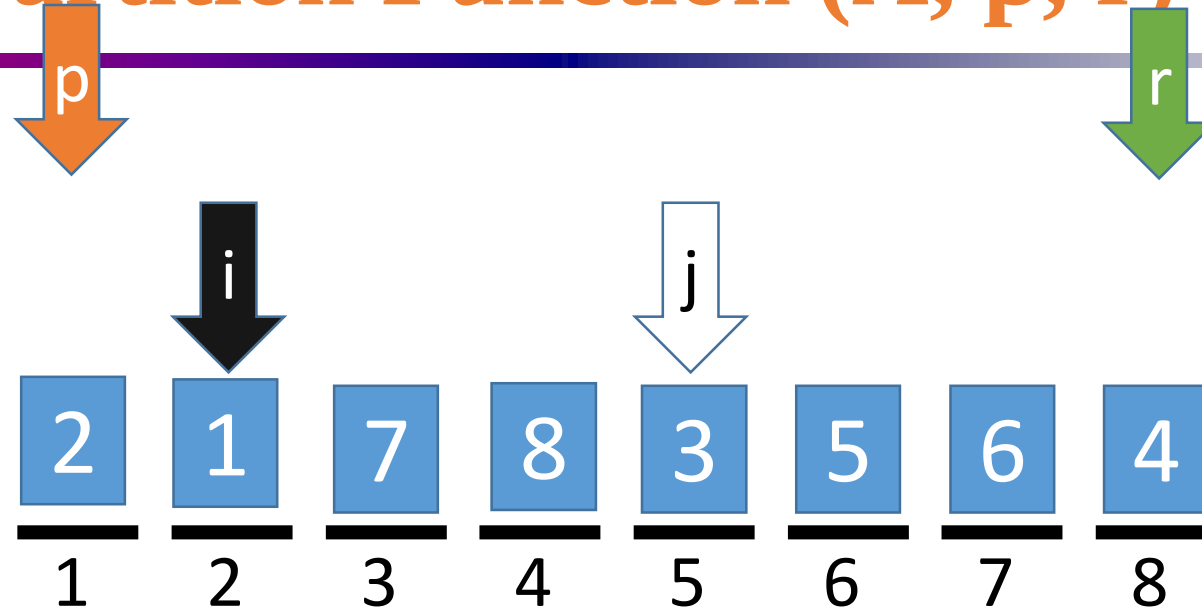
Pivot x = 4

Partition Function (A, p, r)



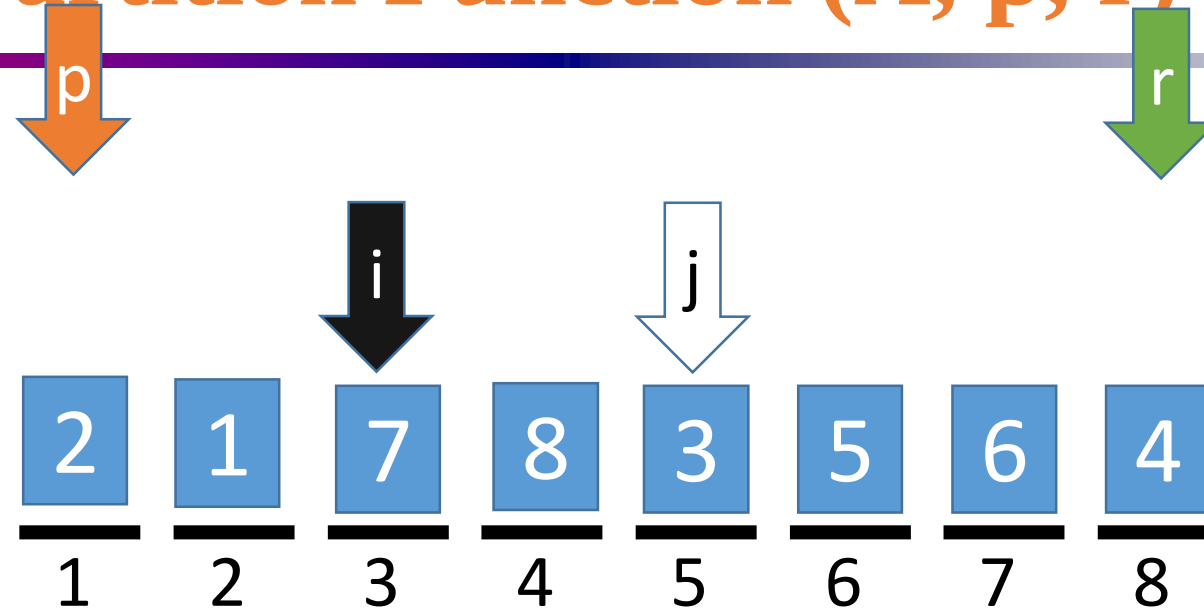
Pivot $x =$ 4

Partition Function (A, p, r)



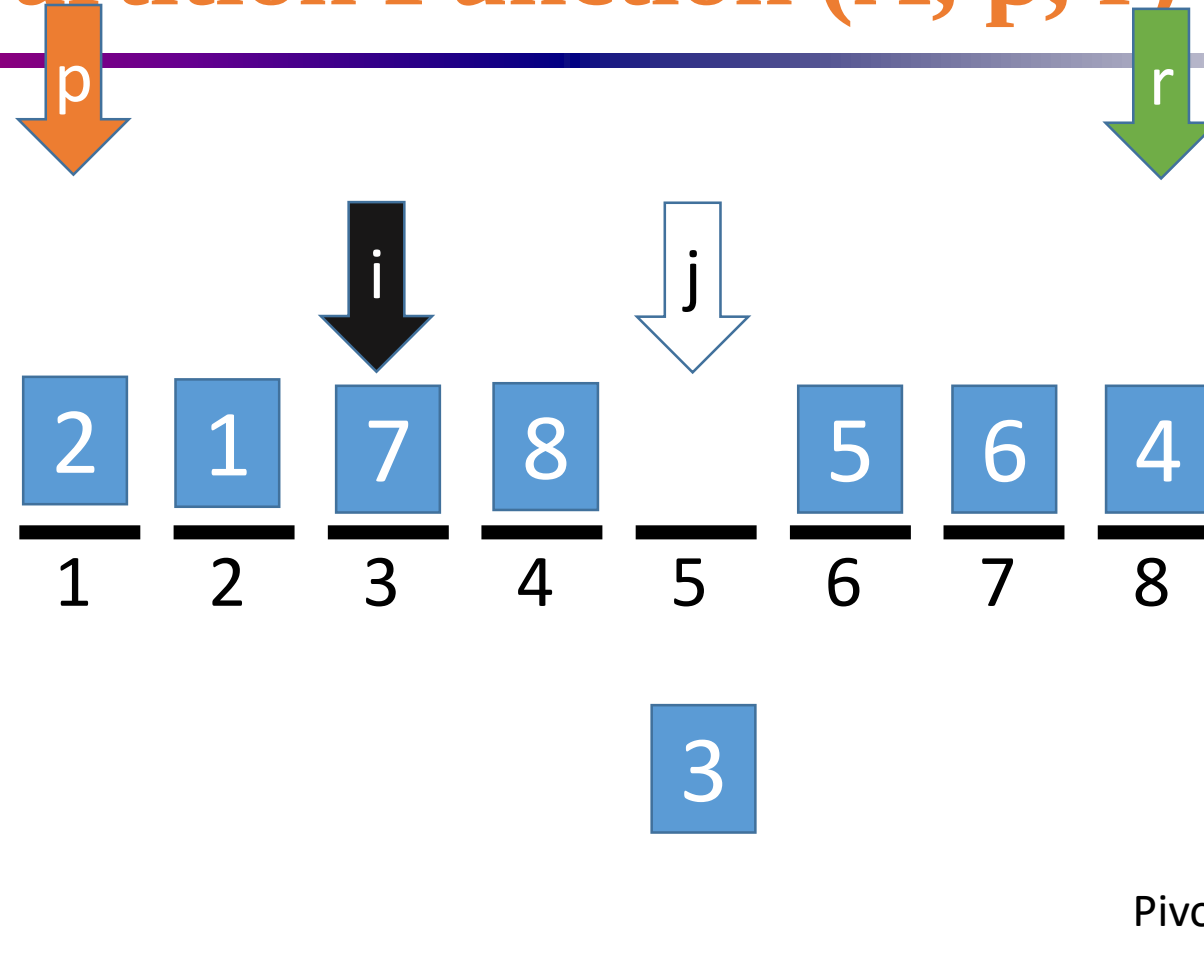
Pivot $x =$ 4

Partition Function (A, p, r)

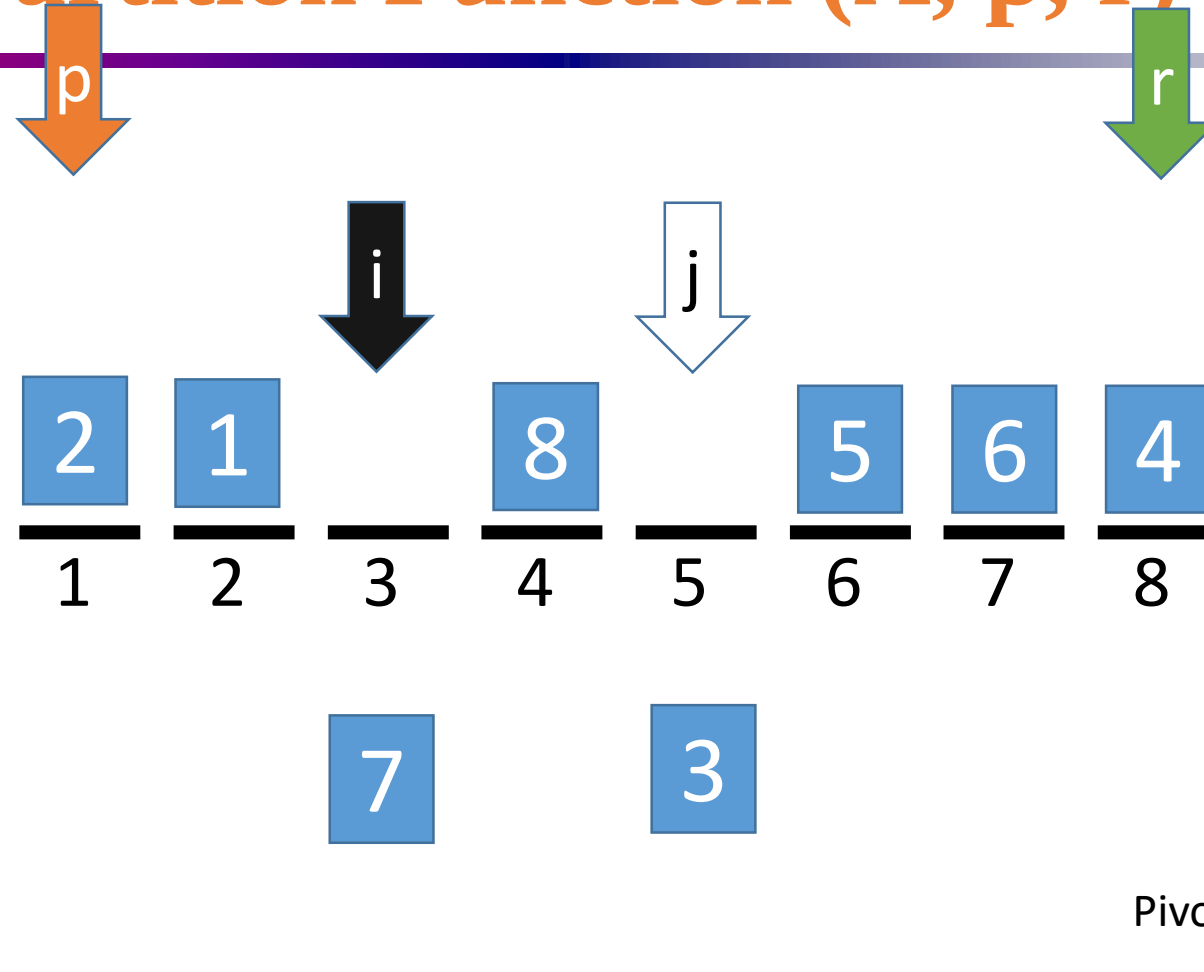


Pivot $x =$ 4

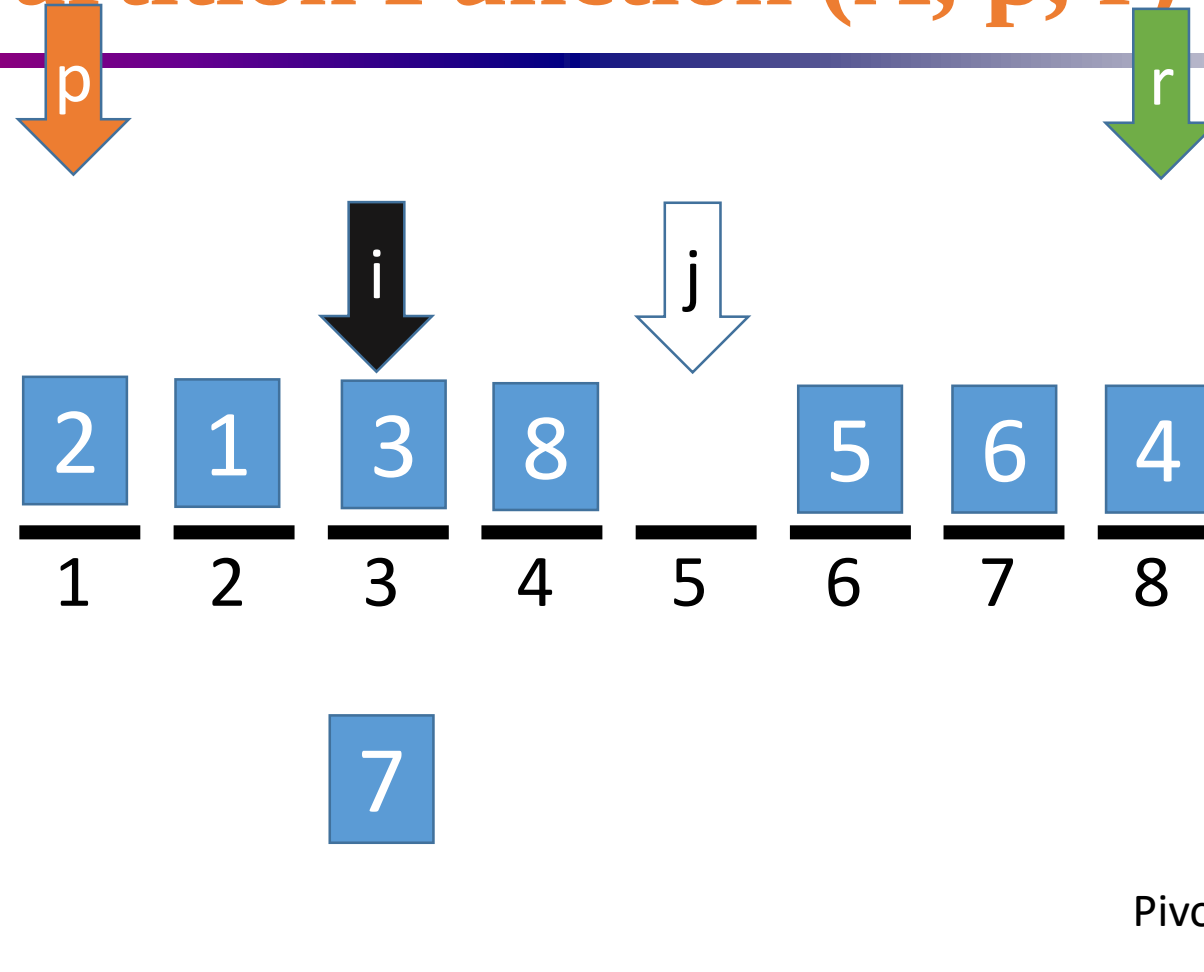
Partition Function (A, p, r)



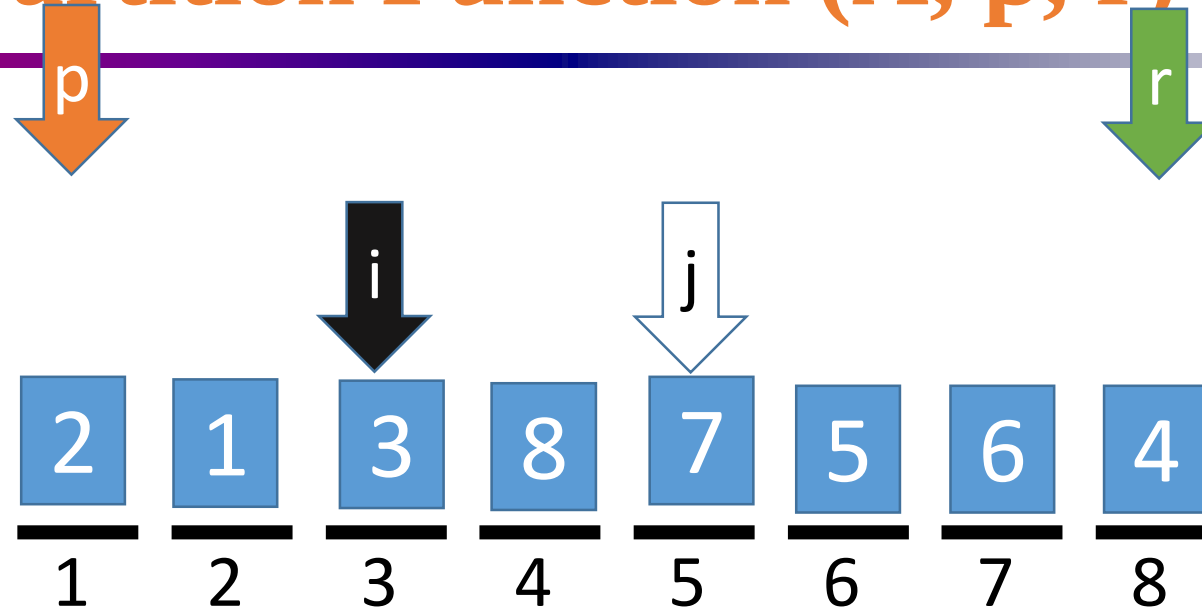
Partition Function (A, p, r)



Partition Function (A, p, r)

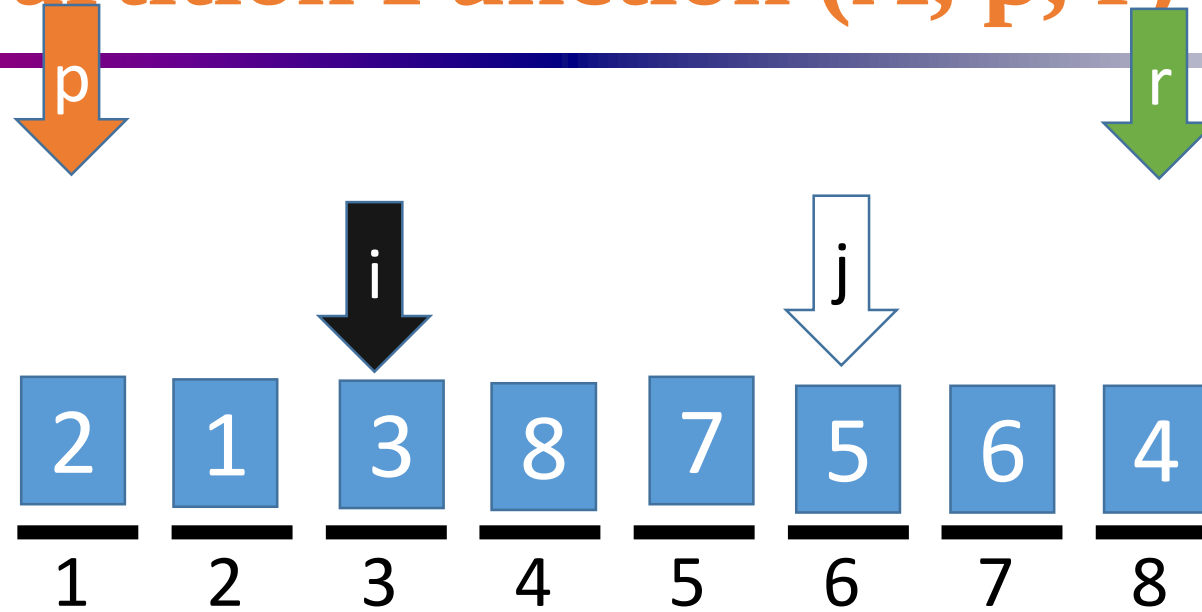


Partition Function (A, p, r)



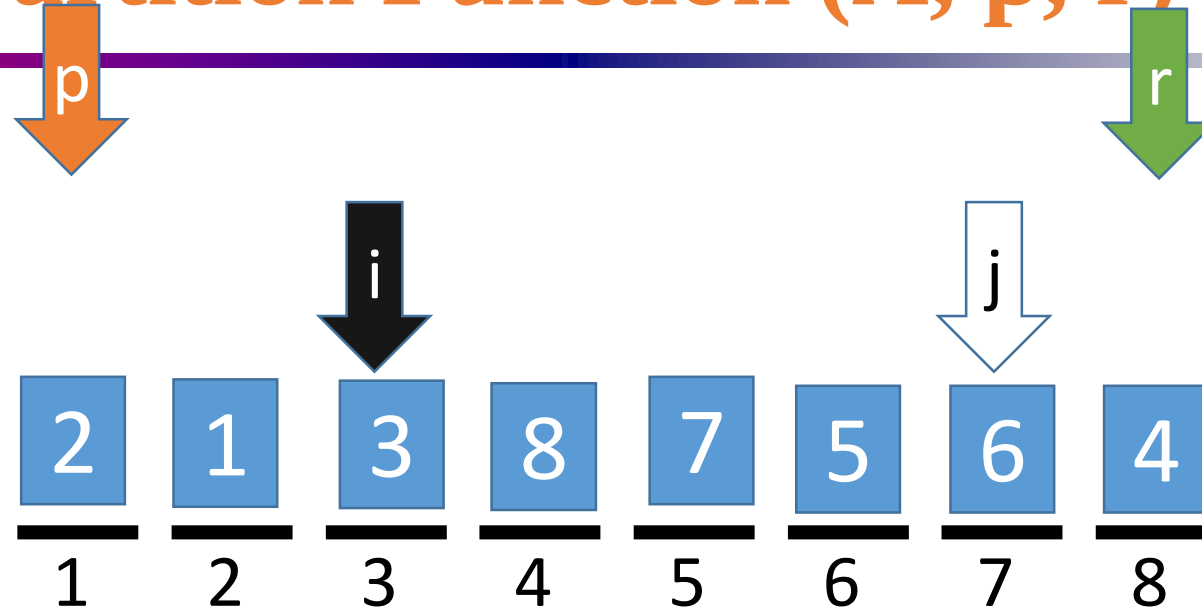
Pivot x = 4

Partition Function (A, p, r)



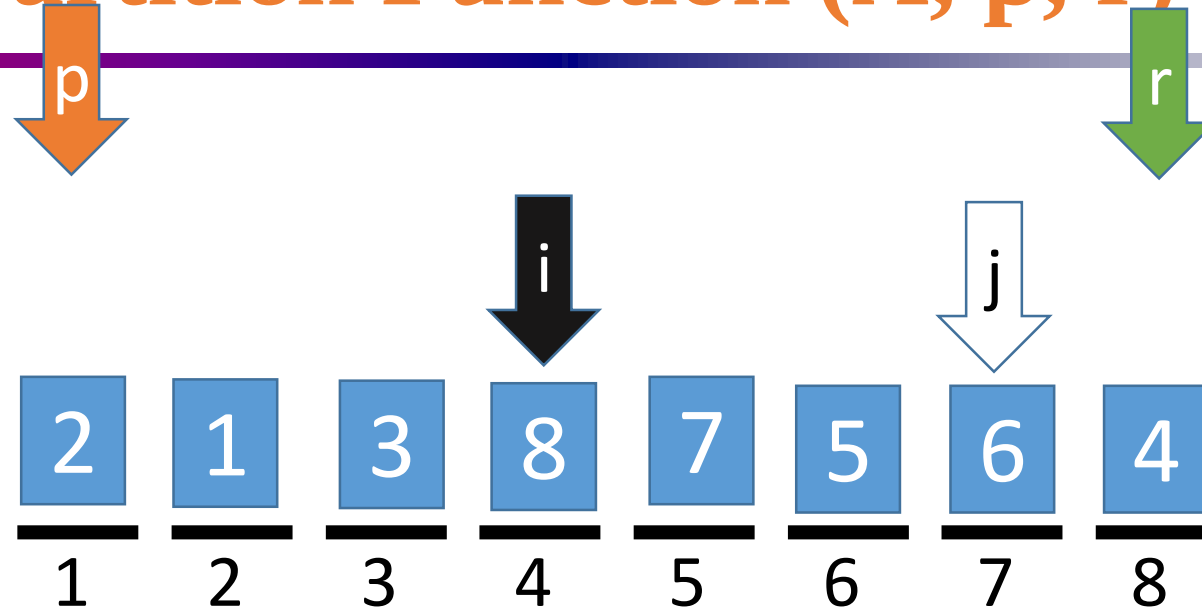
Pivot x = 4

Partition Function (A, p, r)



Pivot x = 4

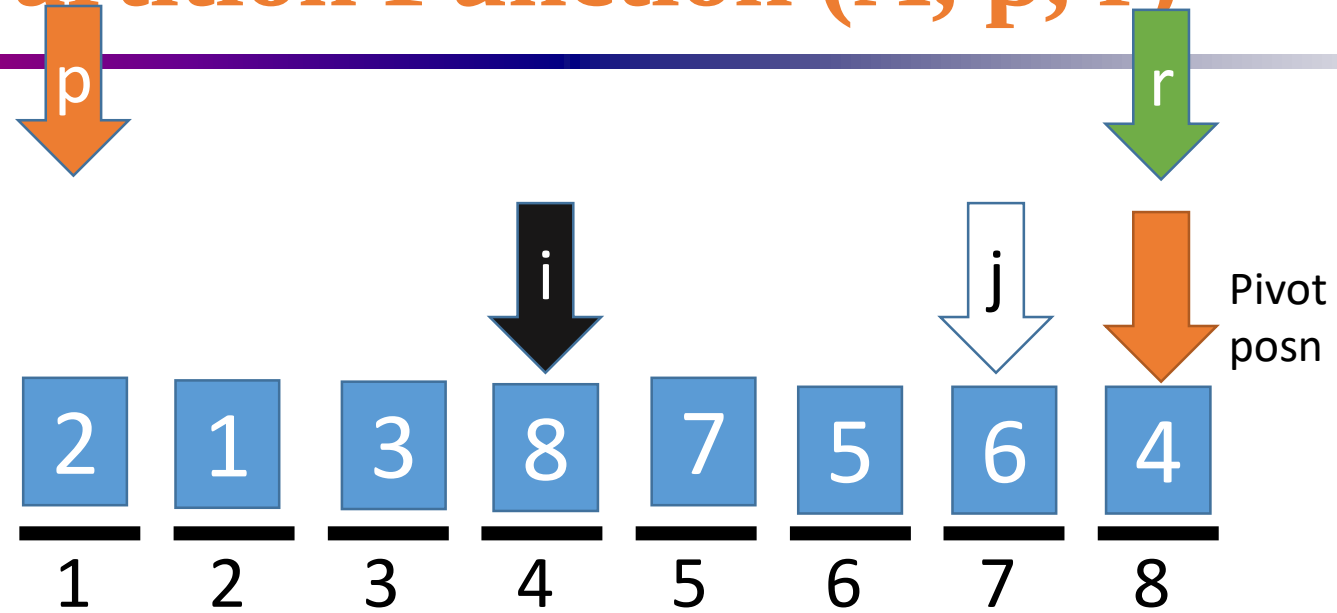
Partition Function (A, p, r)



Pivot x =

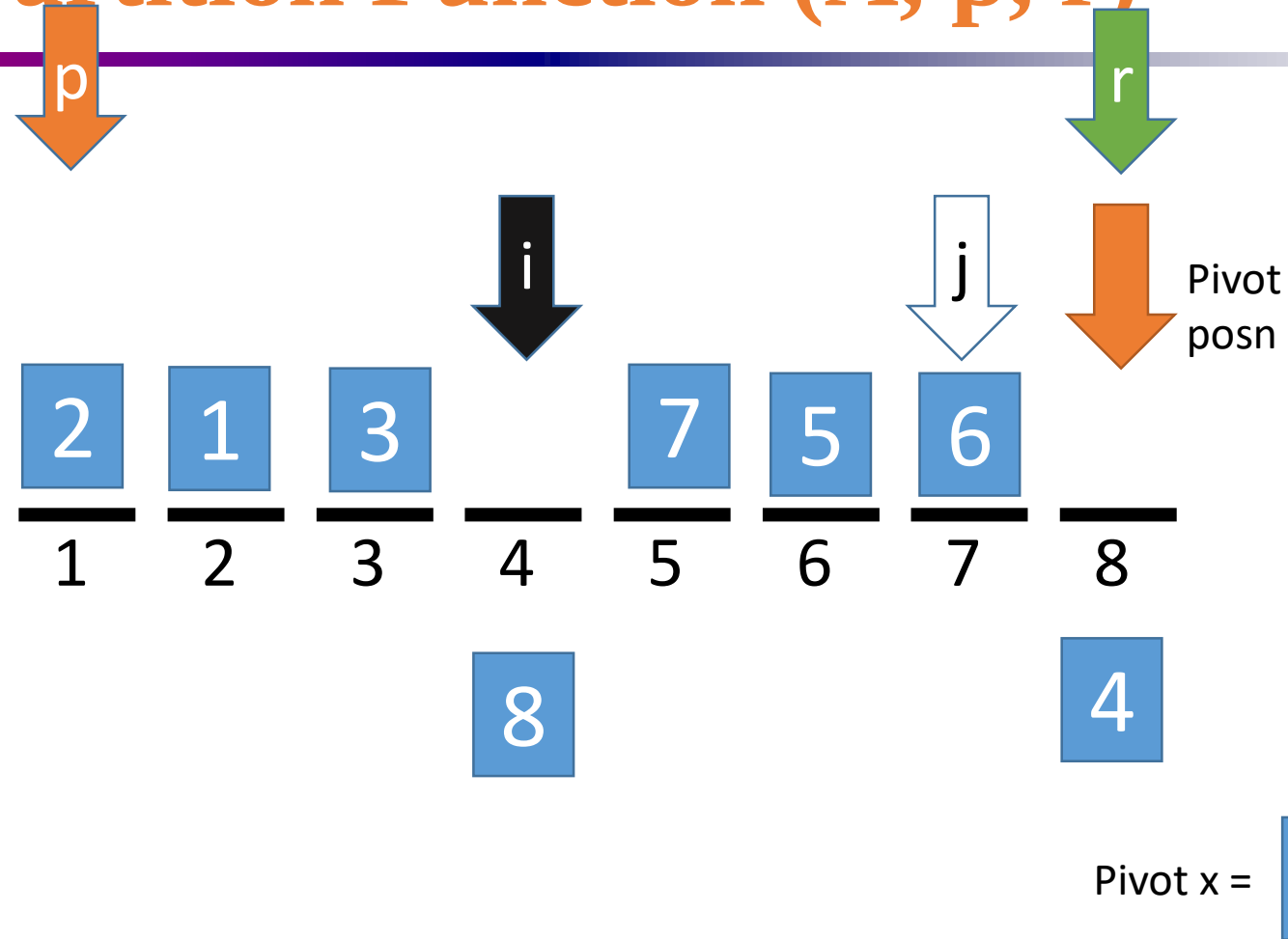
4

Partition Function (A, p, r)

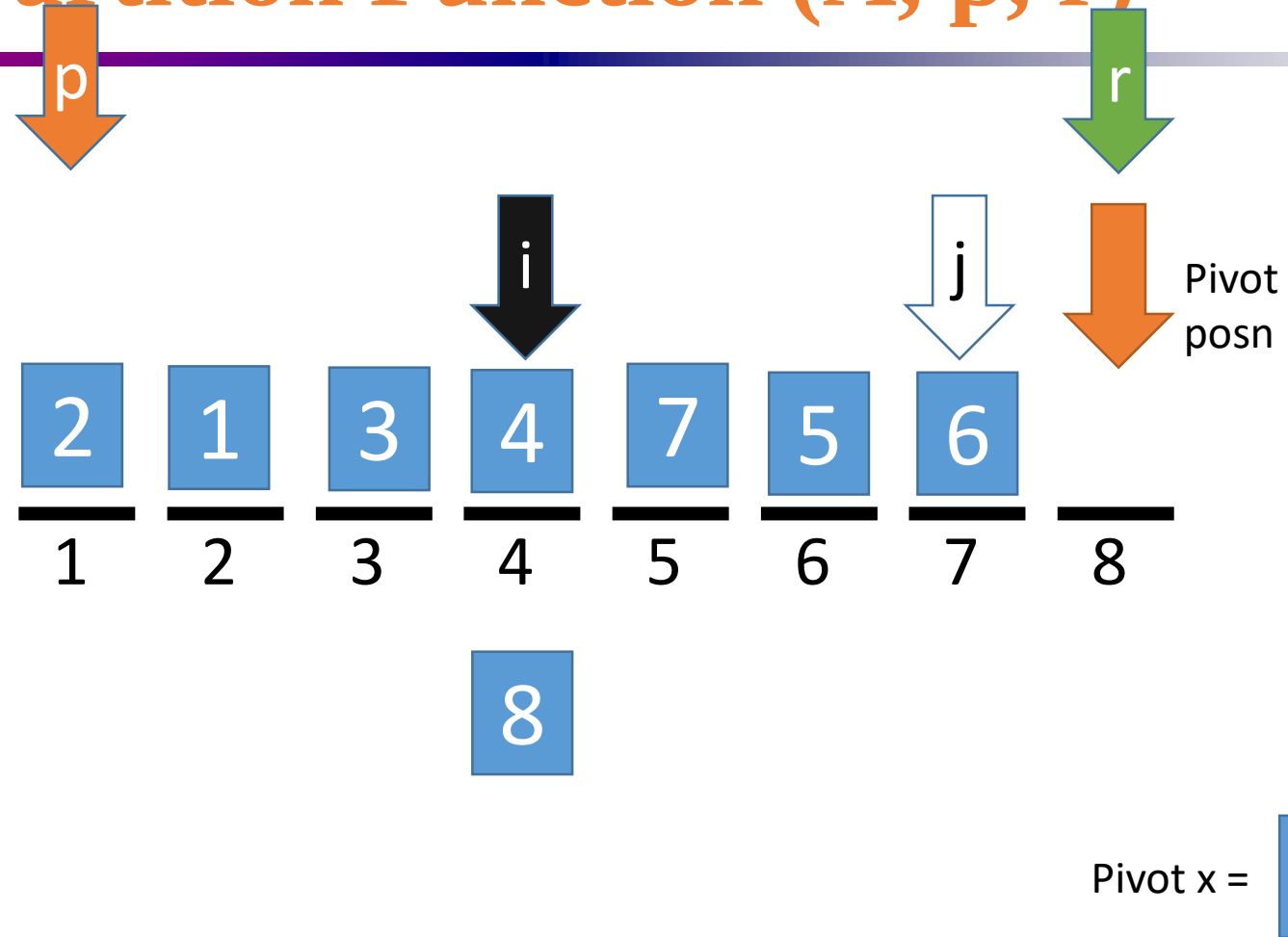


Pivot $x =$ 4

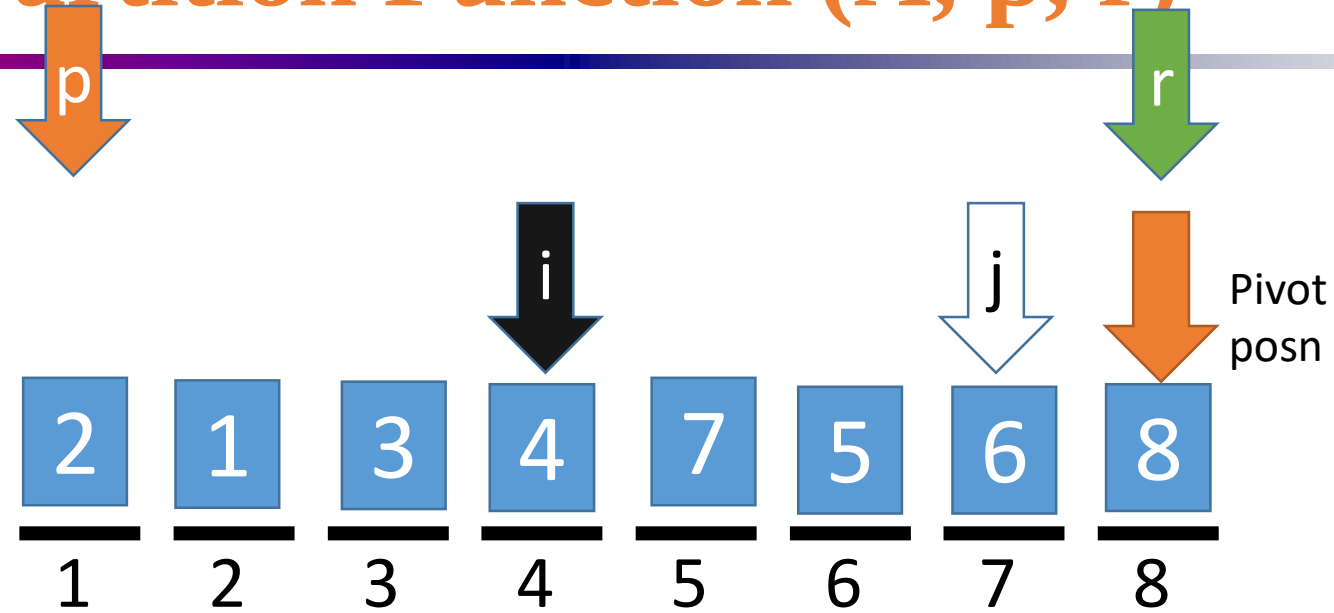
Partition Function (A, p, r)



Partition Function (A, p, r)

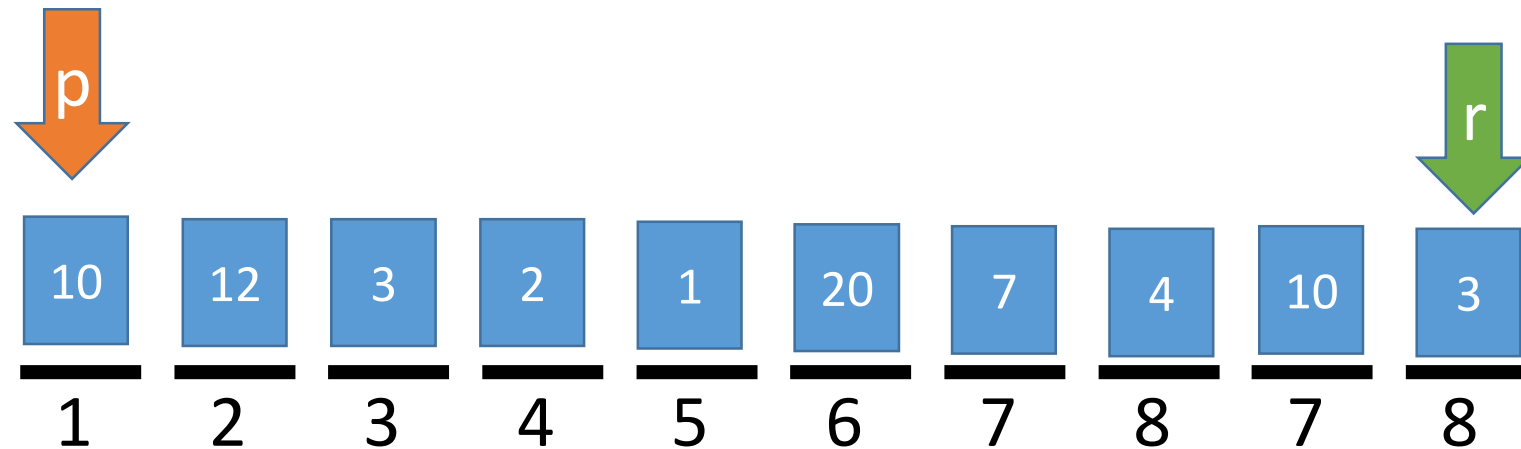


Partition Function (A, p, r)

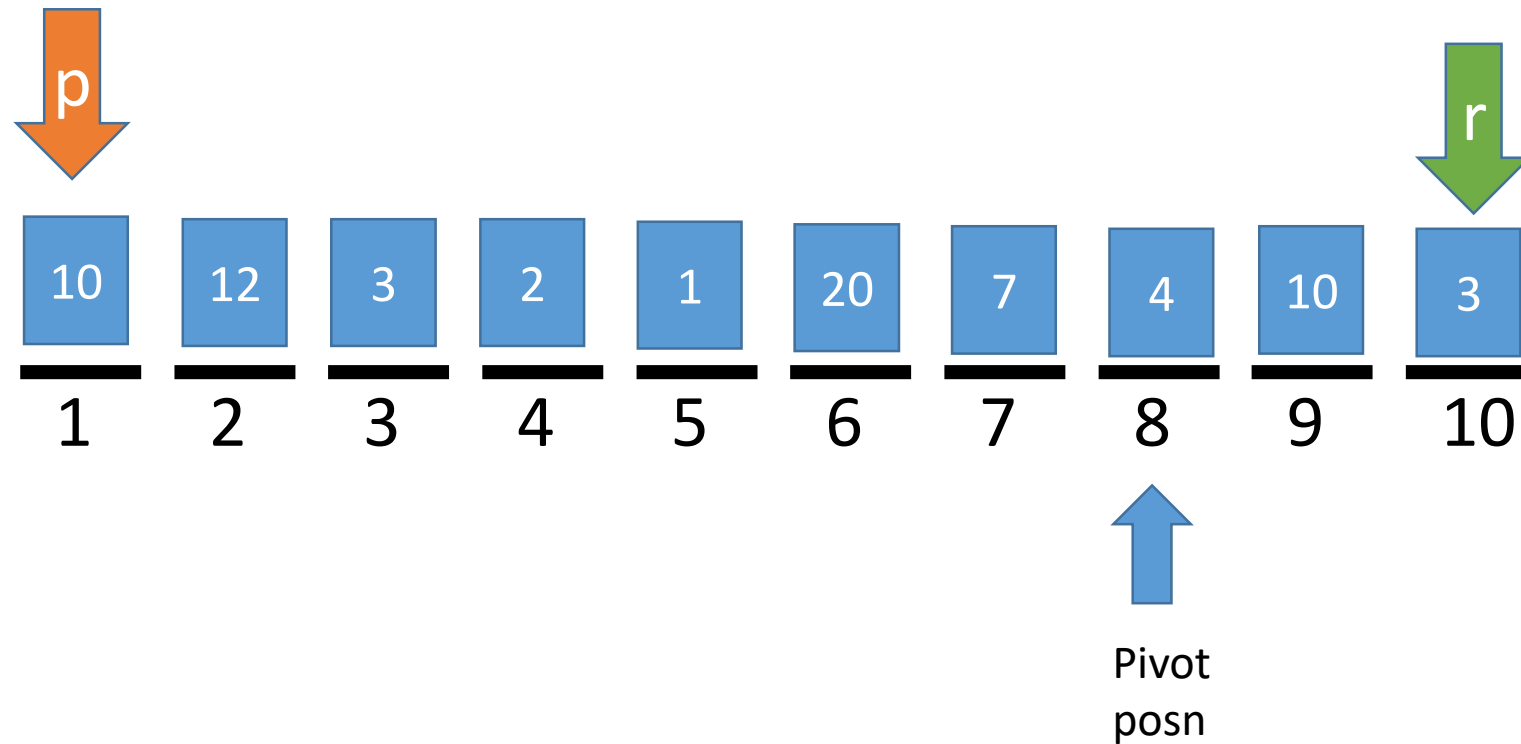


Pivot $x =$ 4

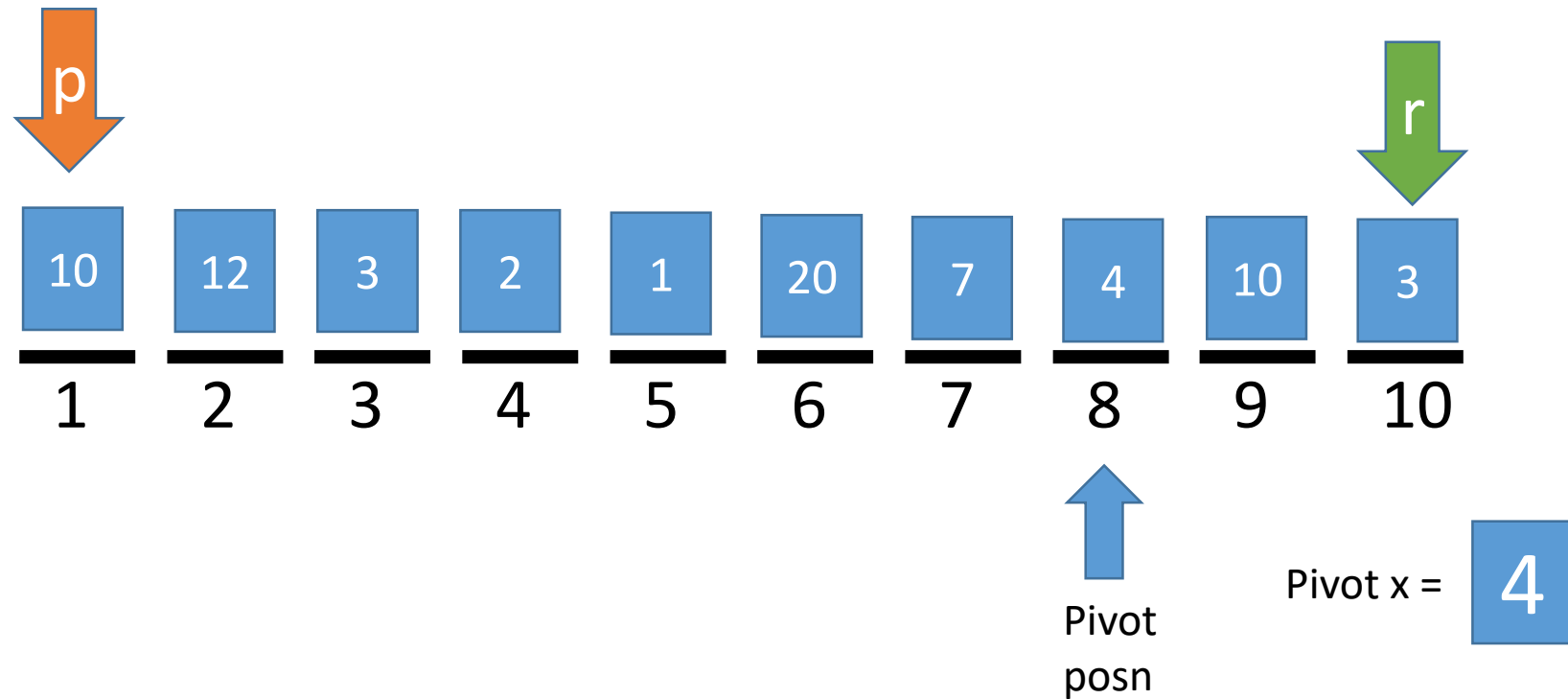
Partition Function (A, p, r)



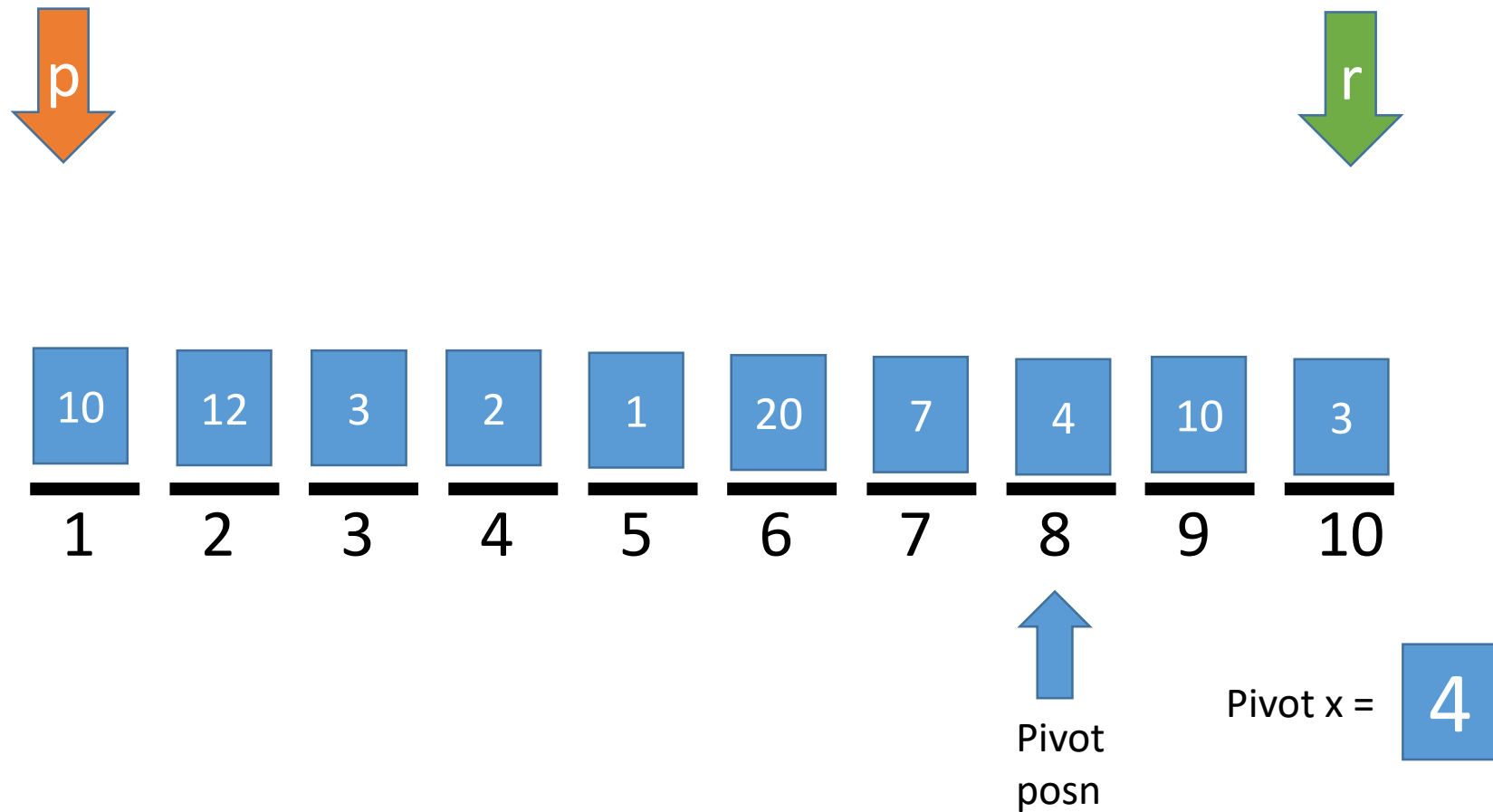
Partition Function (A, p, r)



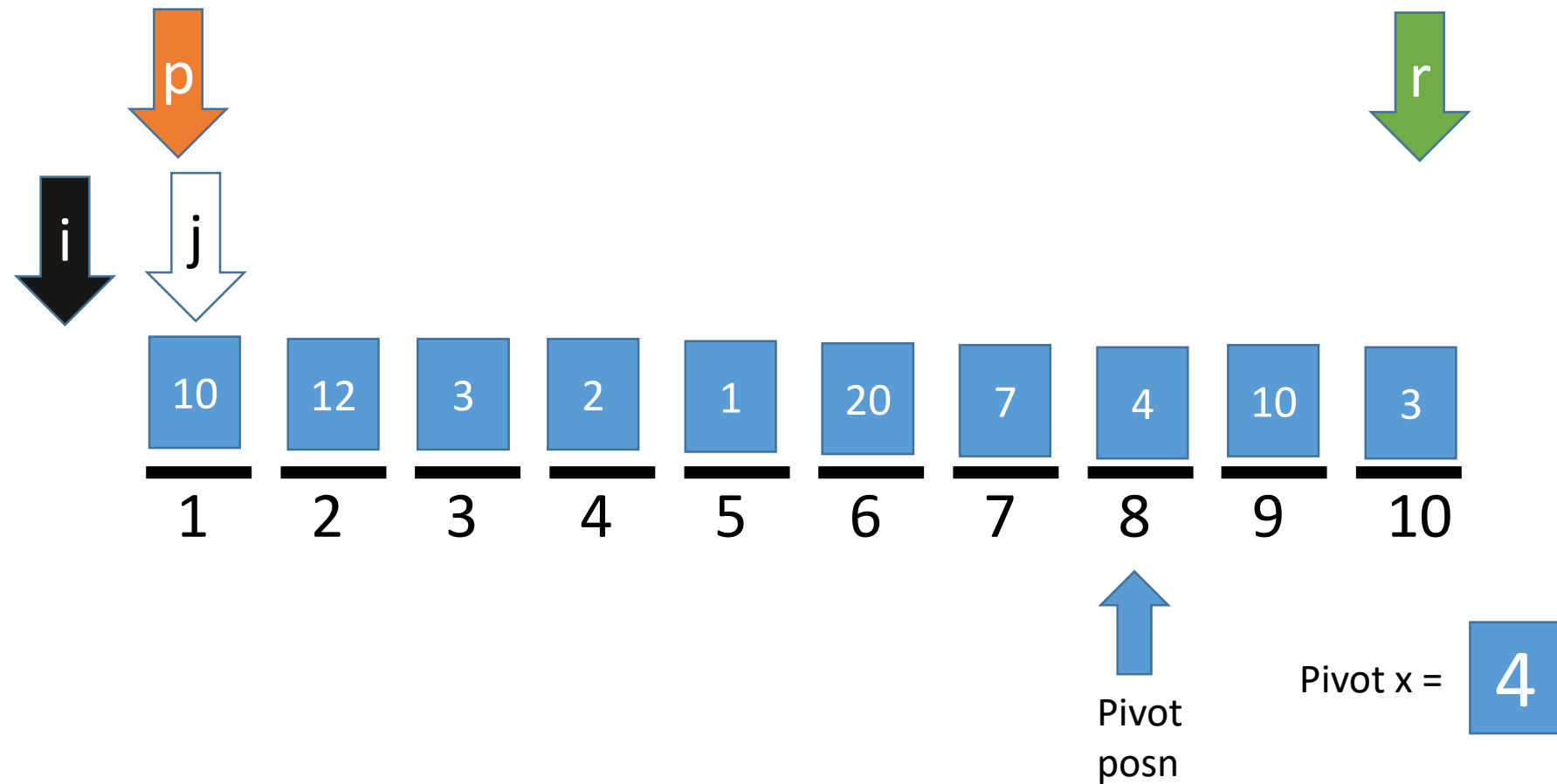
Partition Function (A, p, r)



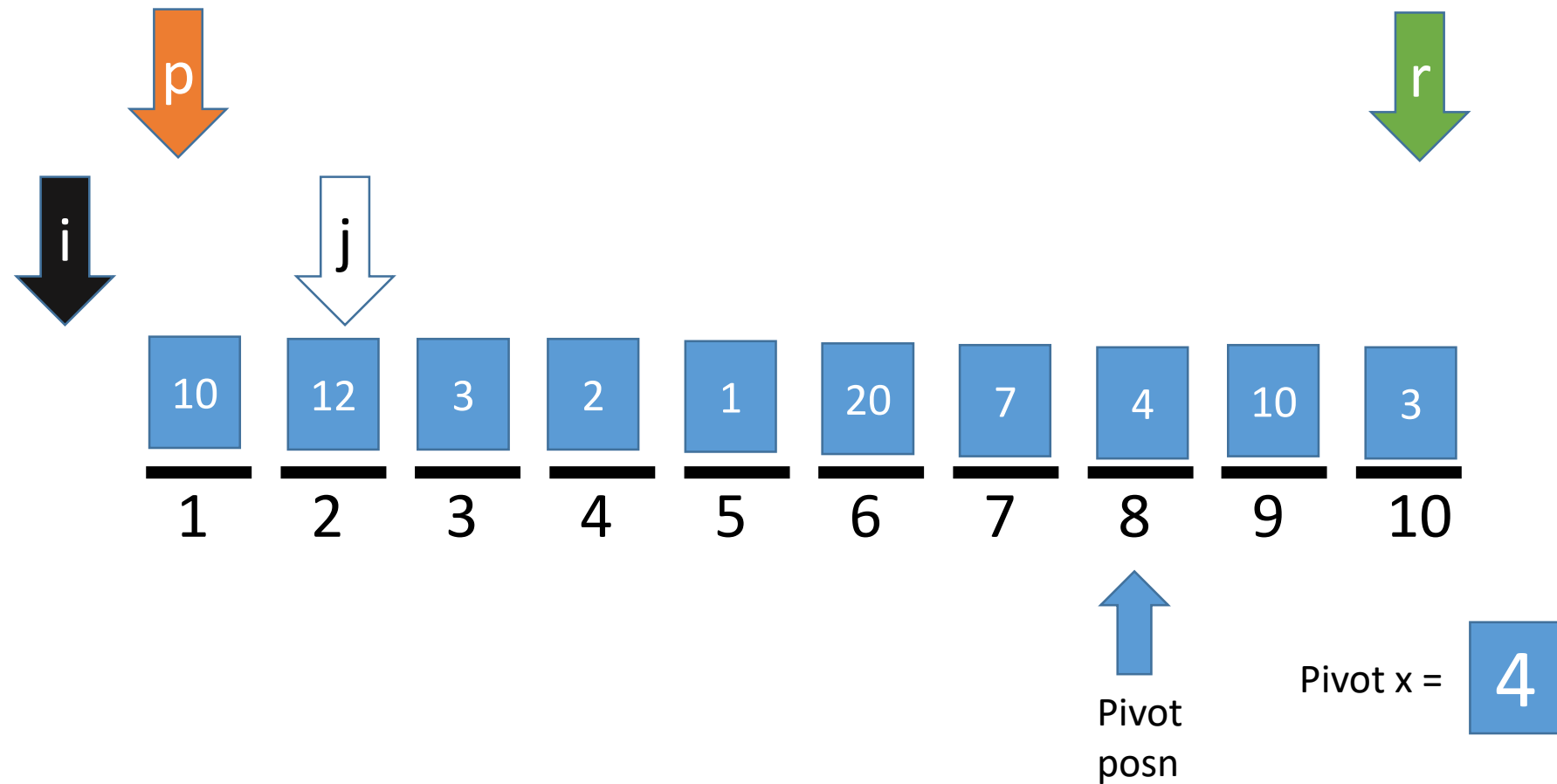
Partition Function (A, p, r)



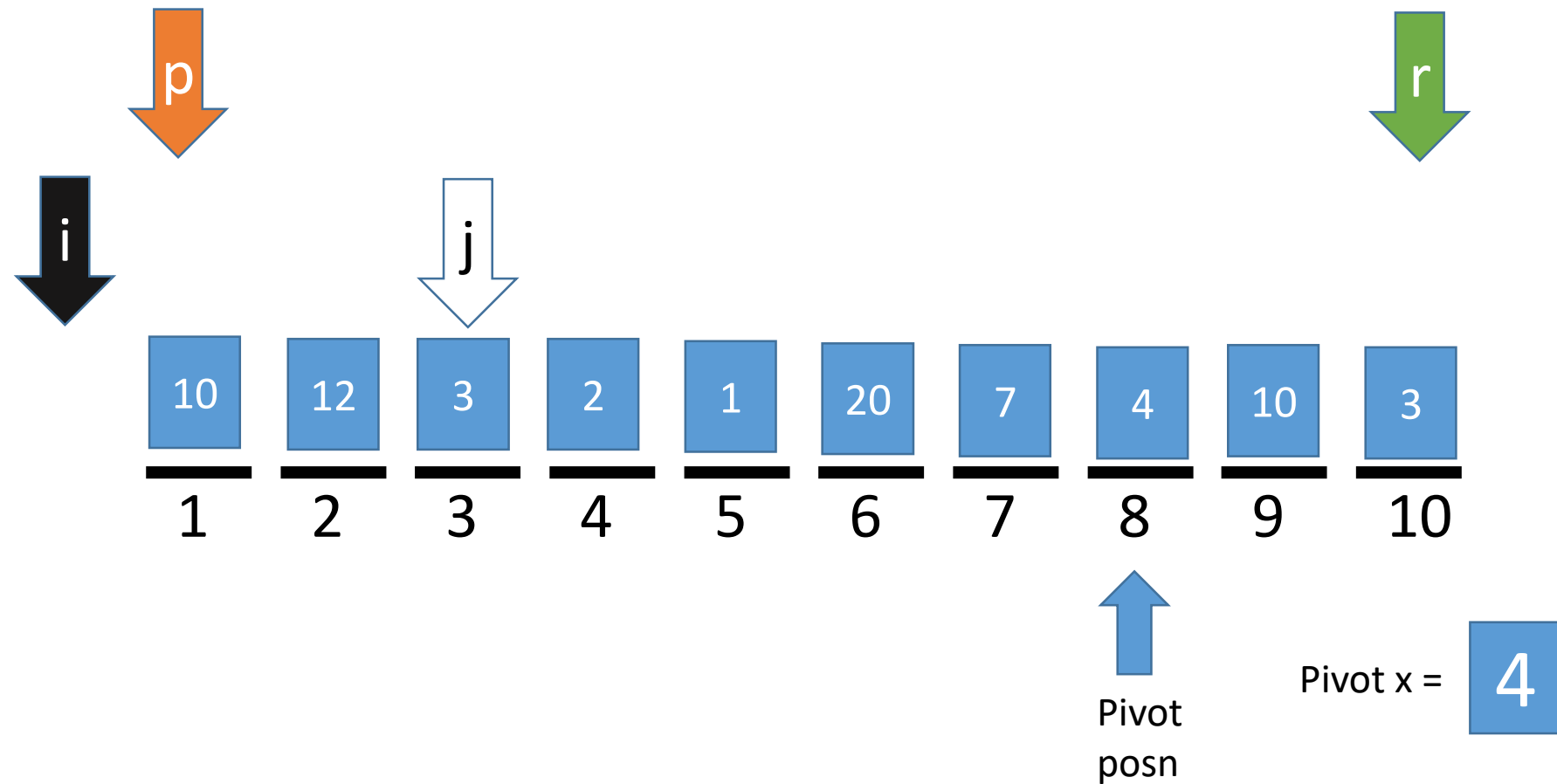
Partition Function (A, p, r)



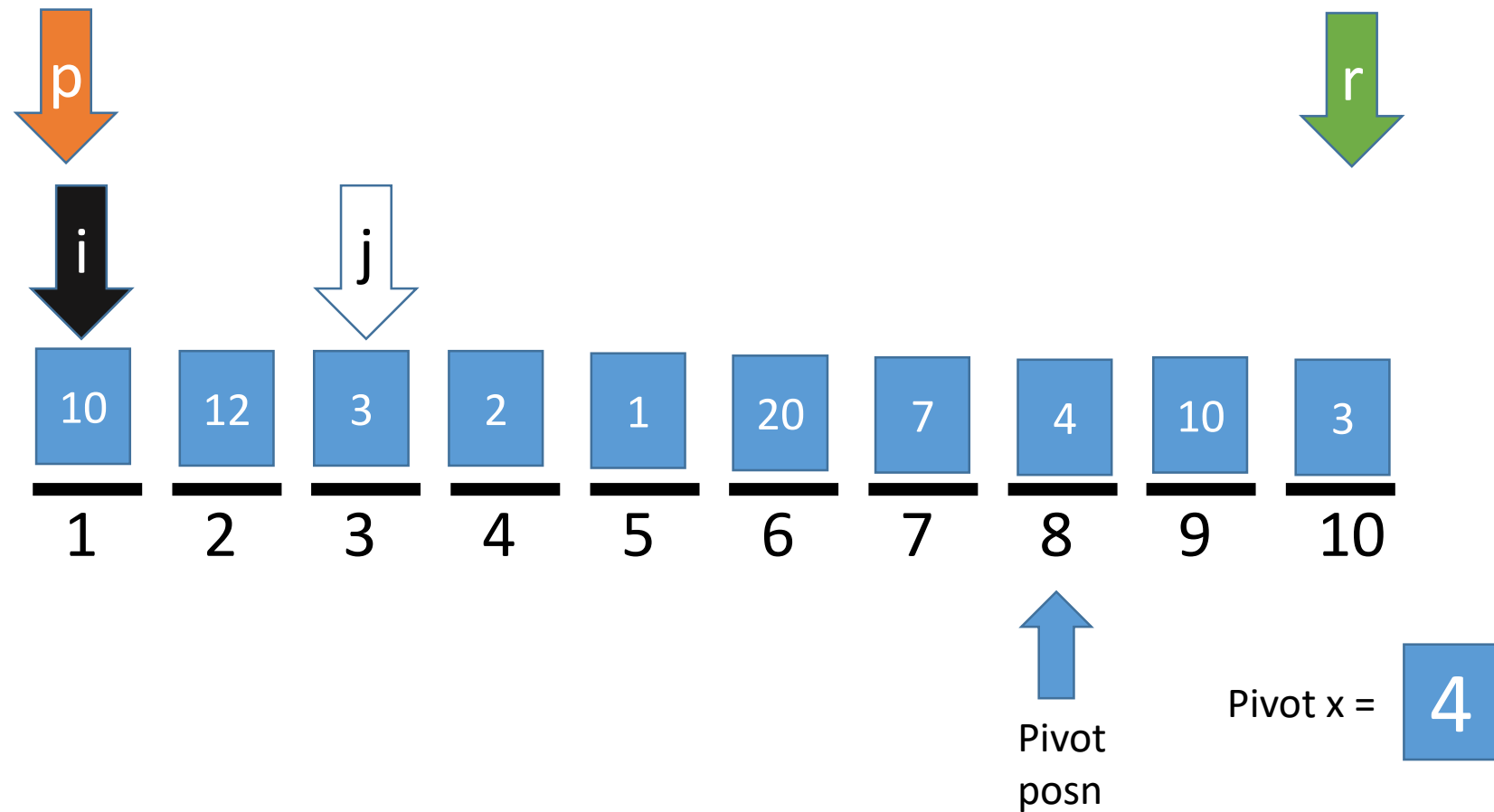
Partition Function (A, p, r)



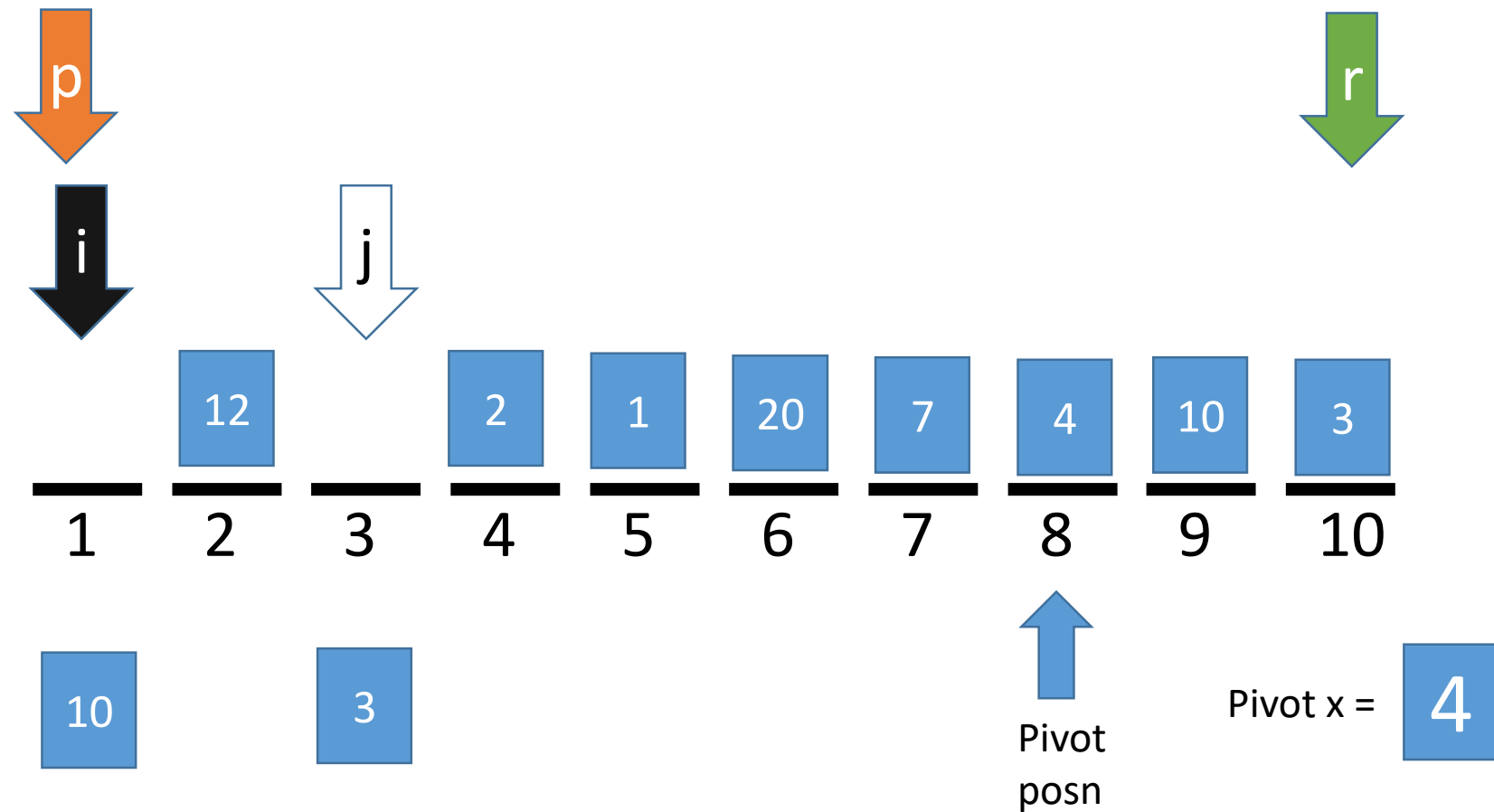
Partition Function (A, p, r)



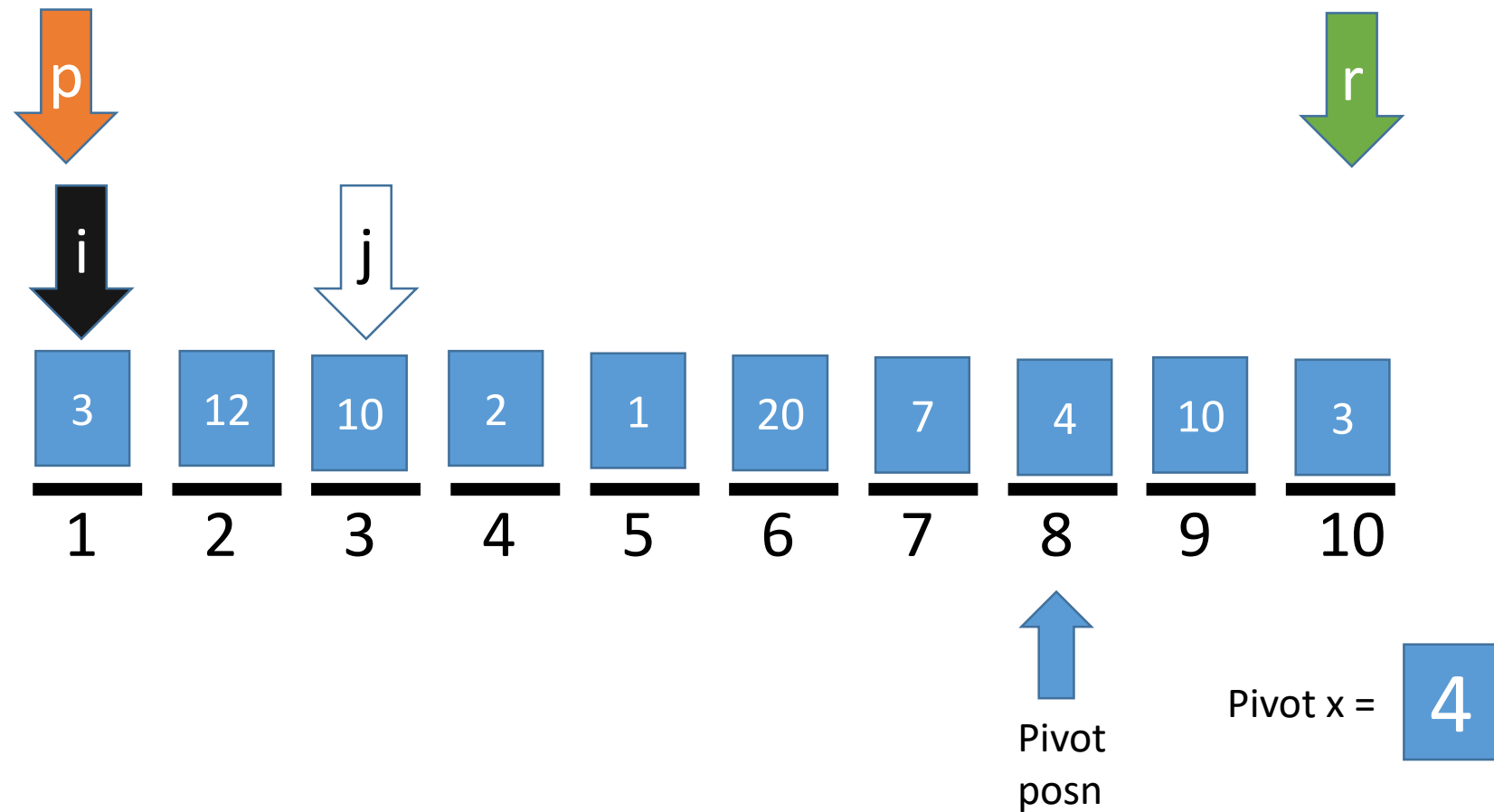
Partition Function (A, p, r)



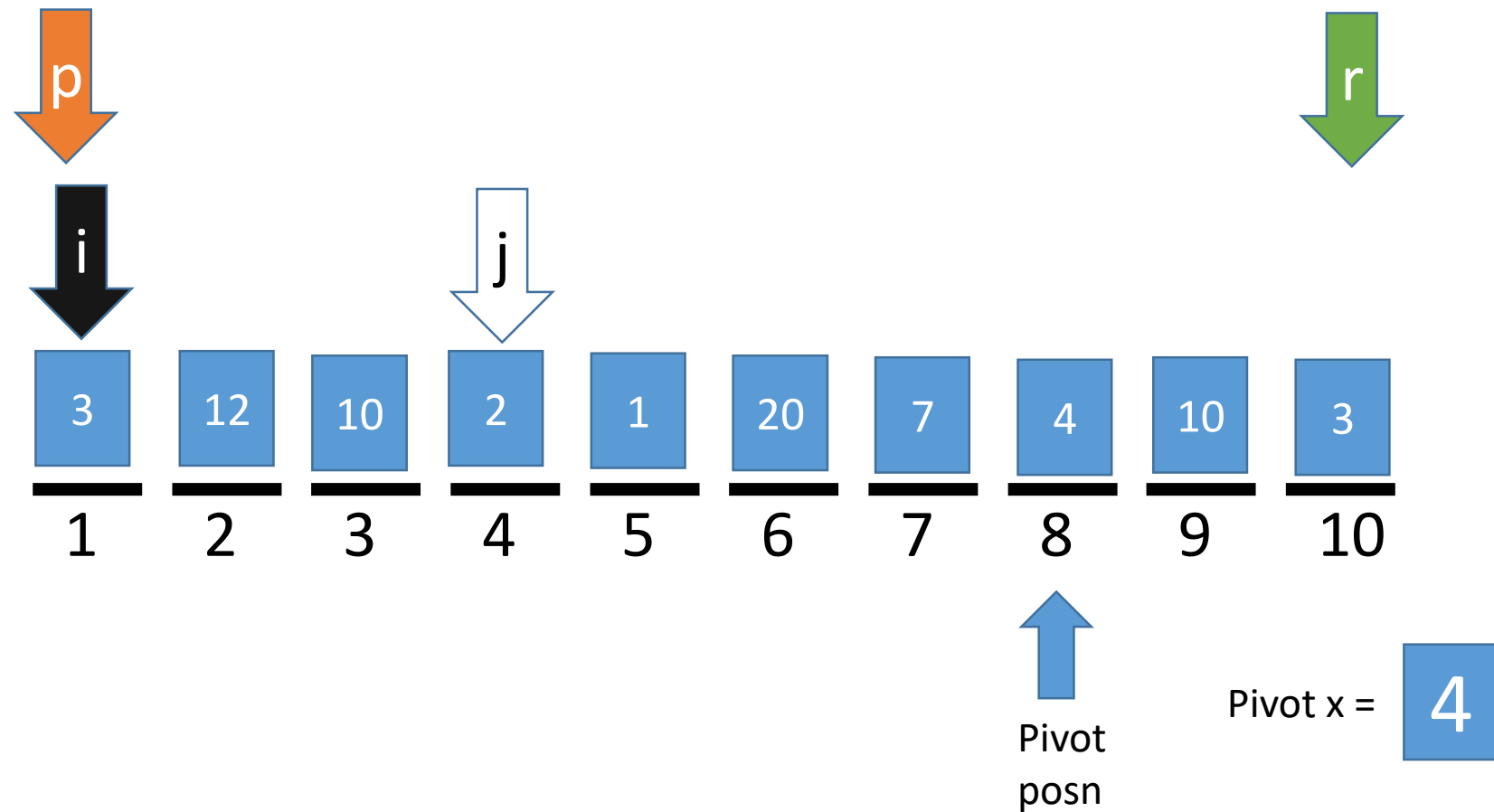
Partition Function (A, p, r)



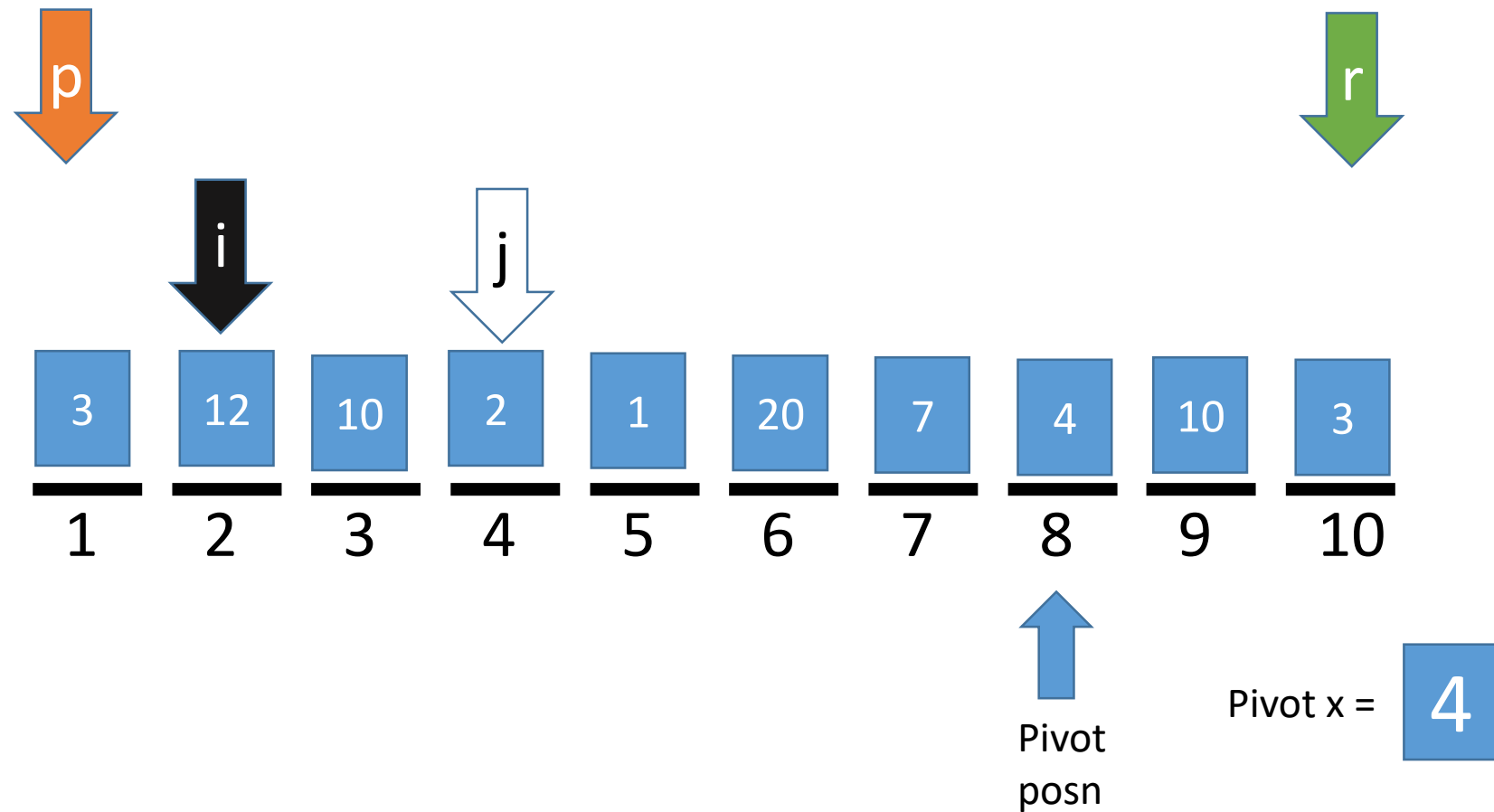
Partition Function (A, p, r)



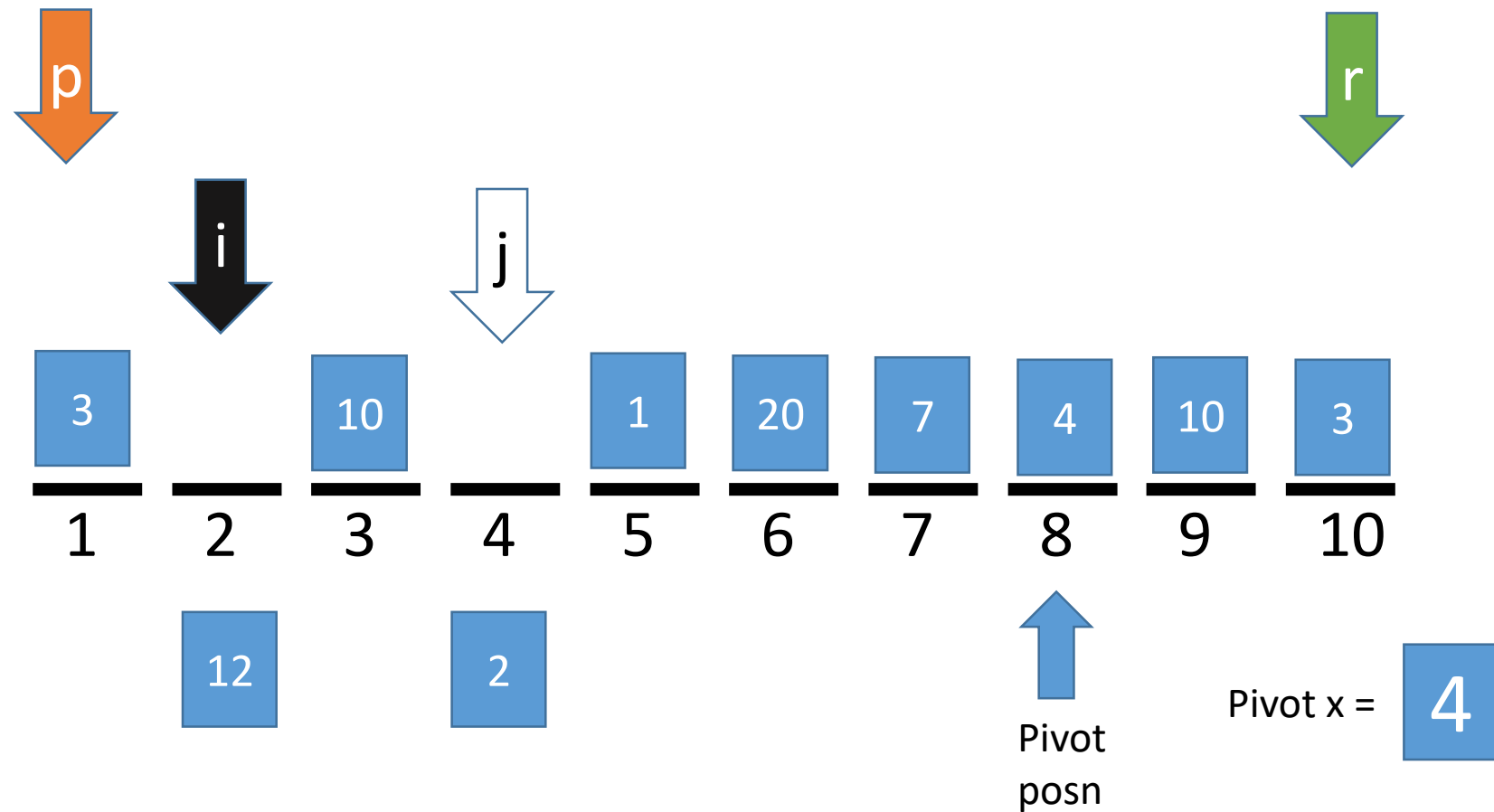
Partition Function (A, p, r)



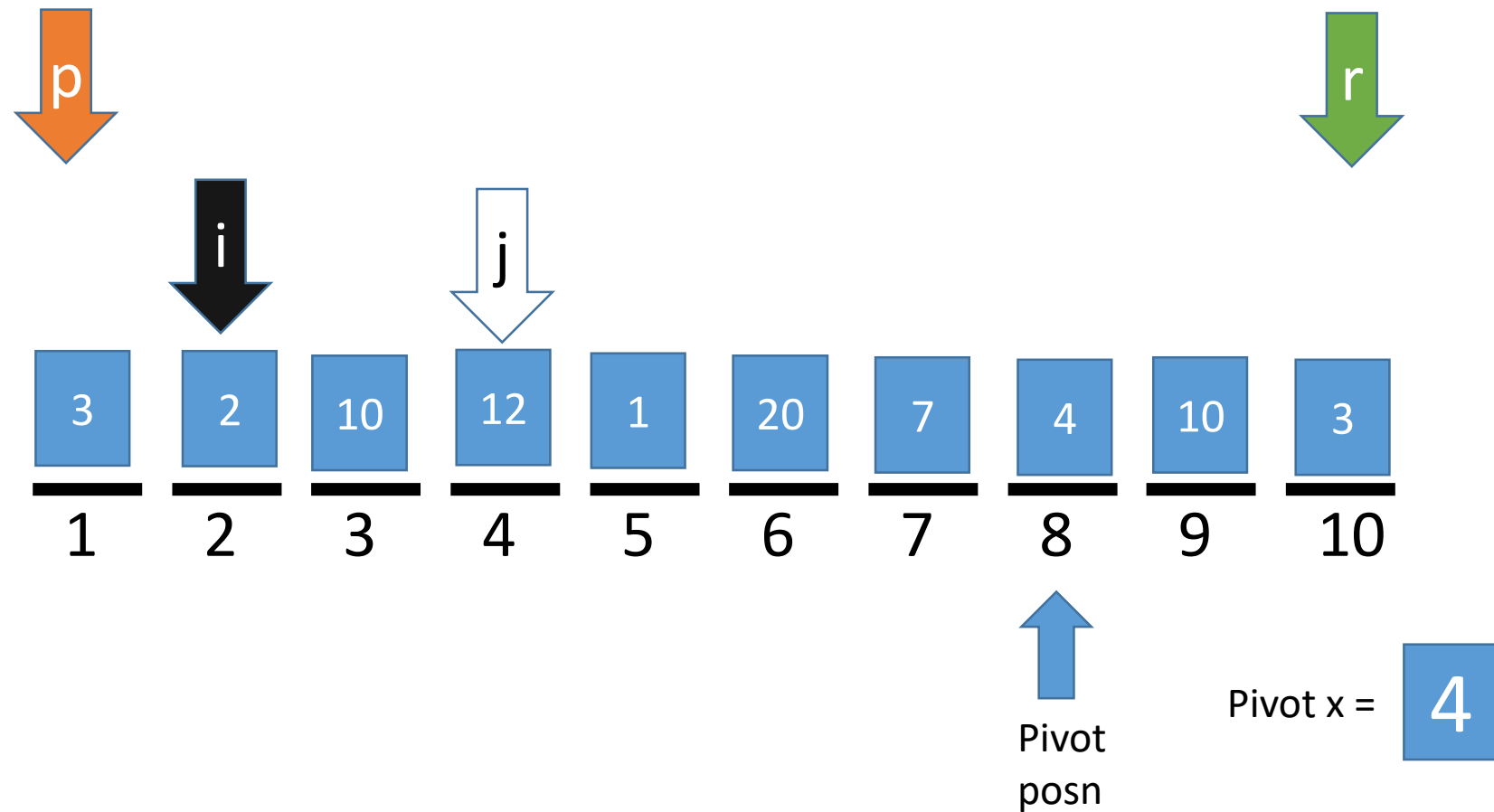
Partition Function (A, p, r)



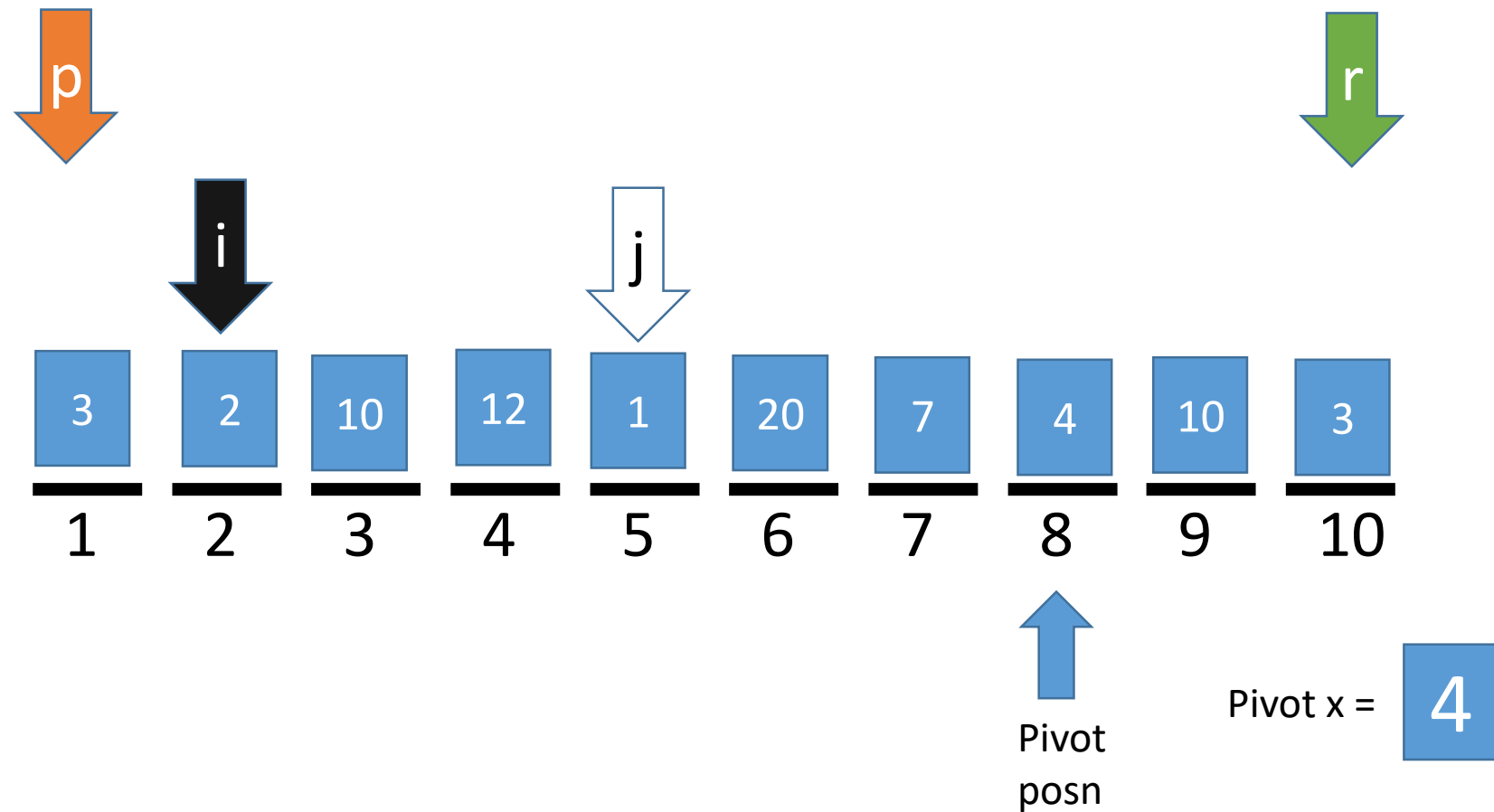
Partition Function (A, p, r)



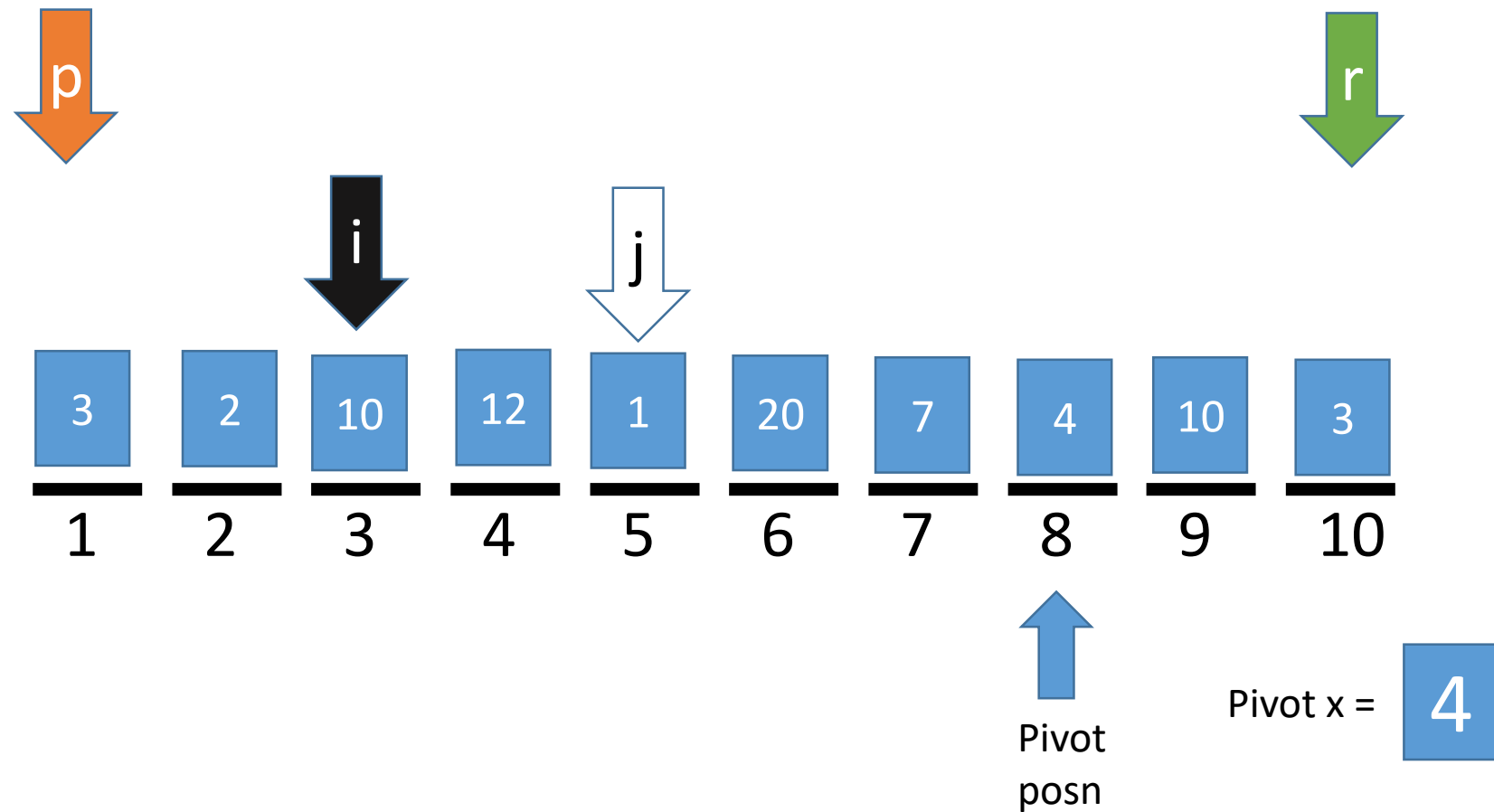
Partition Function (A, p, r)



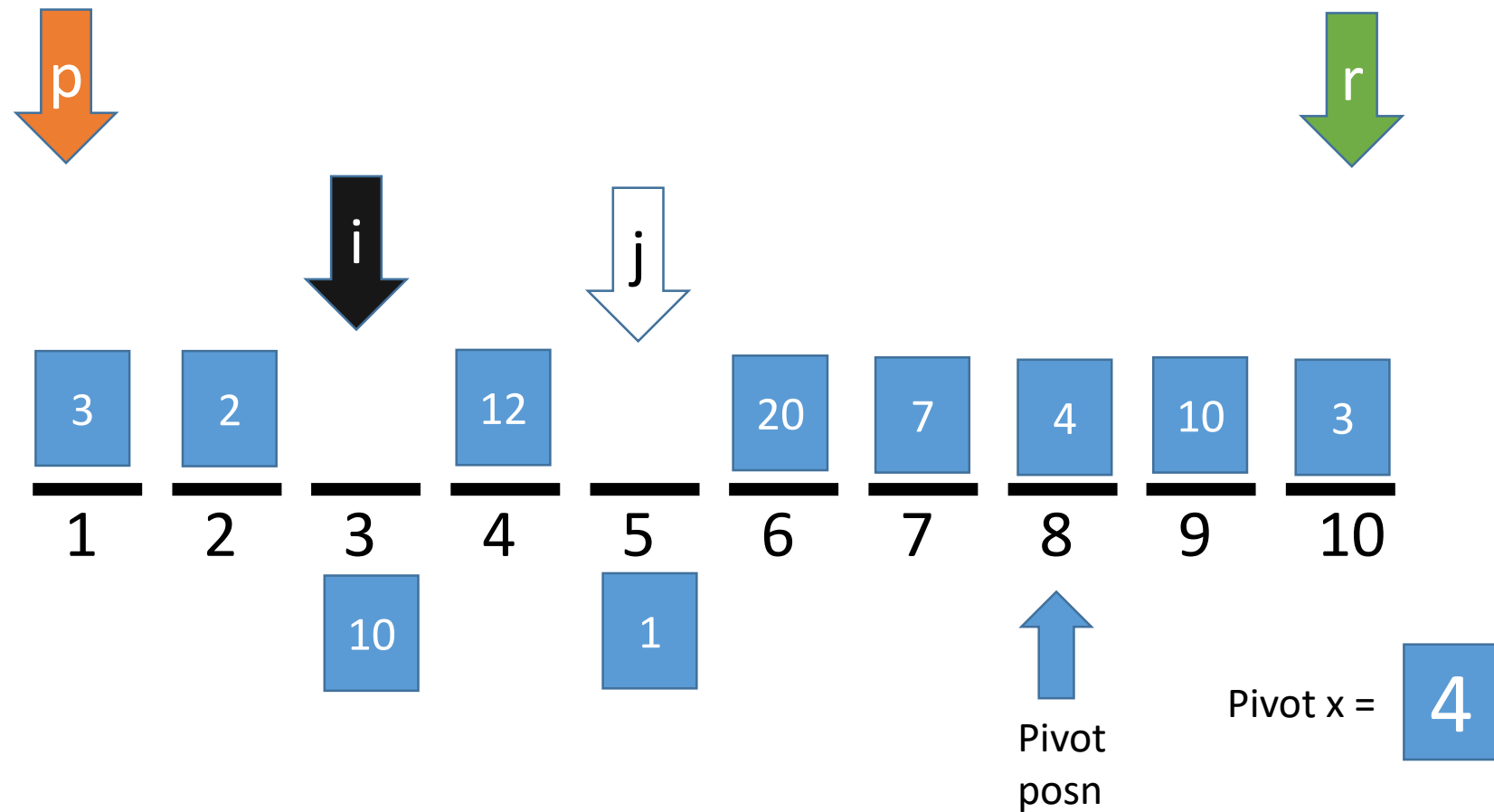
Partition Function (A, p, r)



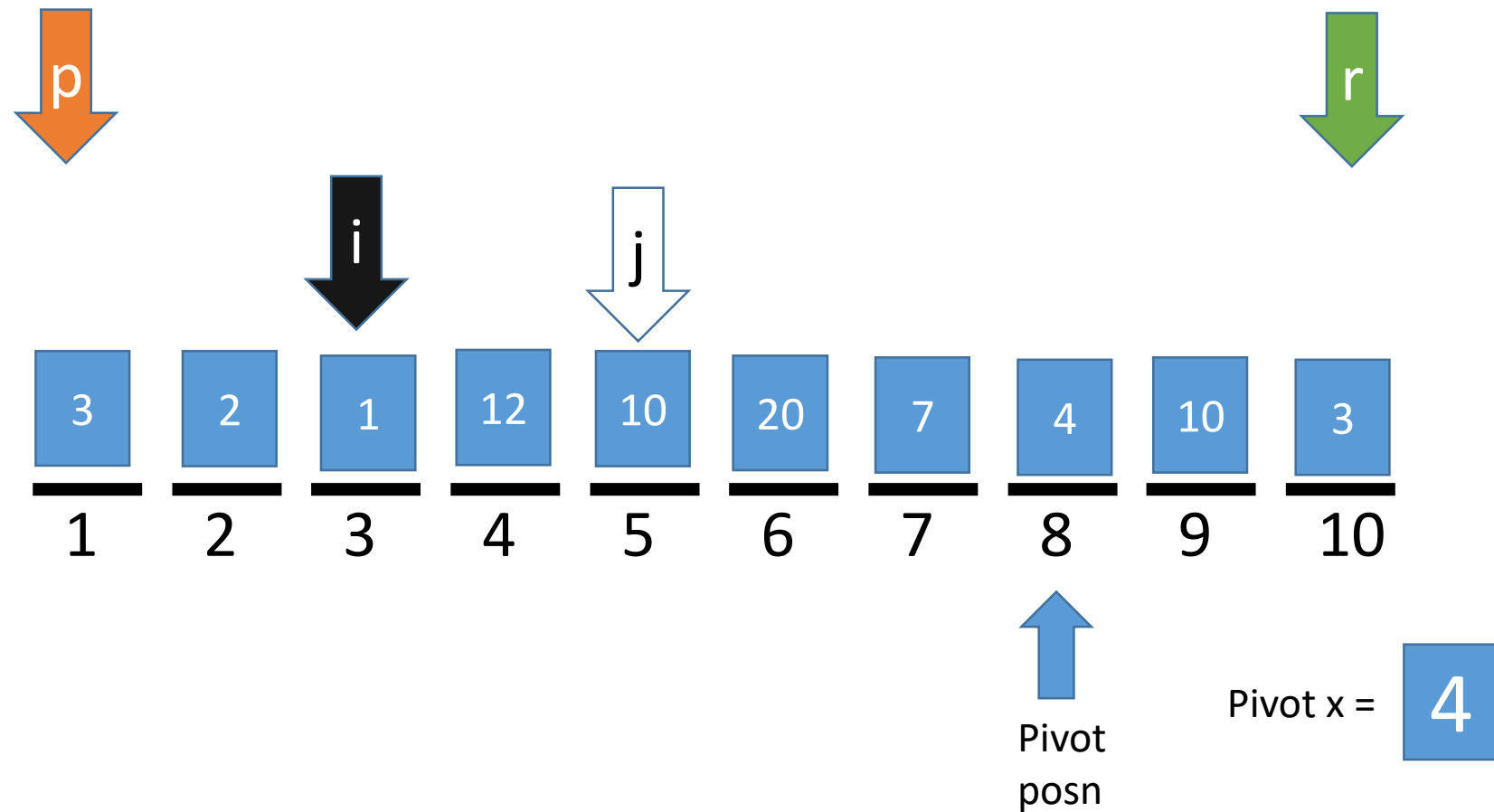
Partition Function (A, p, r)



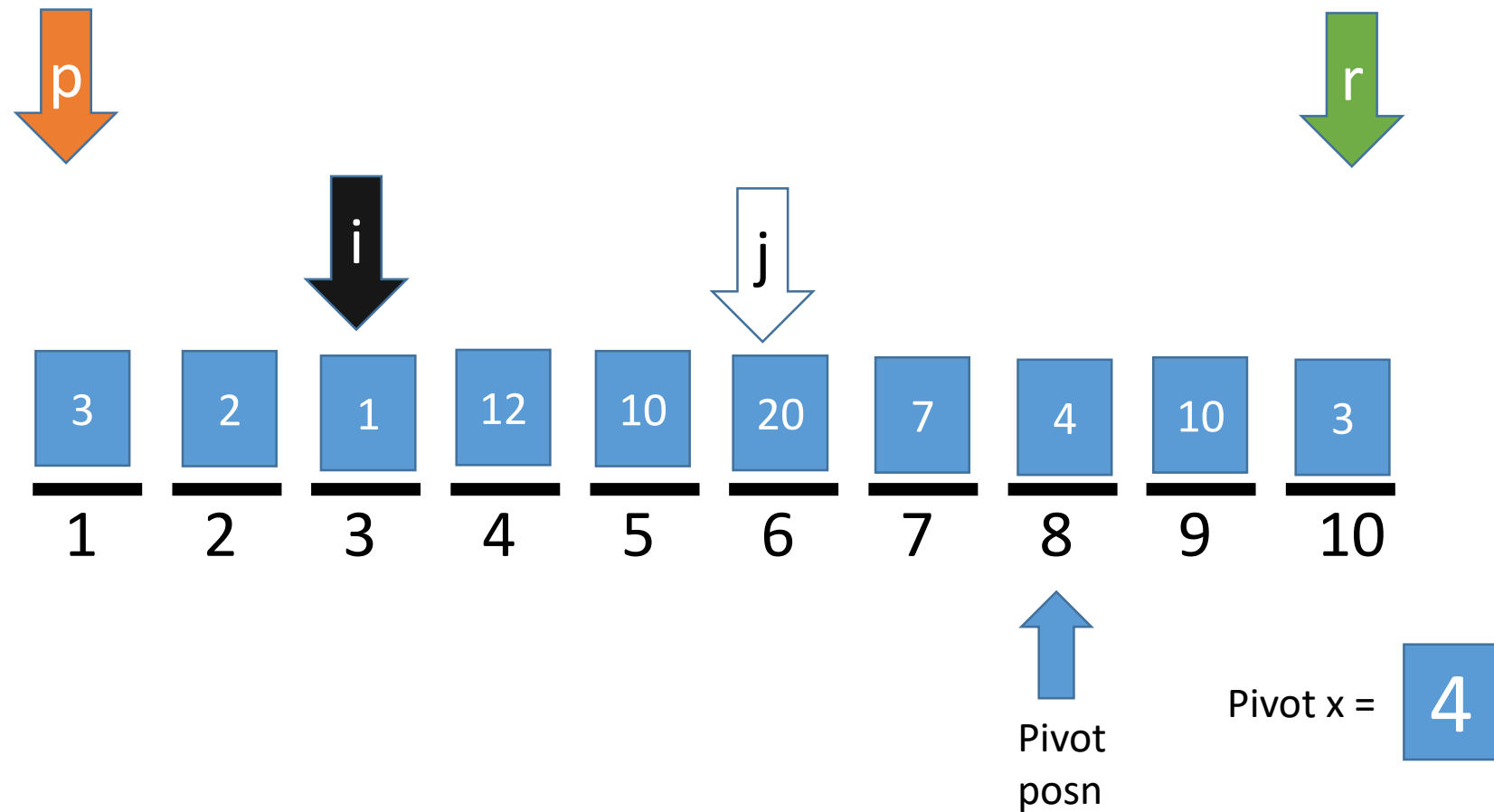
Partition Function (A, p, r)



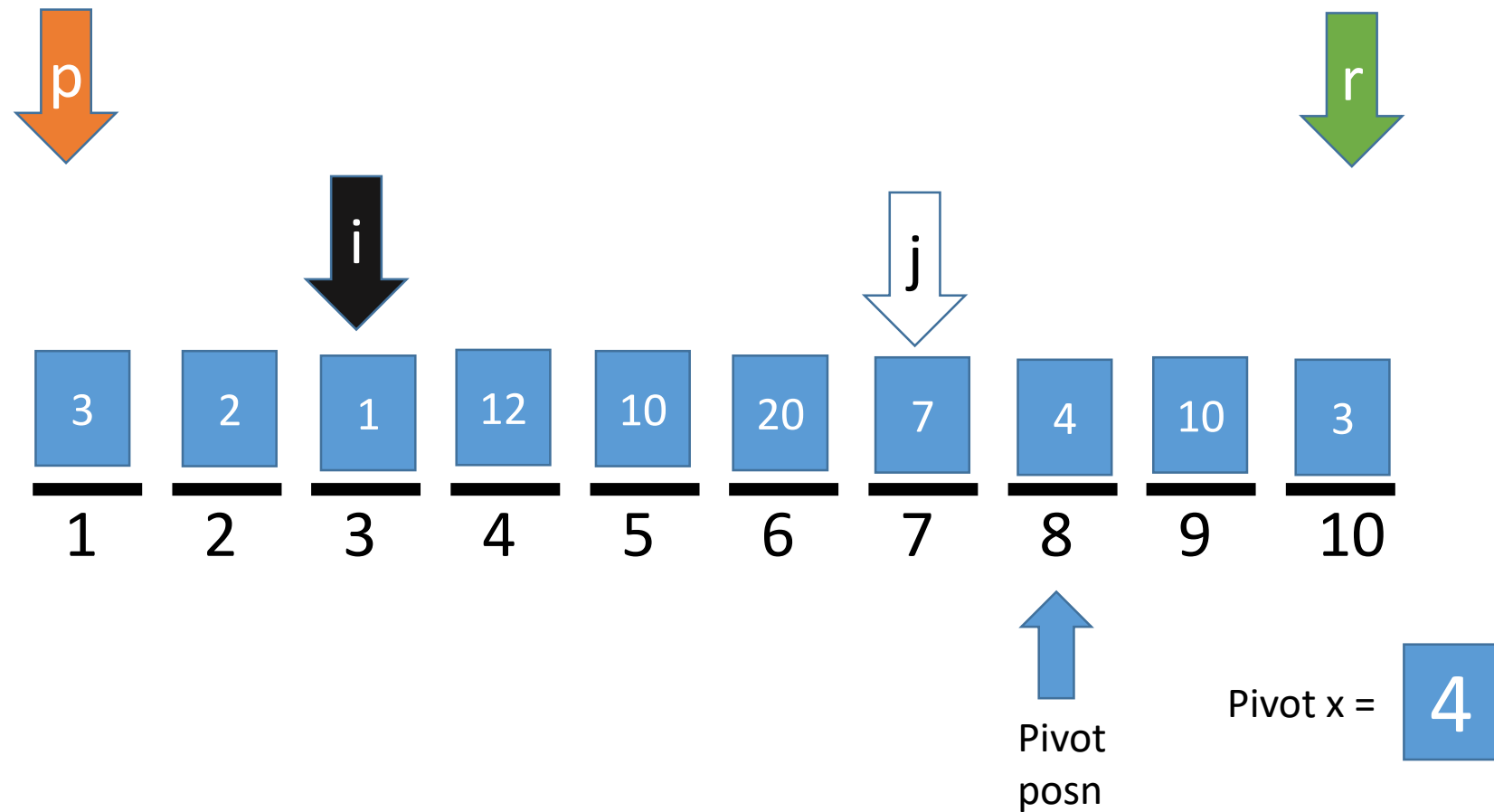
Partition Function (A, p, r)



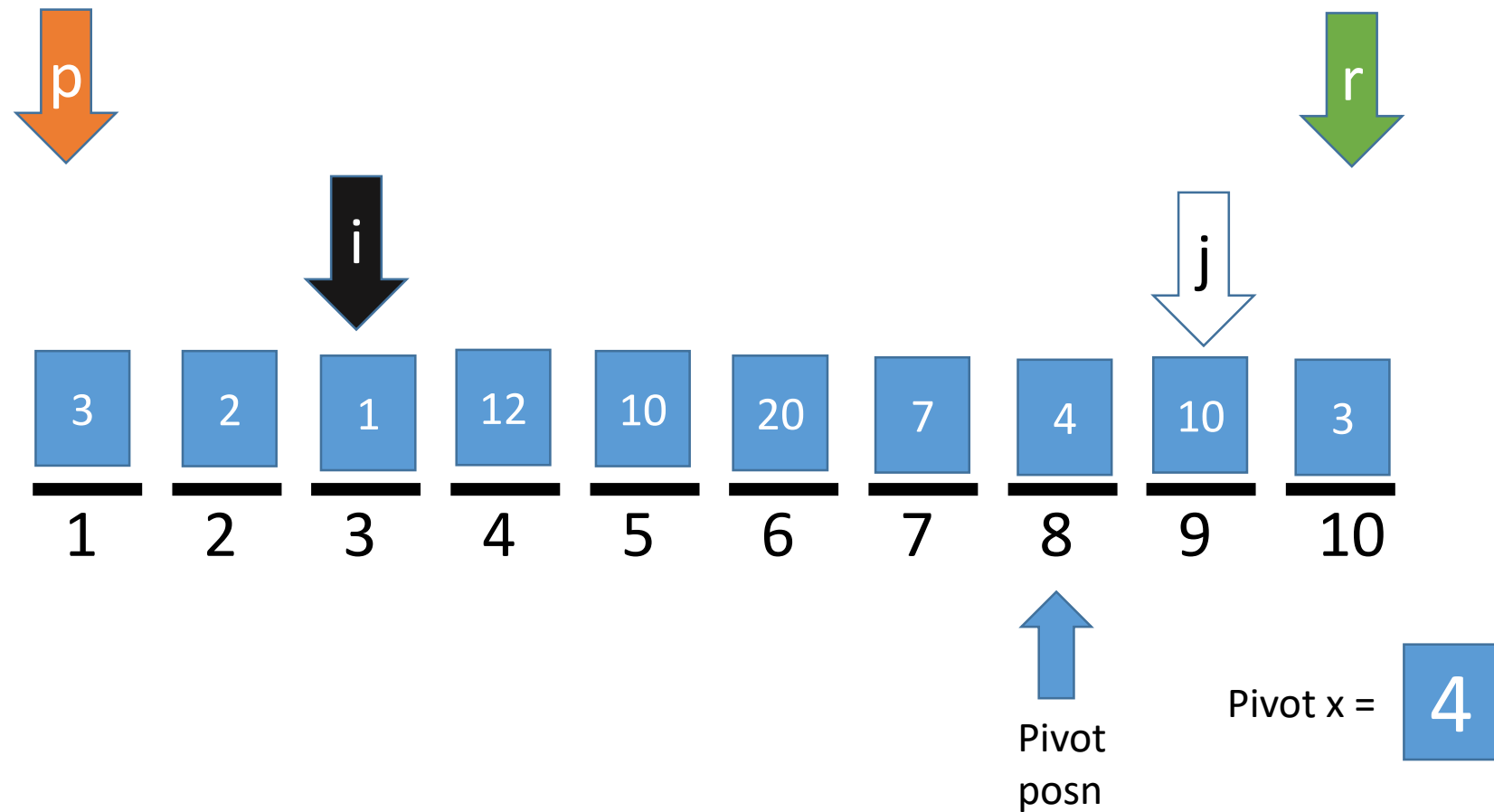
Partition Function (A, p, r)



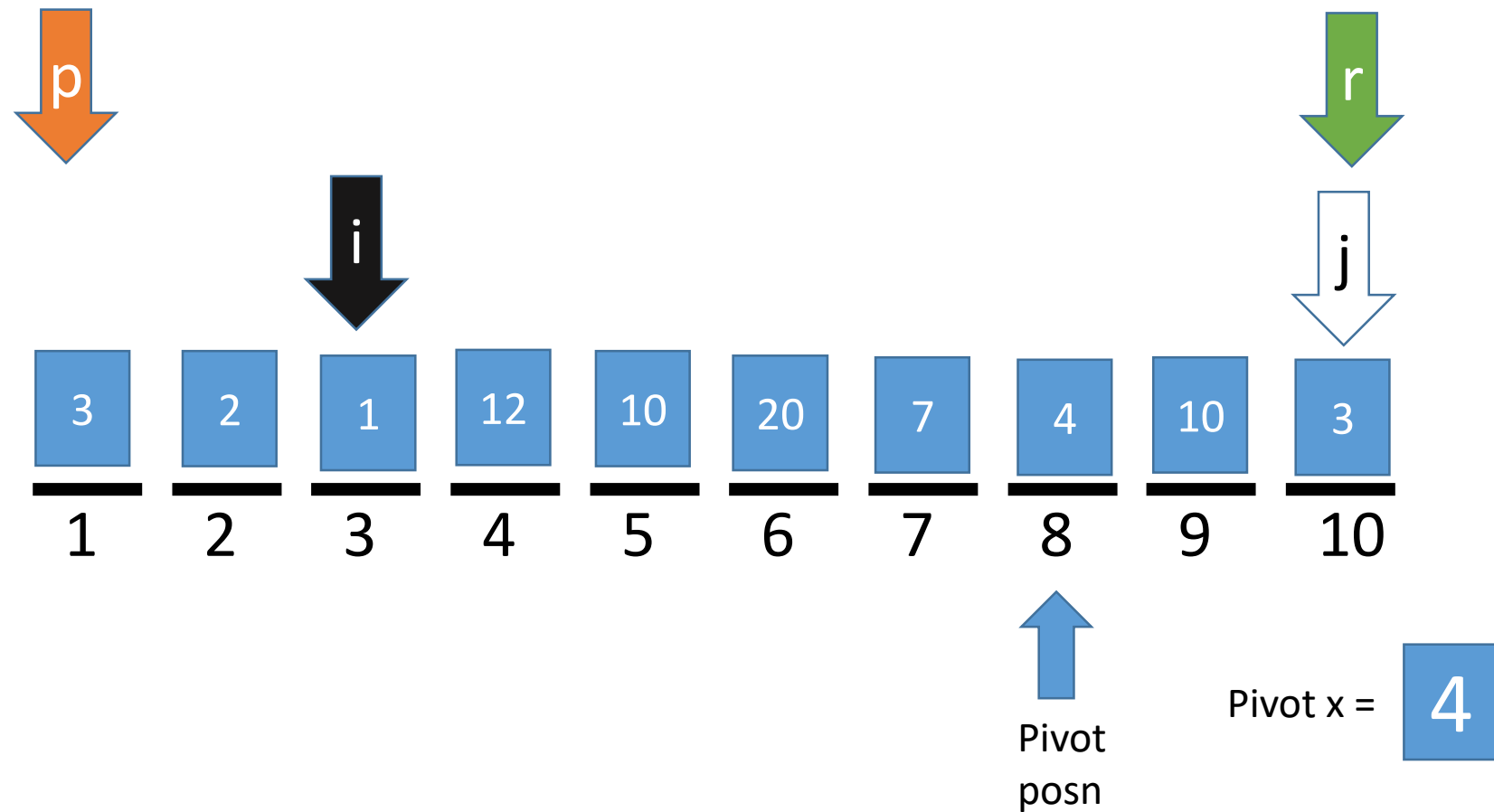
Partition Function (A, p, r)



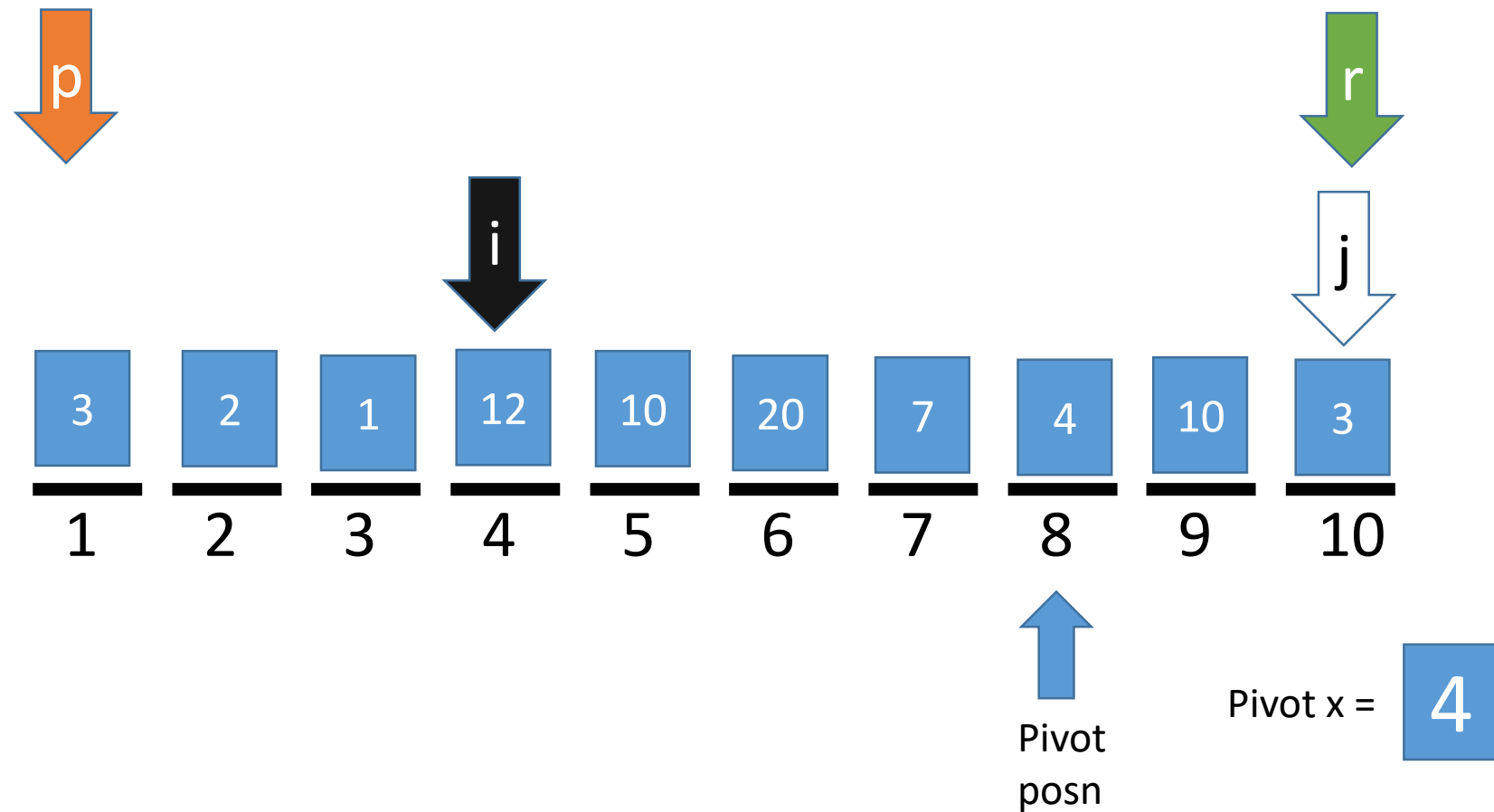
Partition Function (A, p, r)



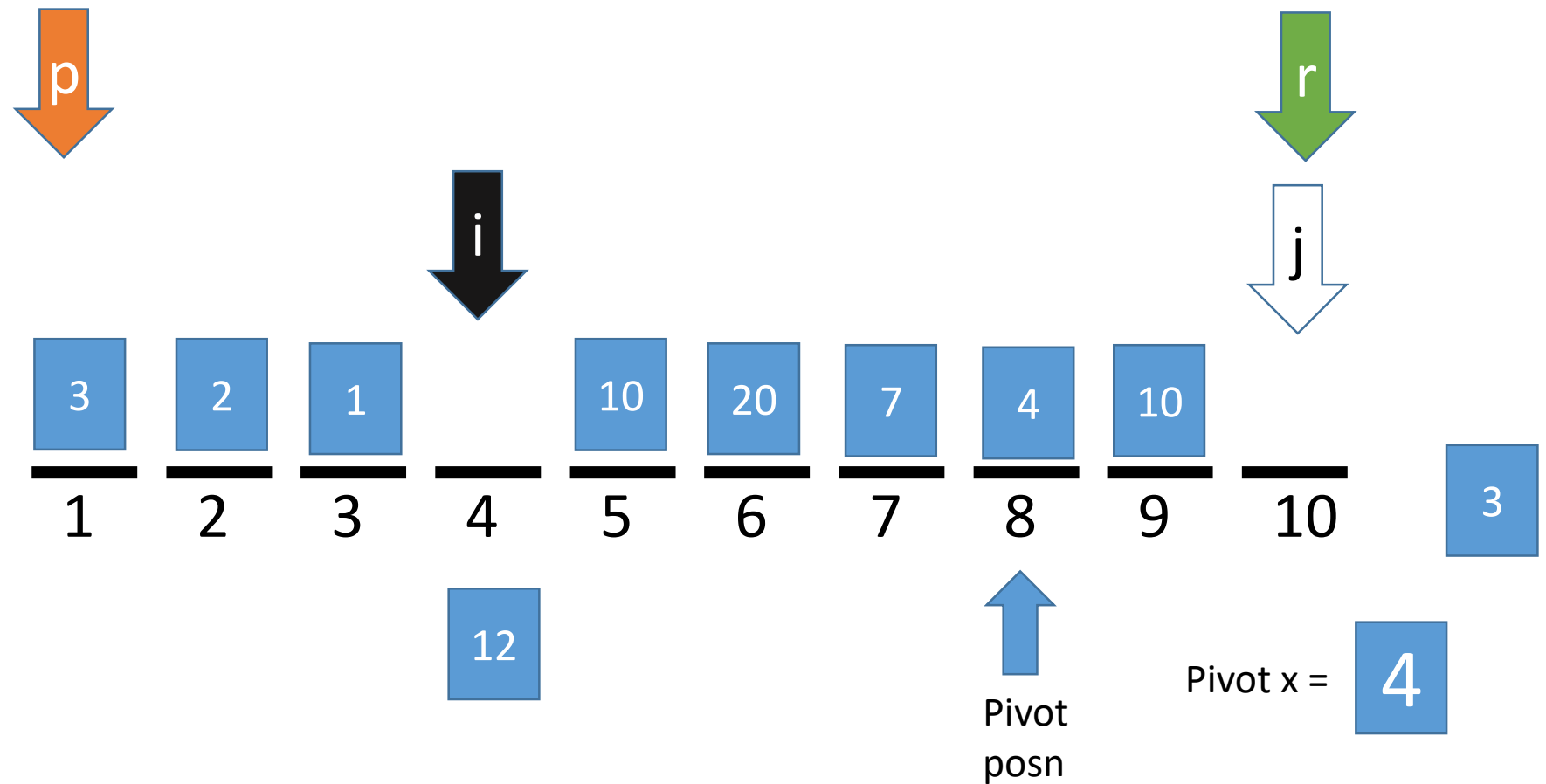
Partition Function (A, p, r)



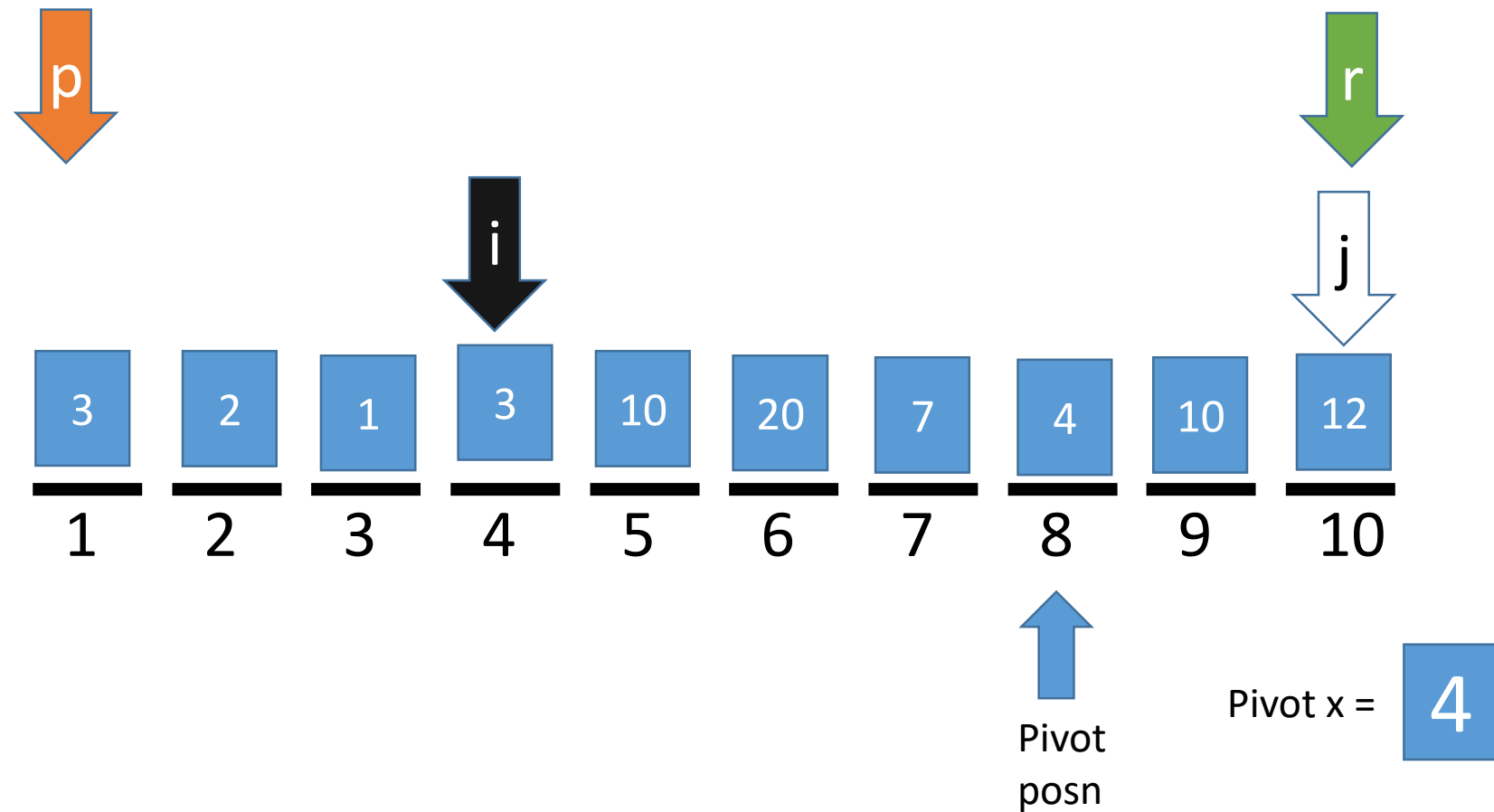
Partition Function (A, p, r)



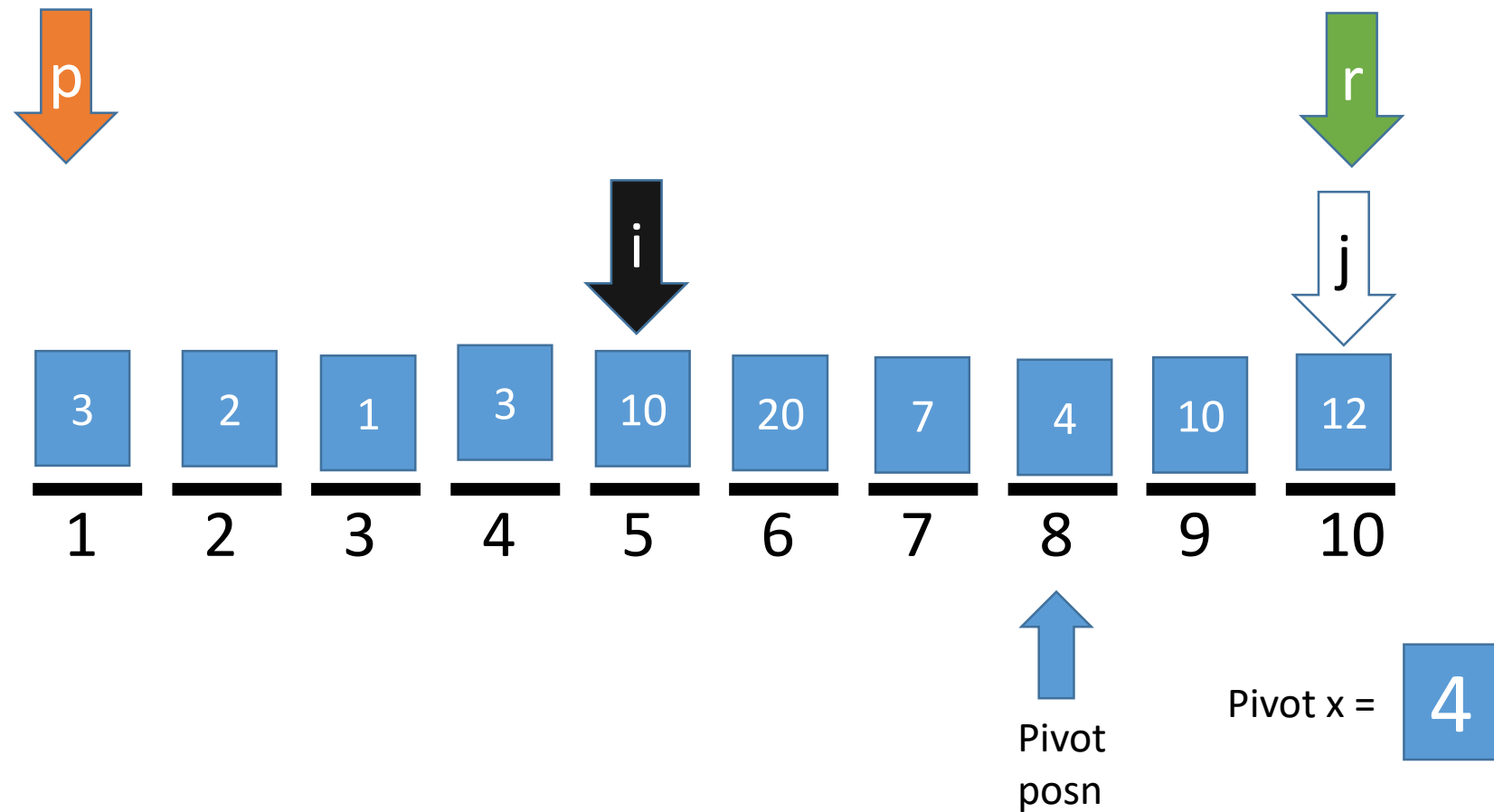
Partition Function (A, p, r)



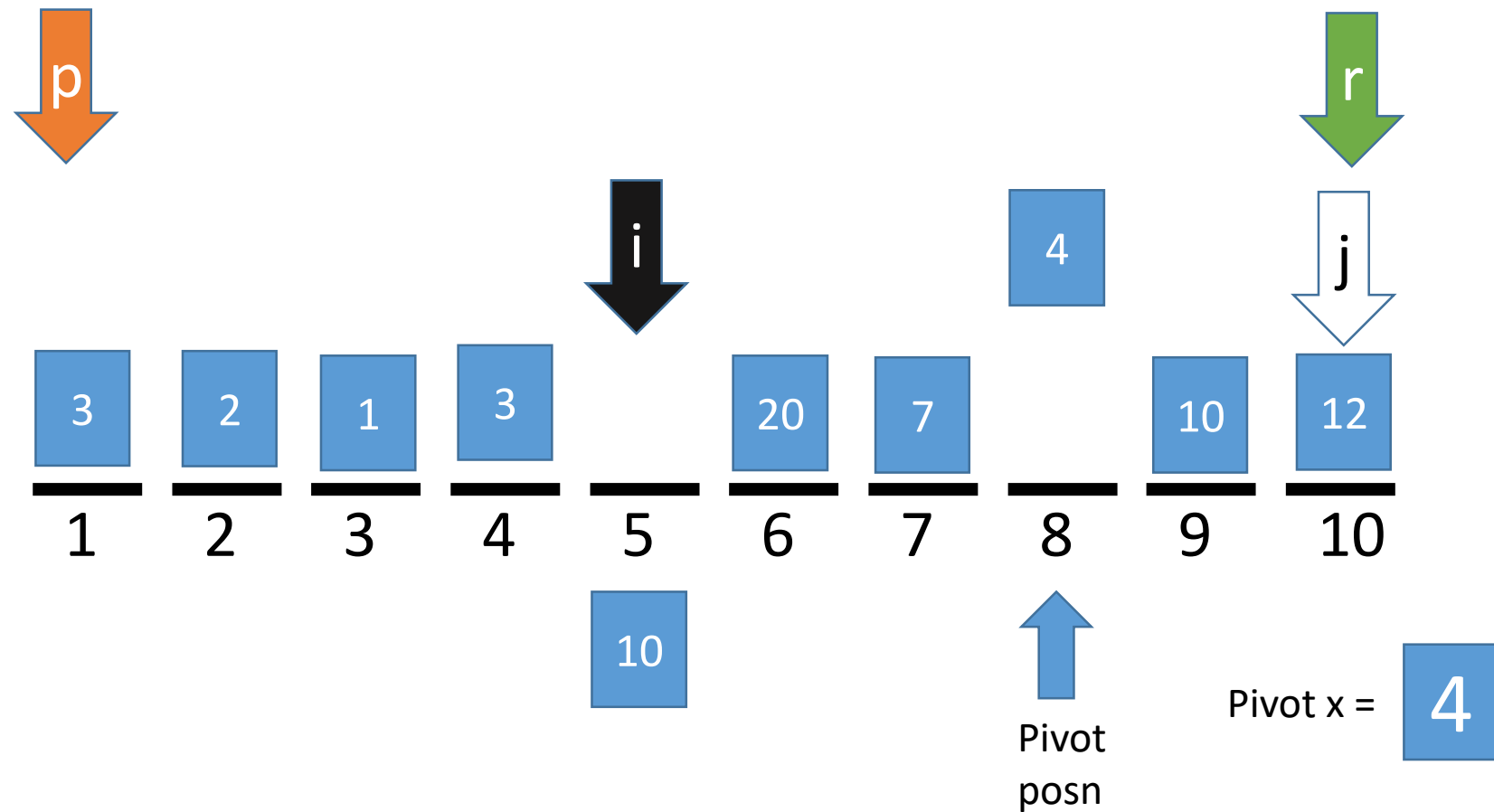
Partition Function (A, p, r)



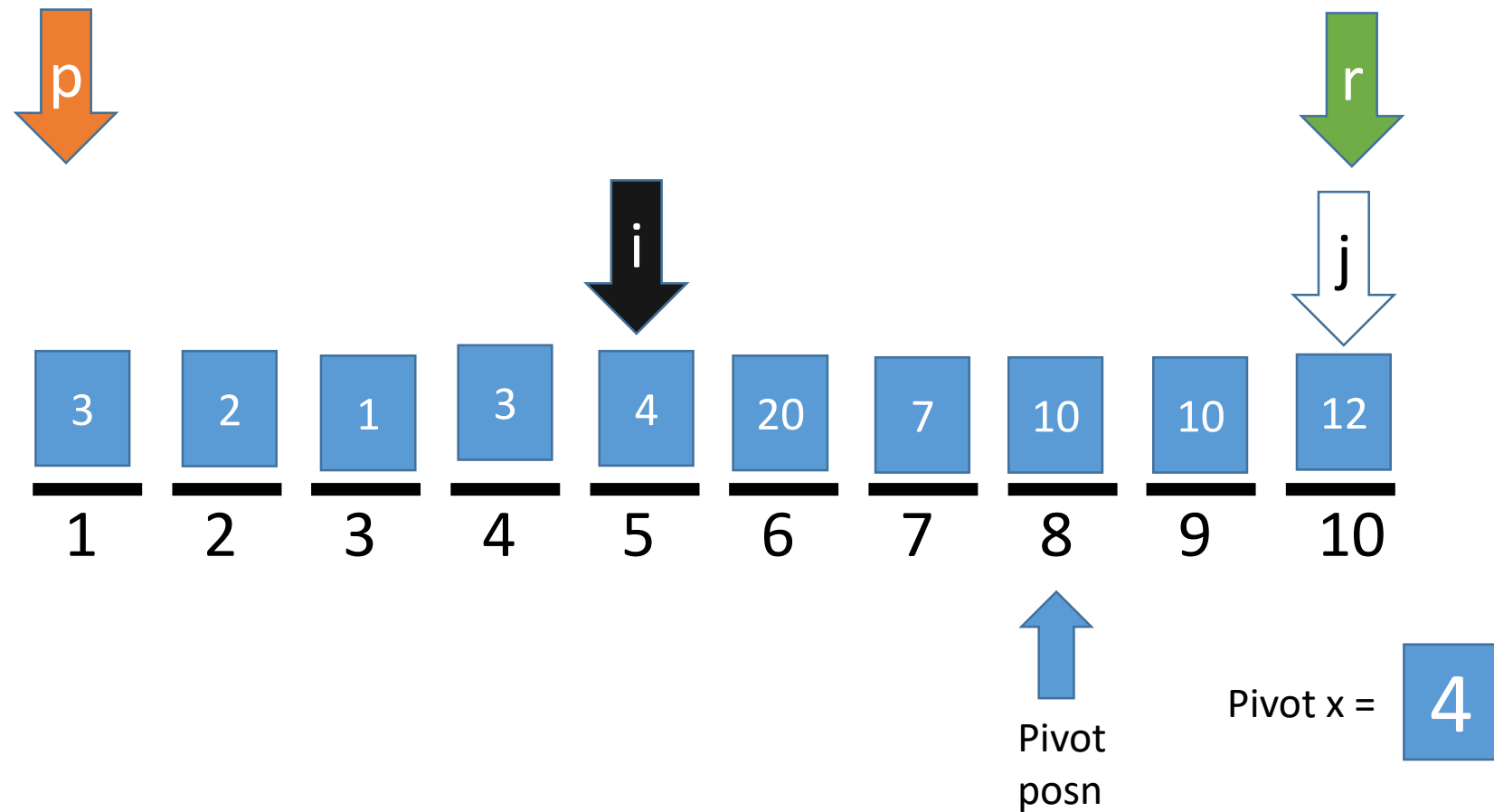
Partition Function (A, p, r)



Partition Function (A, p, r)



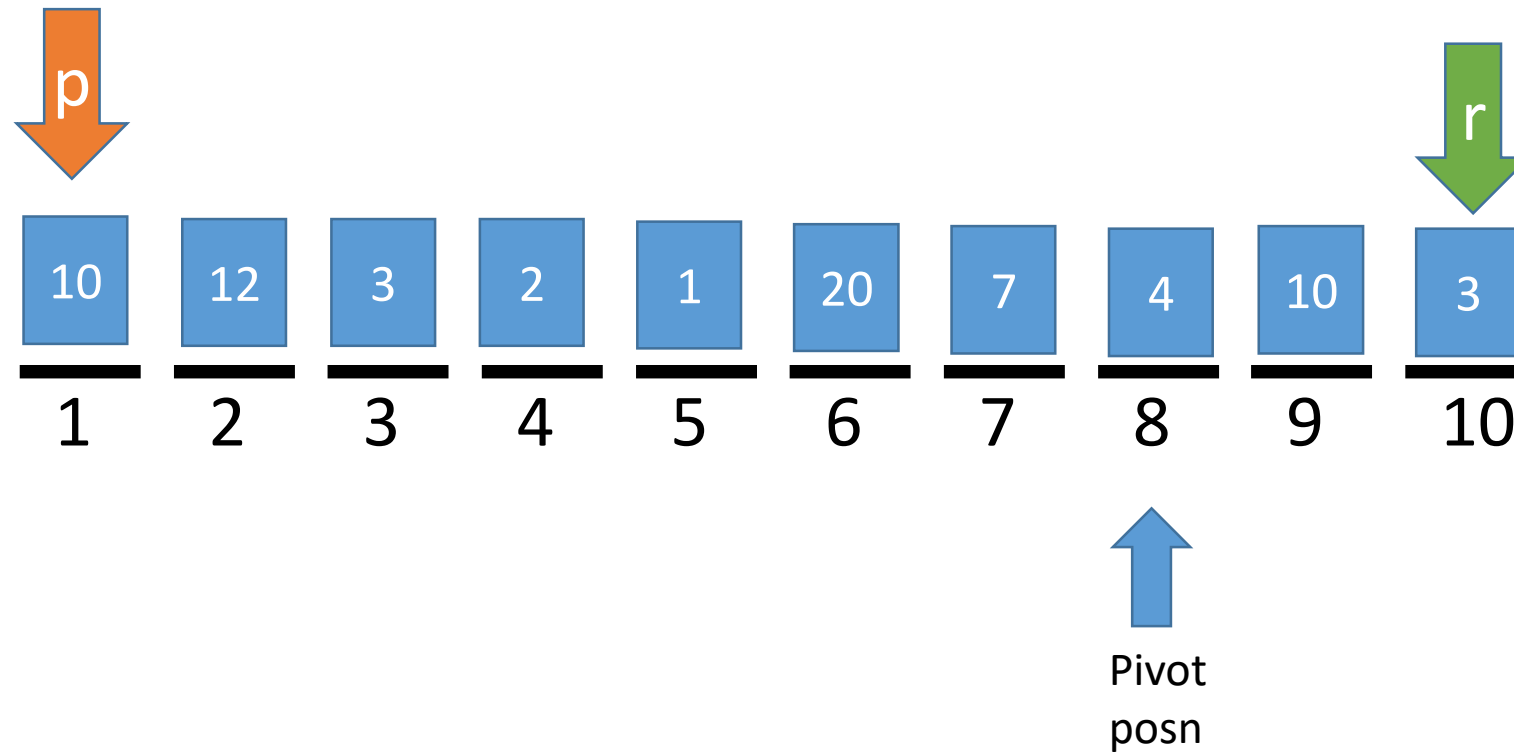
Partition Function (A, p, r)





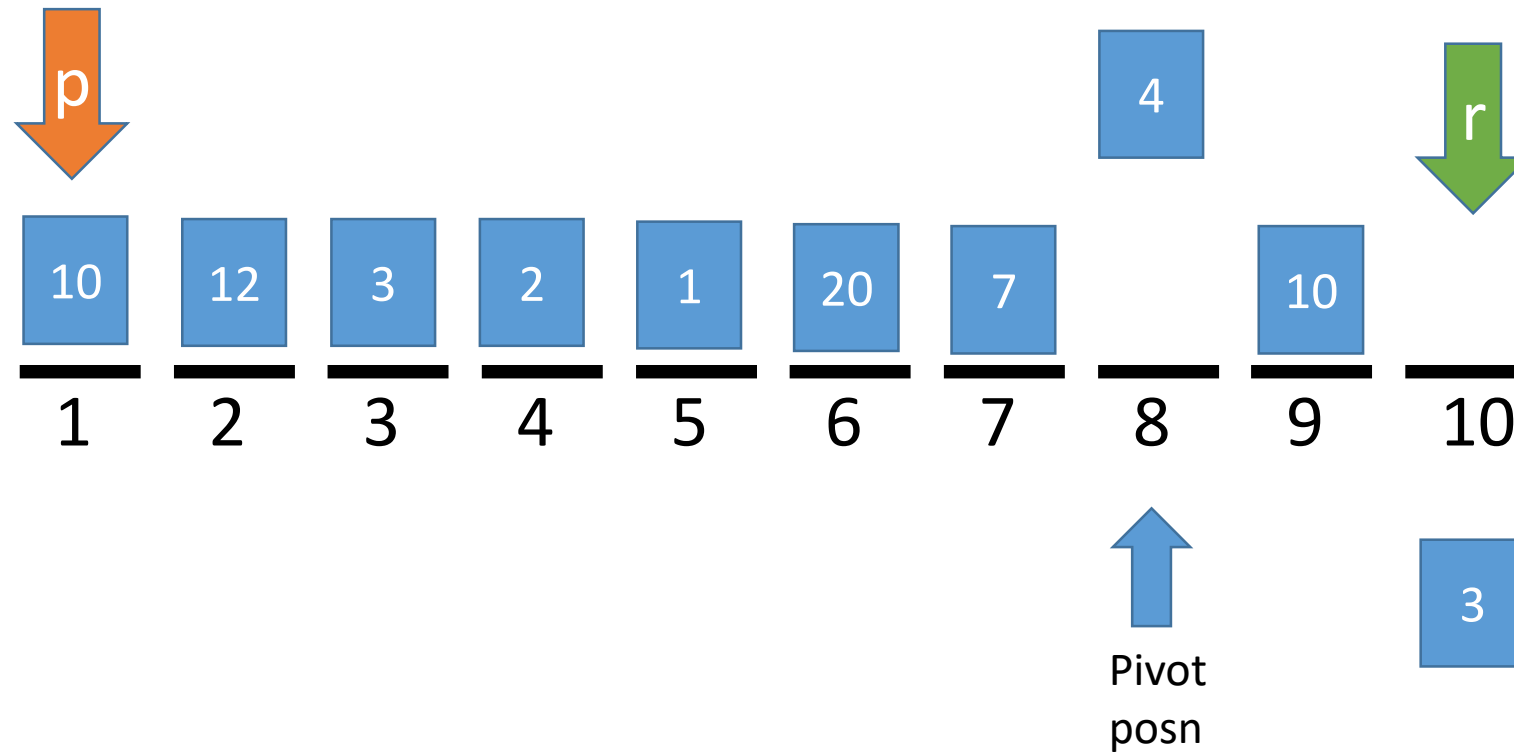
Partition Function (A, p, r)

WE WANT THE PIVOT TO HAVE THE VALUE 4 BUT WANT IT IN THE LAST POSITION
SIMILAR TO THE FIRST ONE



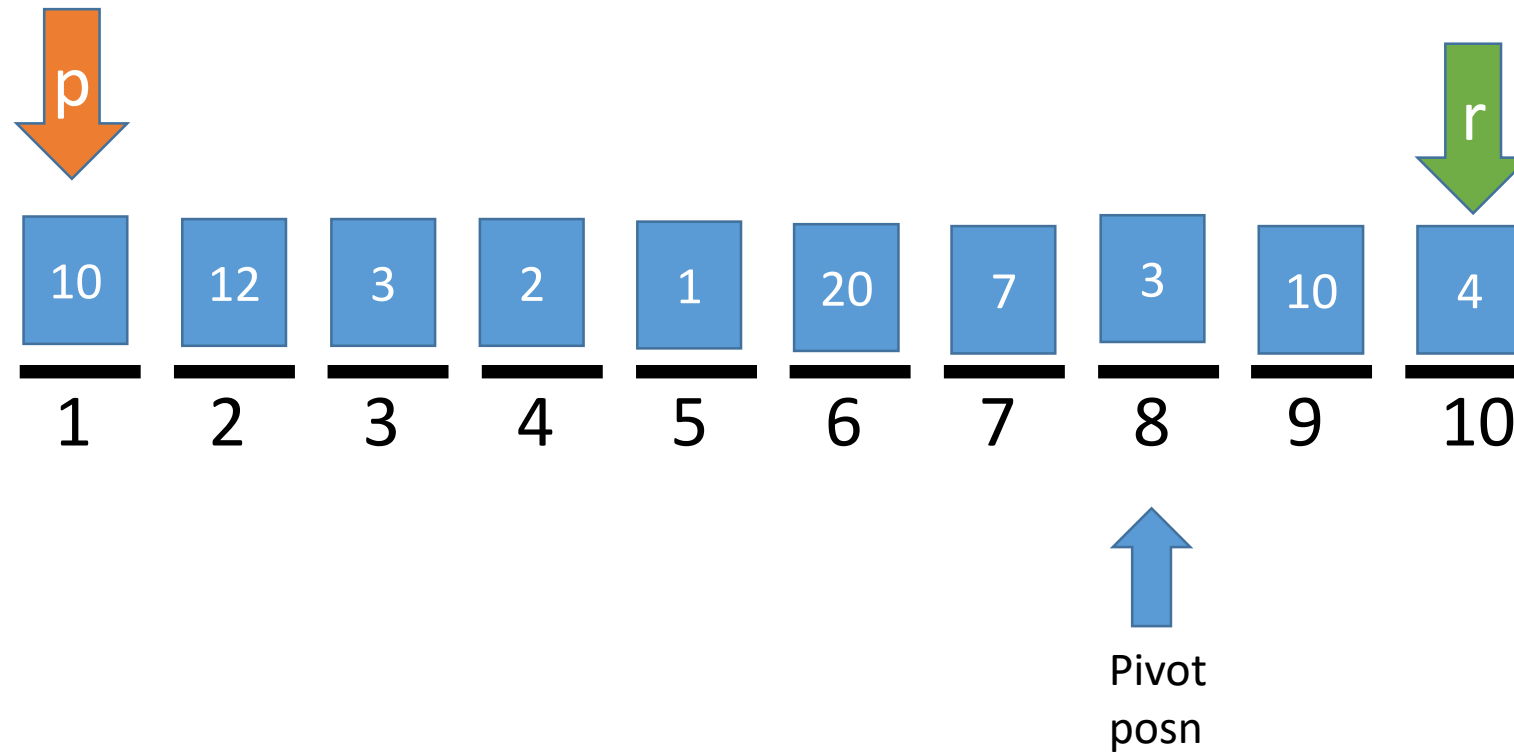
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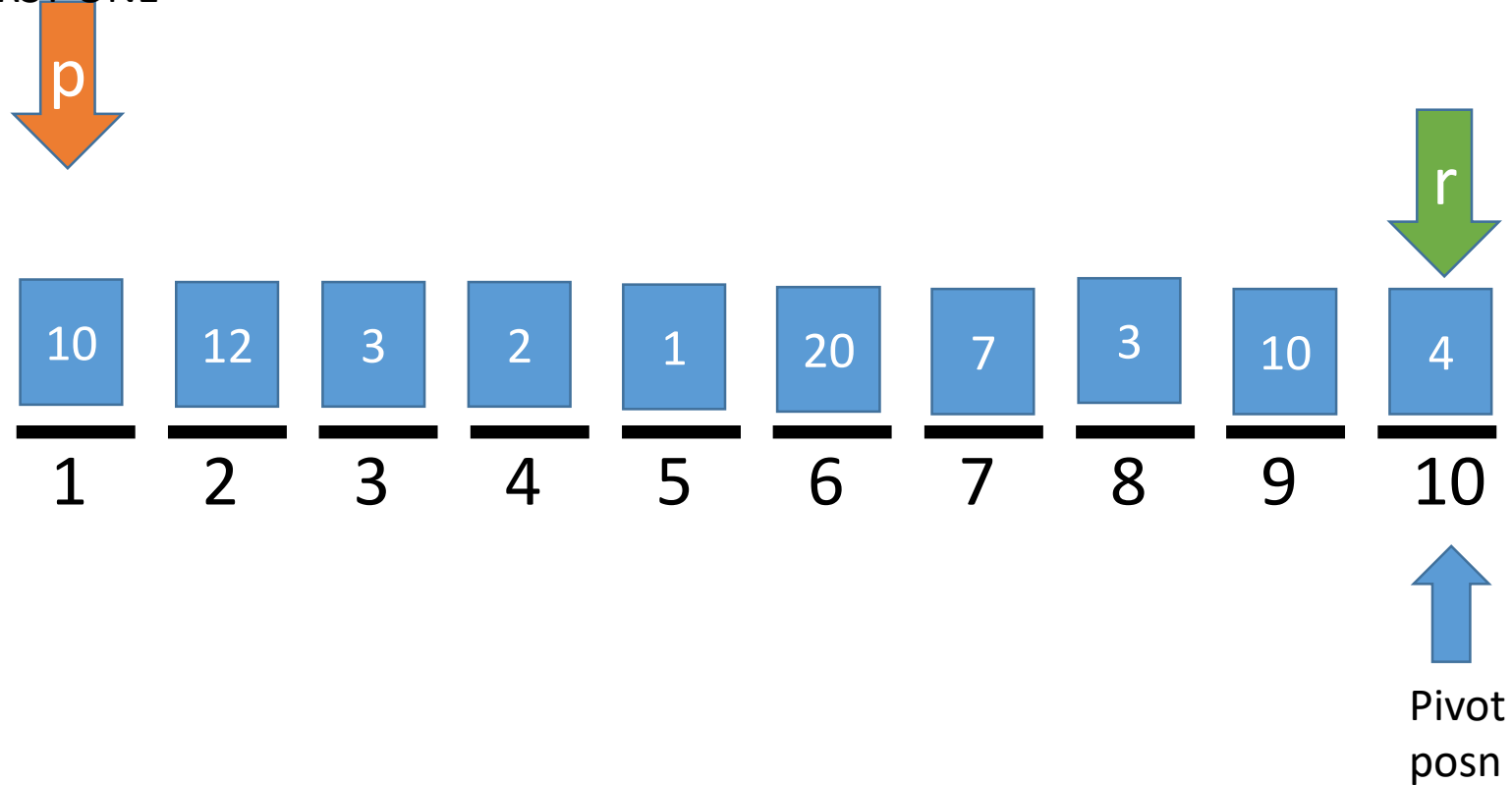
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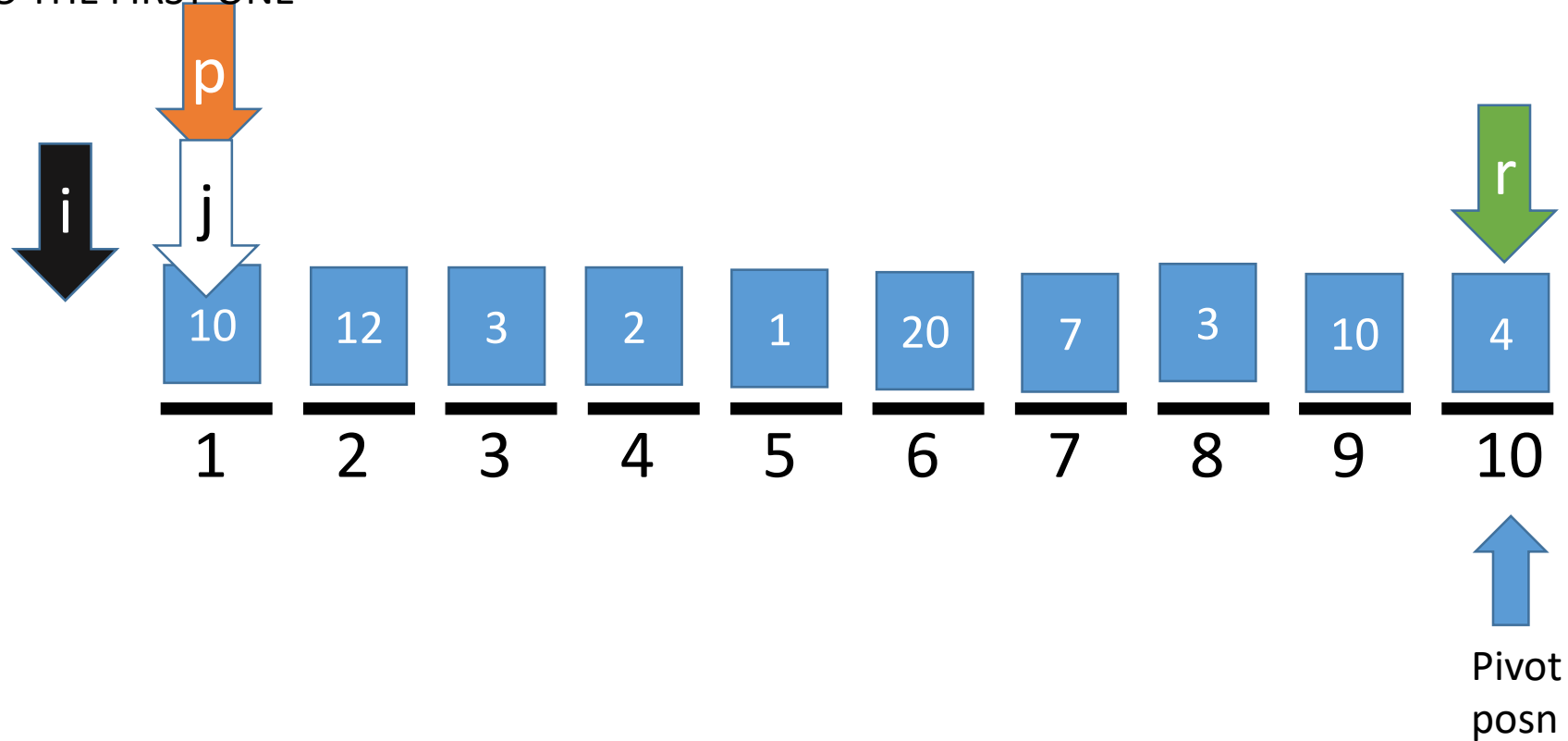
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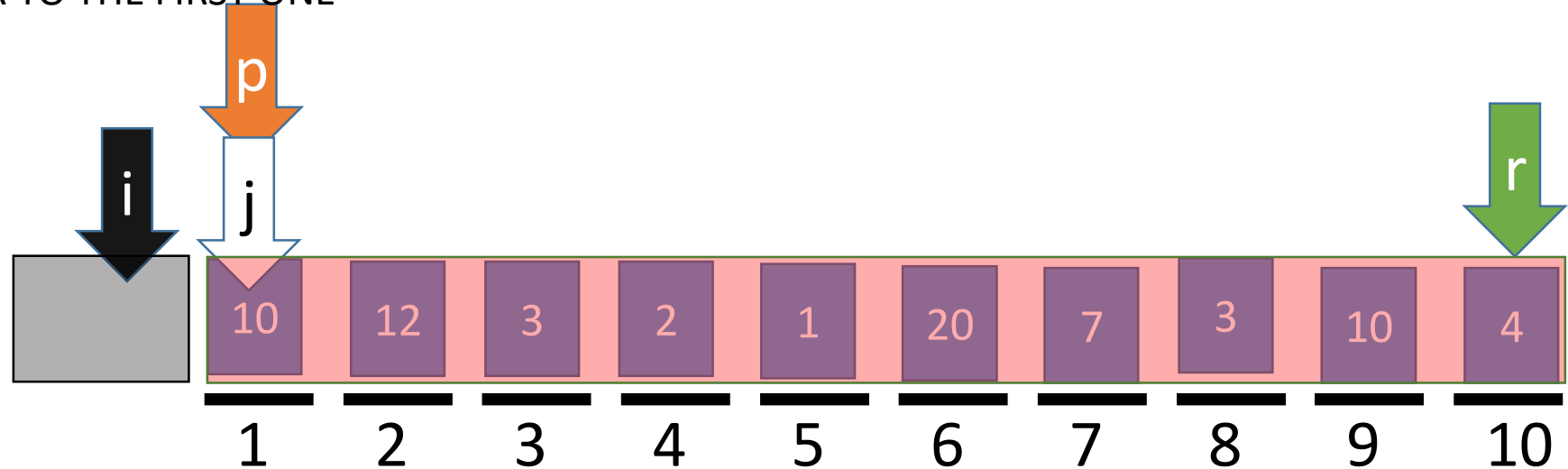
Partition Function (A, p, r)

WE WANT THE PIVOT TO HAVE THE VALUE 4 BUT WANT IT IN THE LAST POSITION
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Partition Function (A, p, r)

WE WANT THE PIVOT TO HAVE THE VALUE 4 BUT WANT IT IN THE LAST POSITION
SIMILAR TO THE FIRST ONE



- Numbers smaller than or equal to pivot (marked by **i at the end**)



- Numbers remaining to be explored (marked by **j at the beginning**)



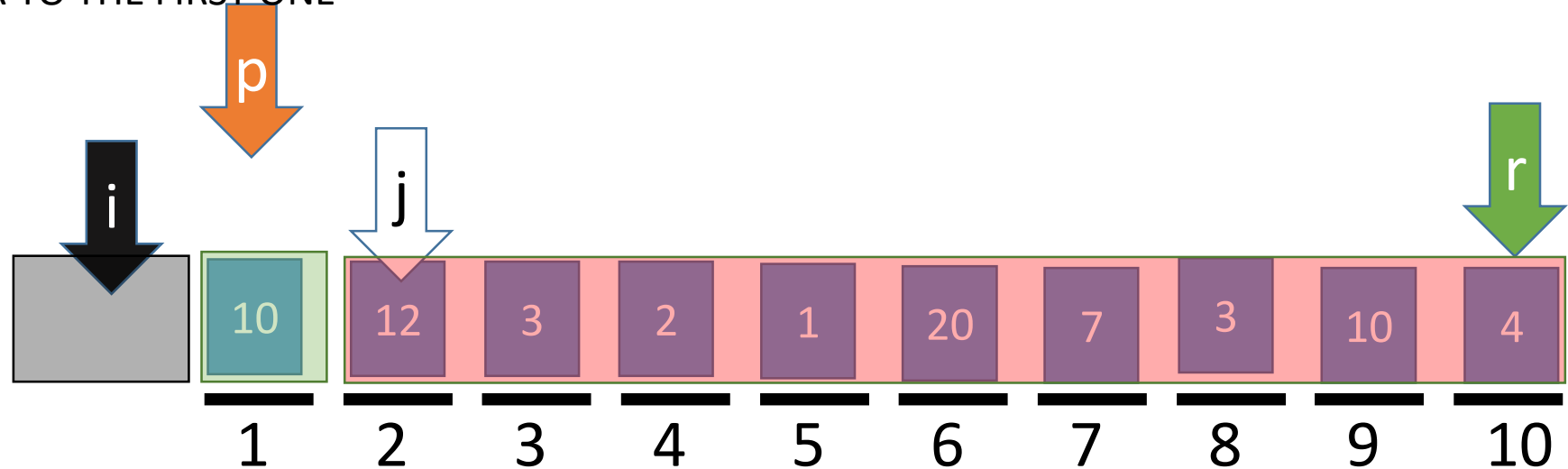
- Numbers greater than the pivot (marked by **j at end**)



Pivot
posn

Partition Function (A, p, r)

WE WANT THE PIVOT TO HAVE THE VALUE 4 BUT WANT IT IN THE LAST POSITION
SIMILAR TO THE FIRST ONE



- Numbers smaller than or equal to pivot (marked by **i at the end**)



- Numbers remaining to be explored (marked by **j at the beginning**)



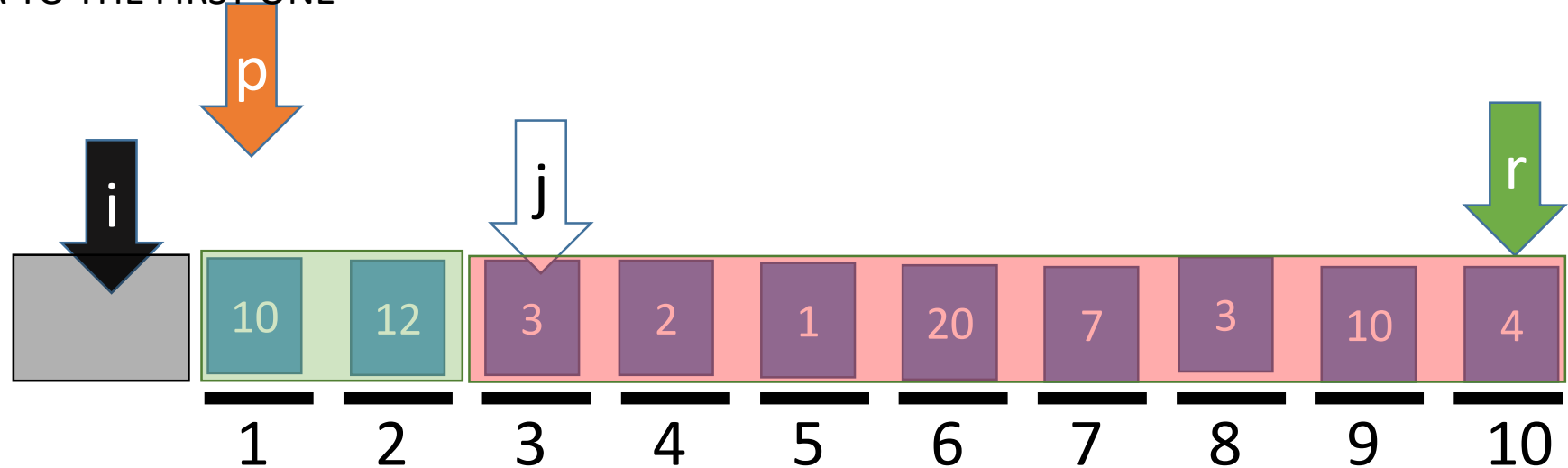
- Numbers greater than the pivot (marked by **j at end**)



Pivot
posn

Partition Function (A, p, r)

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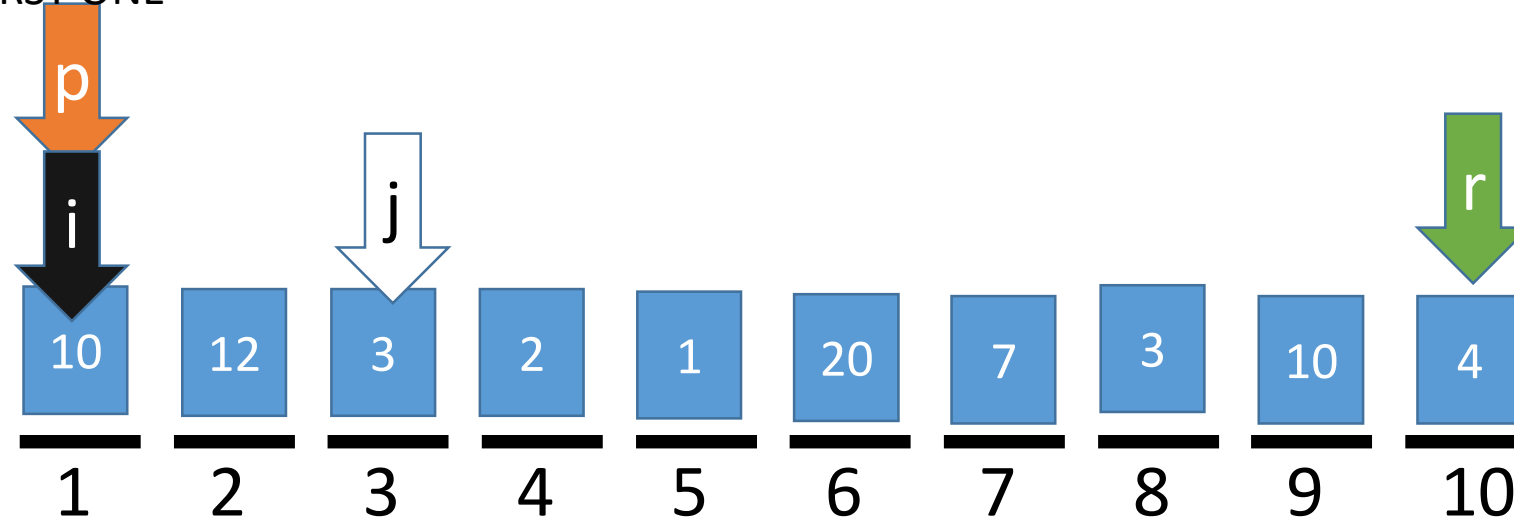
- Numbers greater than the pivot (marked by **j at end**)



Pivot
posn

Partition Function (A, p, r)

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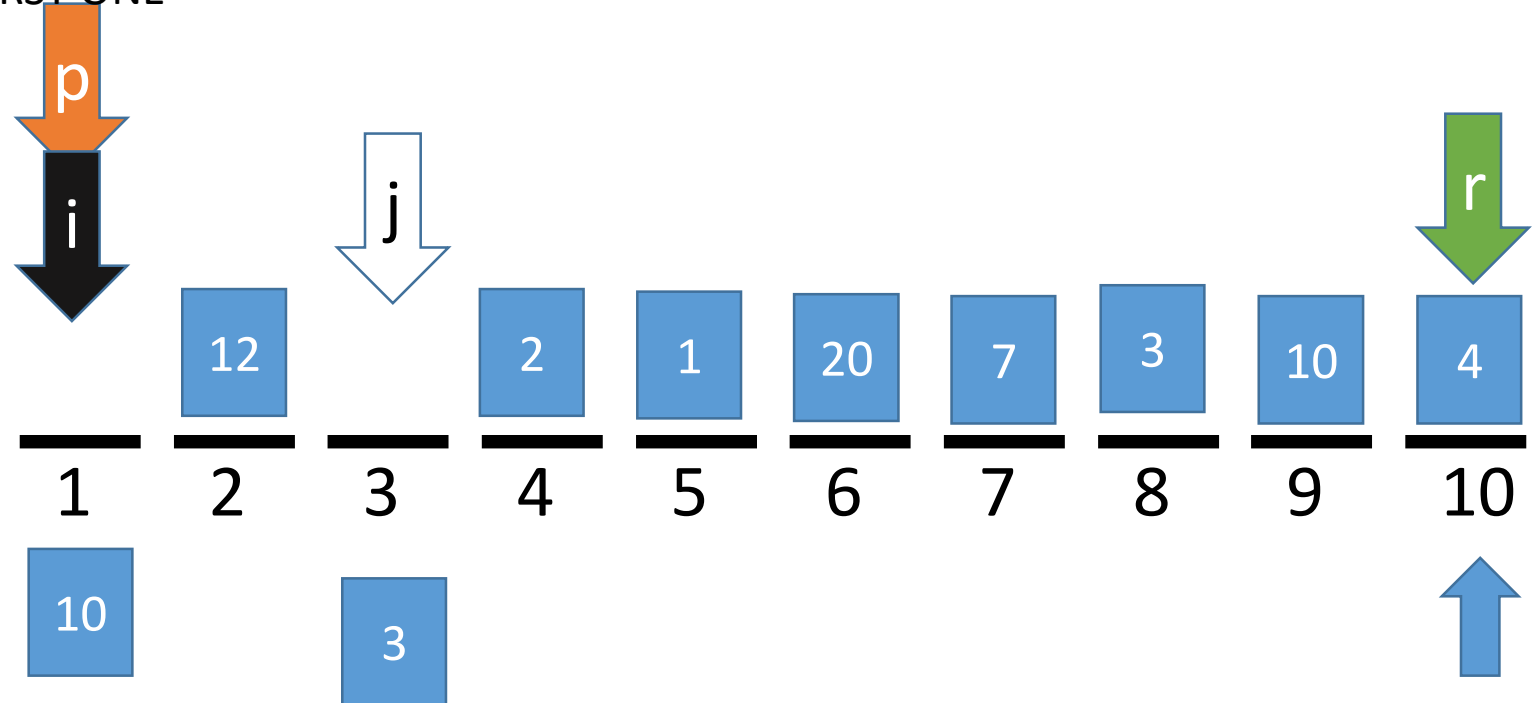
- Numbers greater than the pivot (marked by **j at end**)



Pivot
posn

Partition Function (A, p, r)

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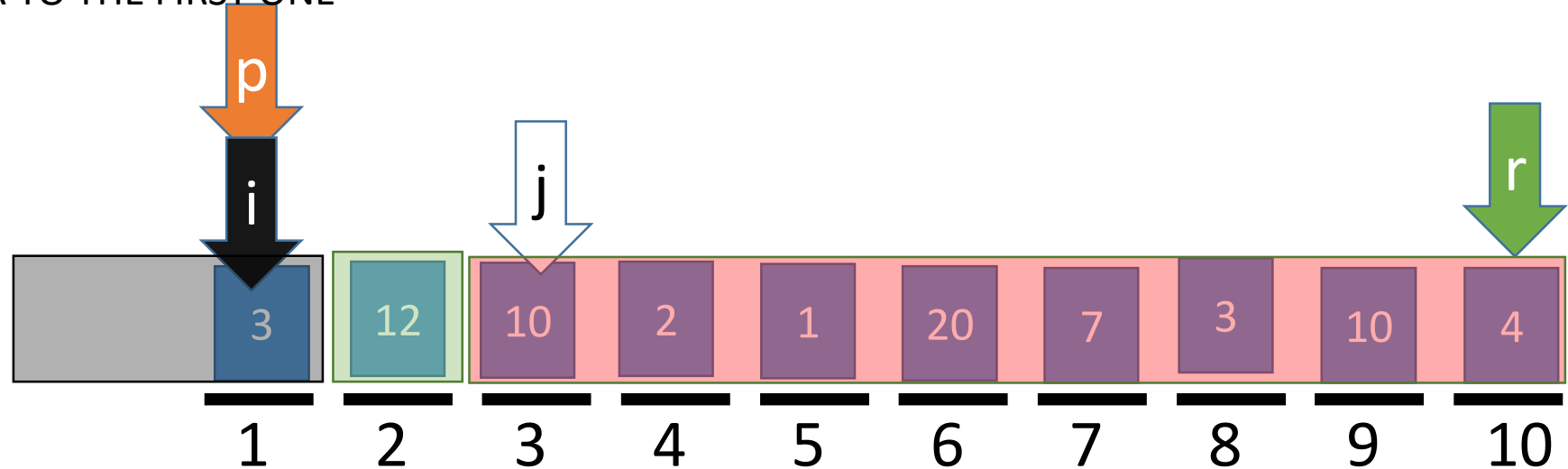
- Numbers greater than the pivot (marked by **j at end**)



Pivot
posn

Partition Function (A, p, r)

WE WANT THE PIVOT TO HAVE THE VALUE 4 BUT WANT IT IN THE LAST POSITION
SIMILAR TO THE FIRST ONE



- Numbers smaller than or equal to pivot (marked by **i at the end**)



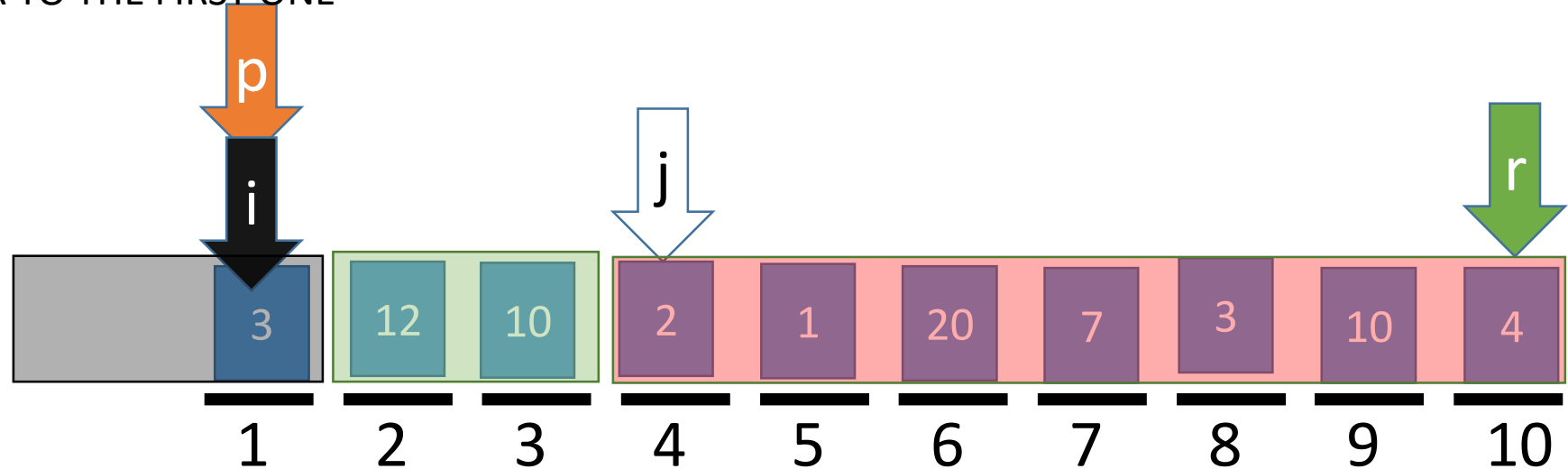
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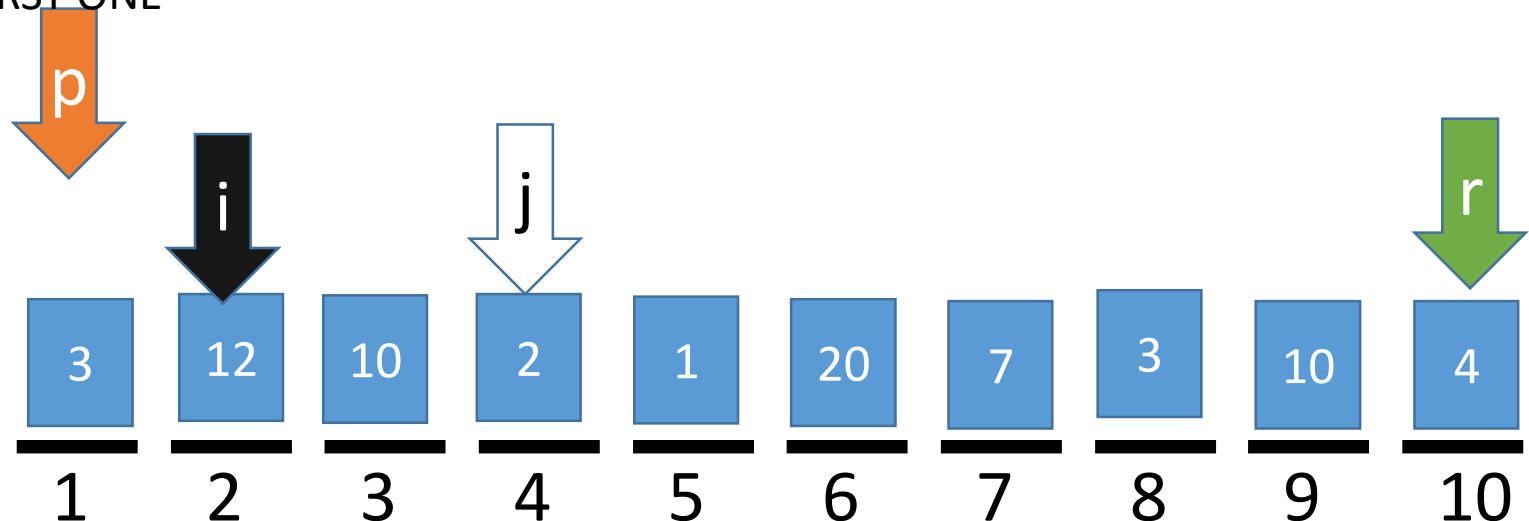
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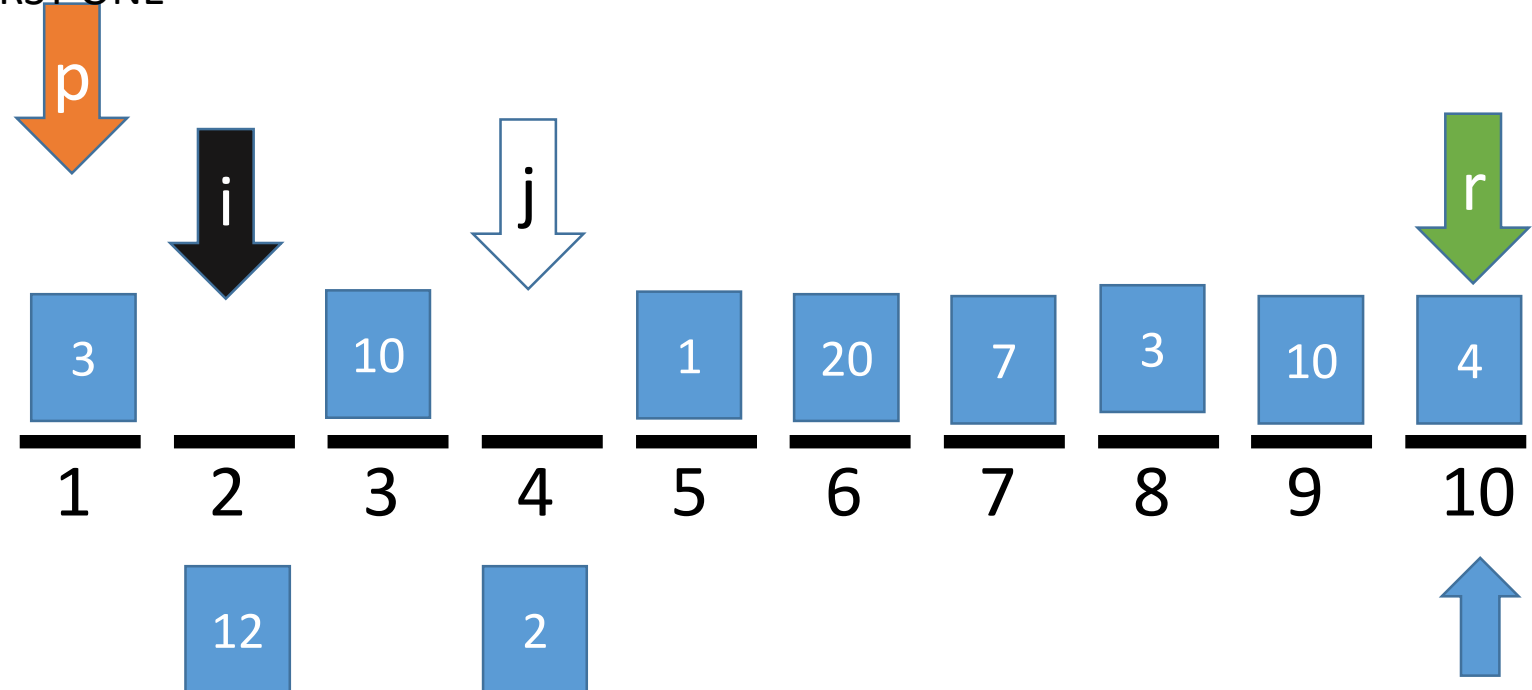
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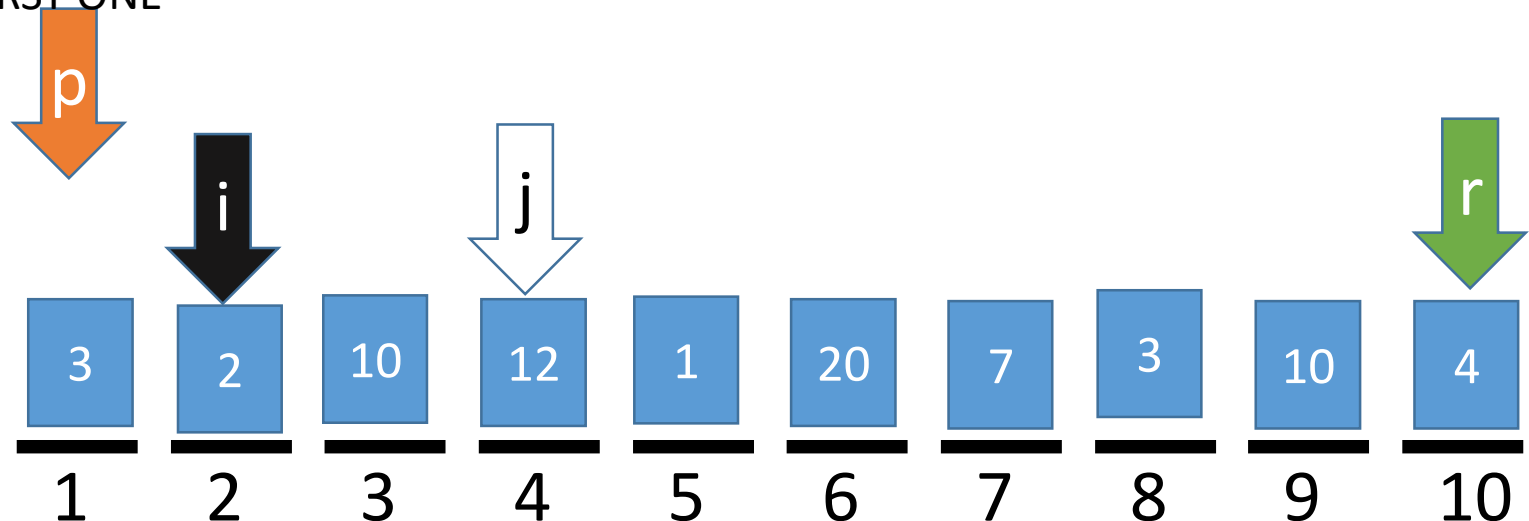
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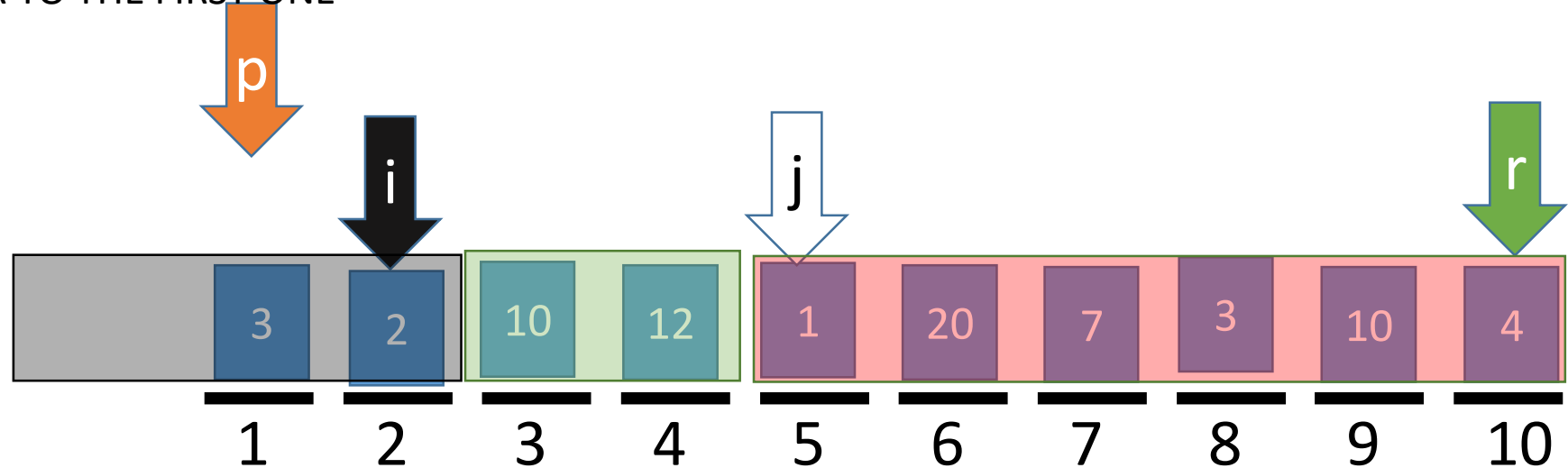
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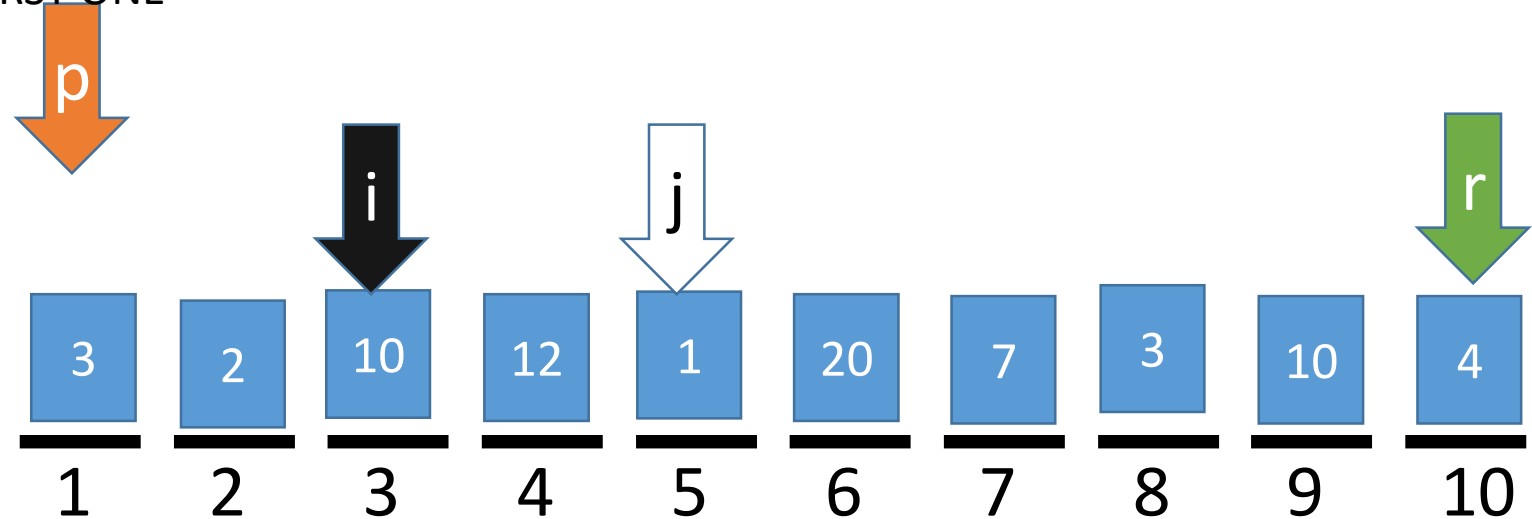
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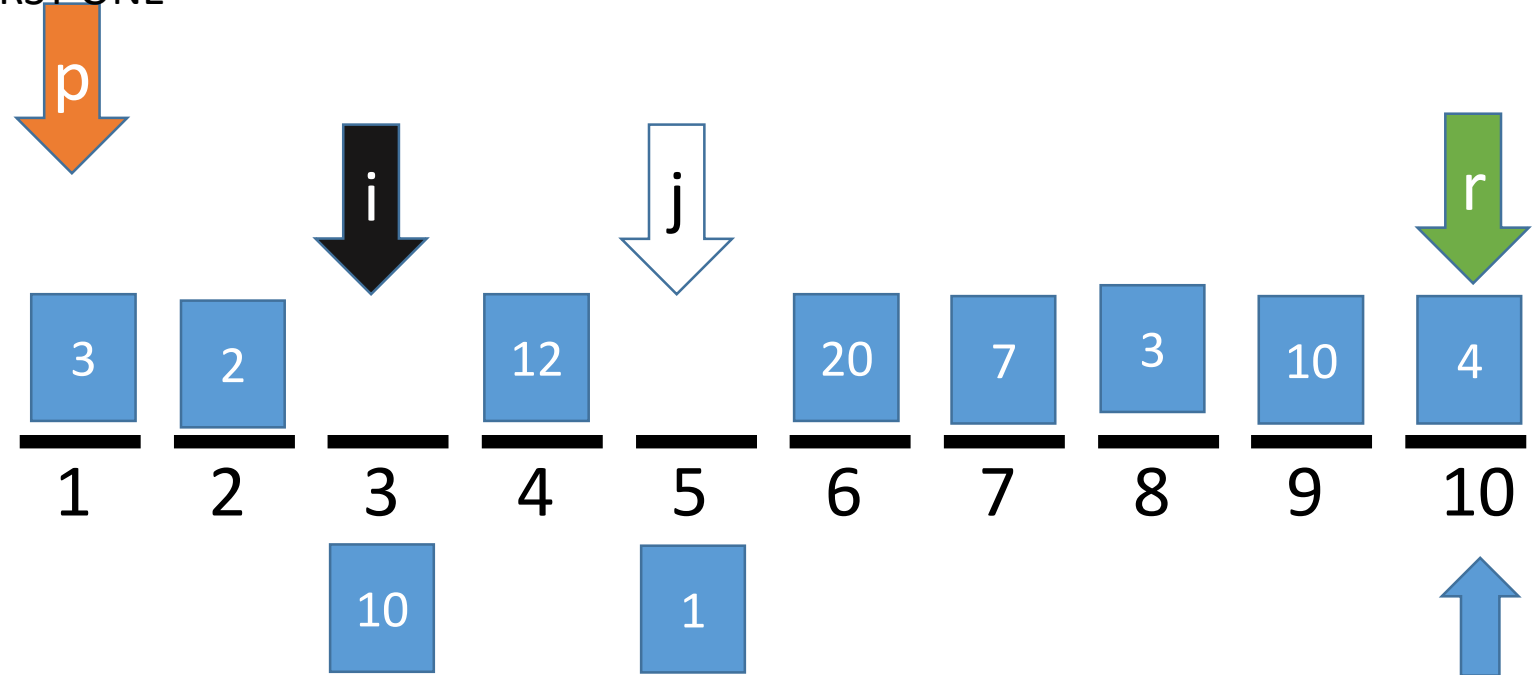
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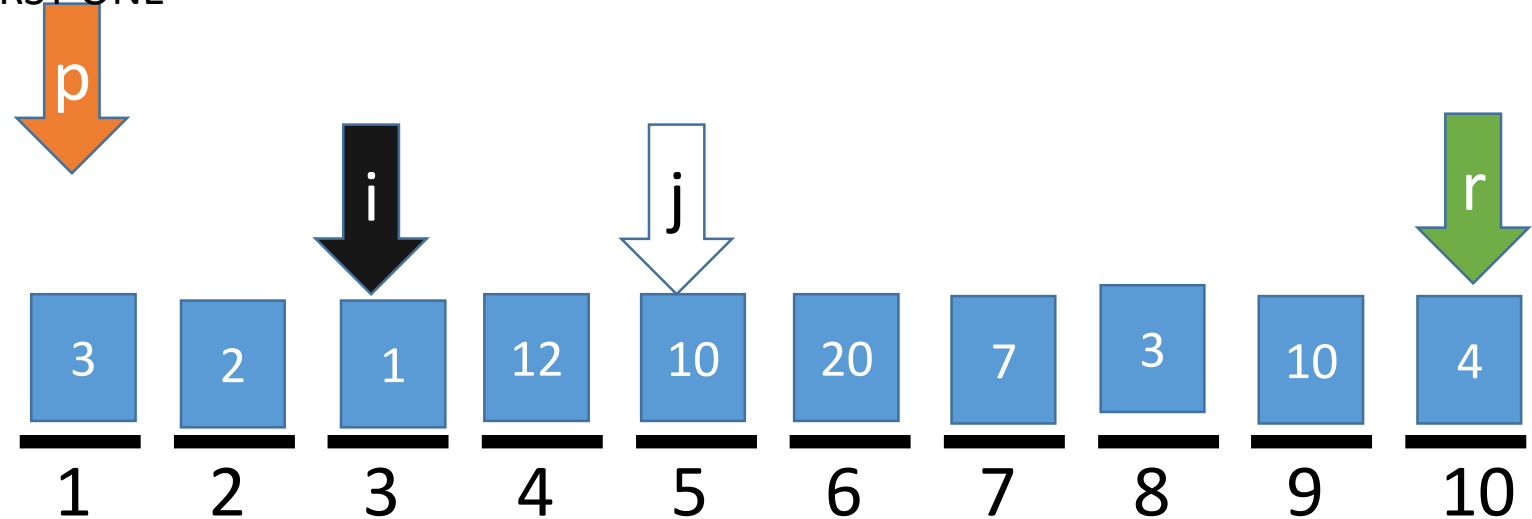
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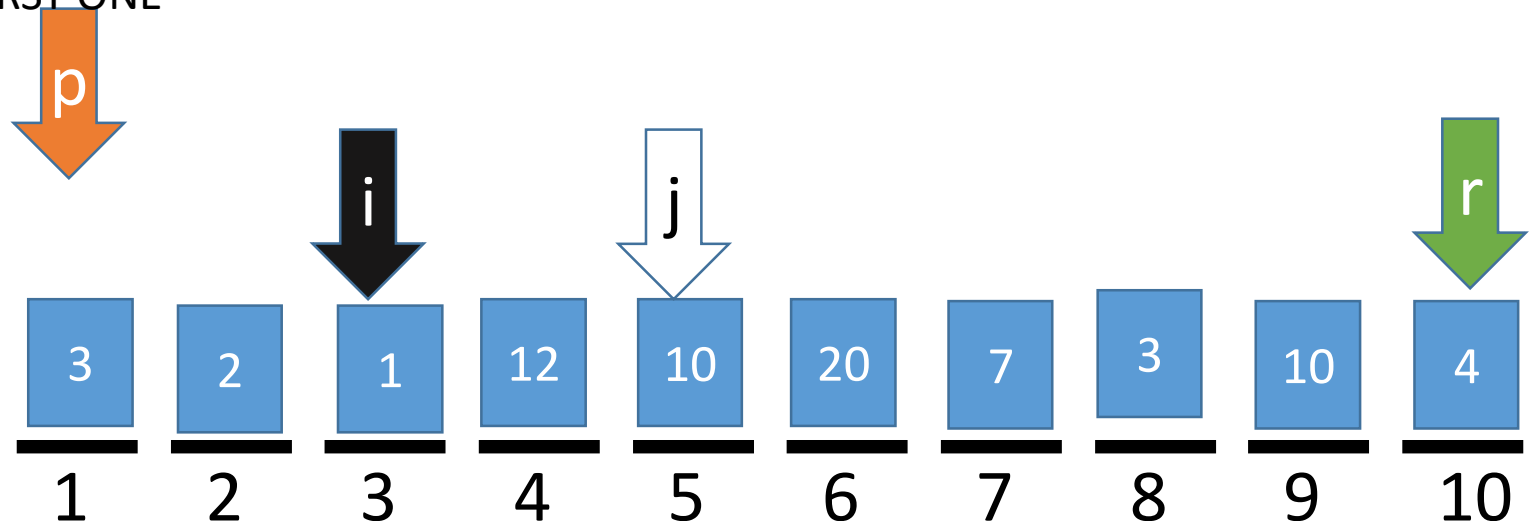
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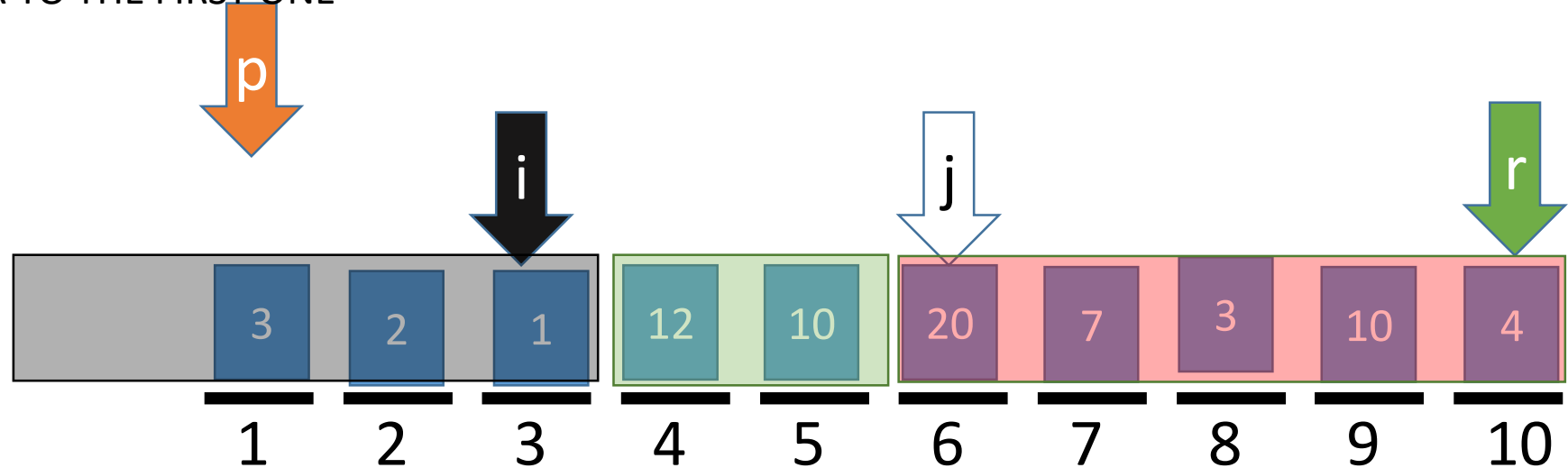
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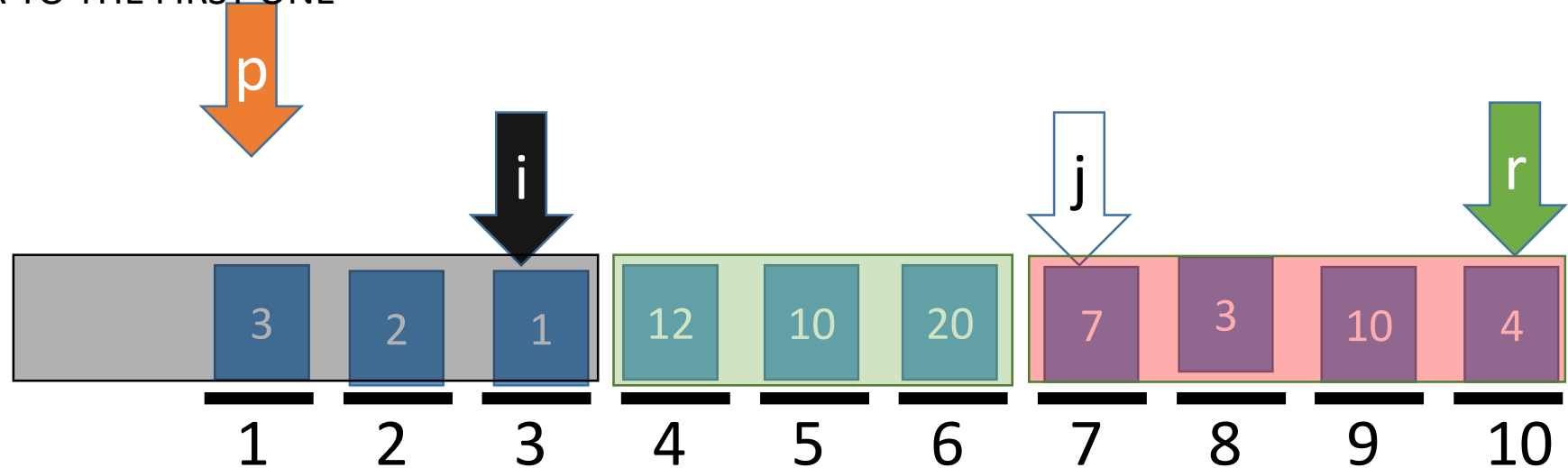
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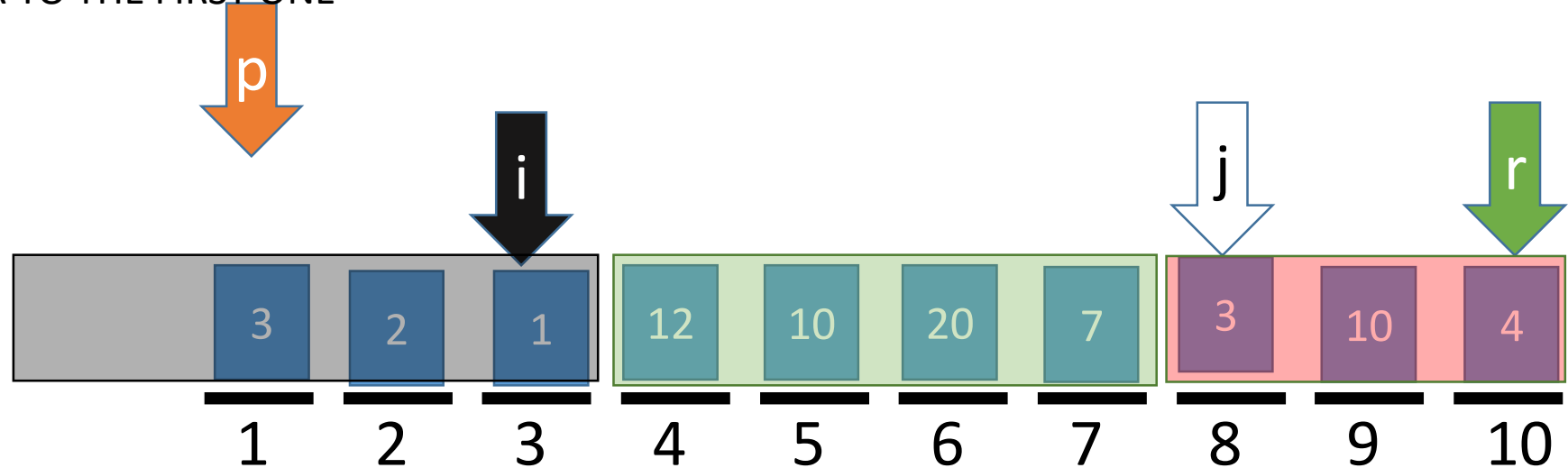
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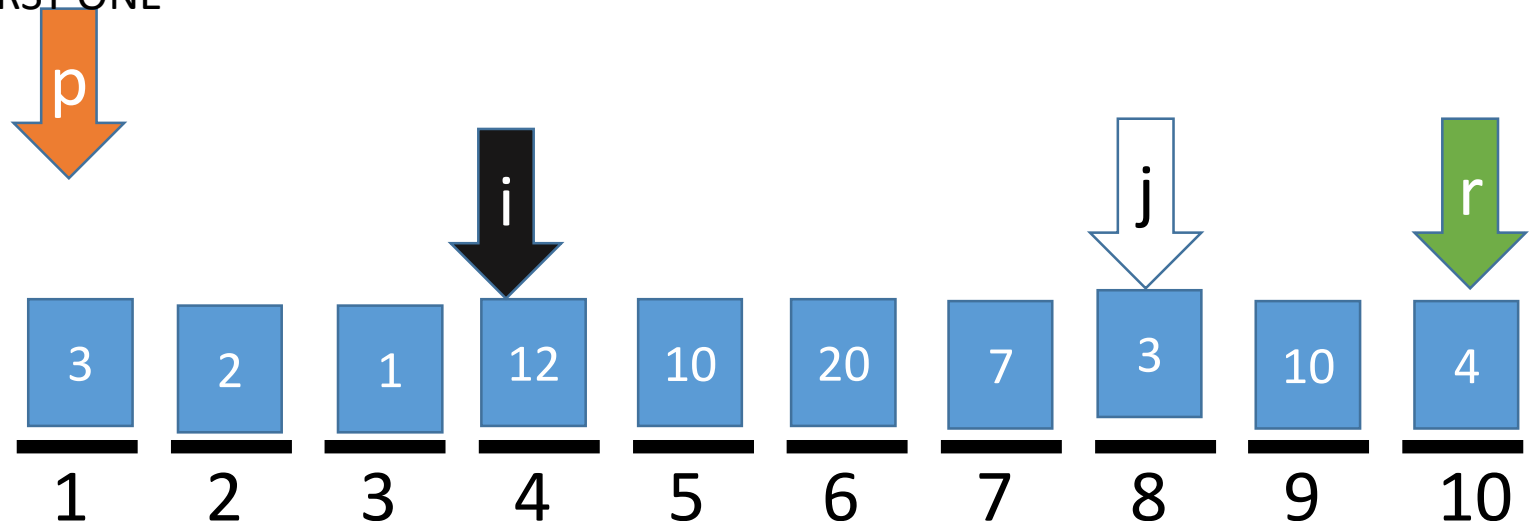
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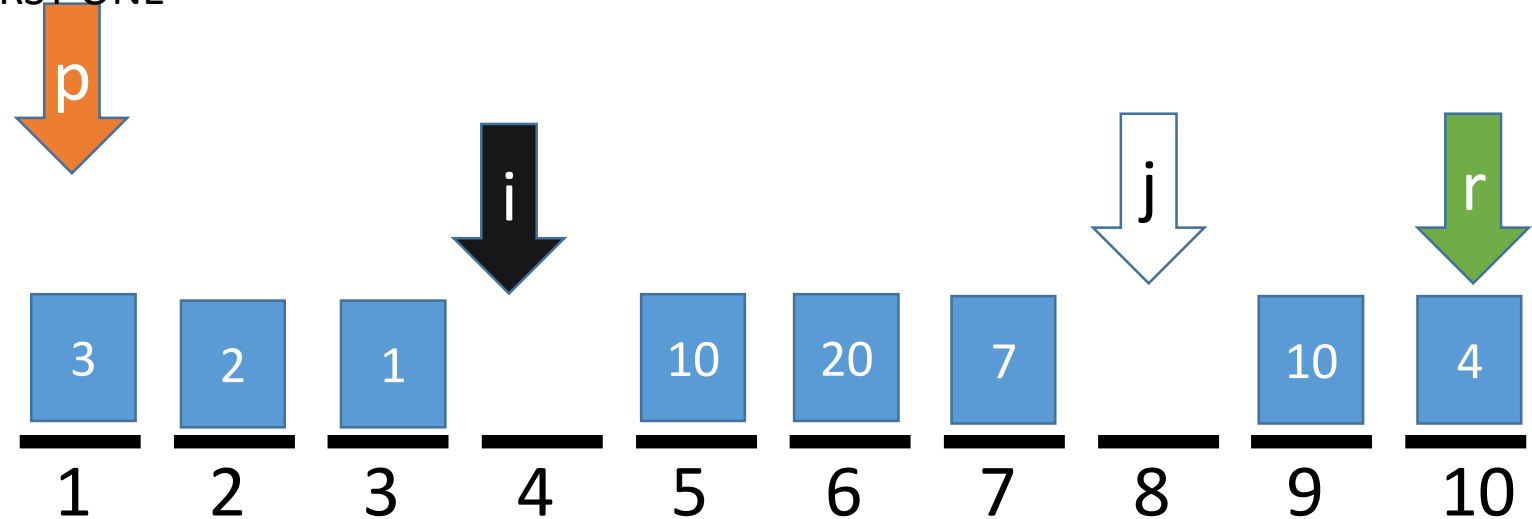
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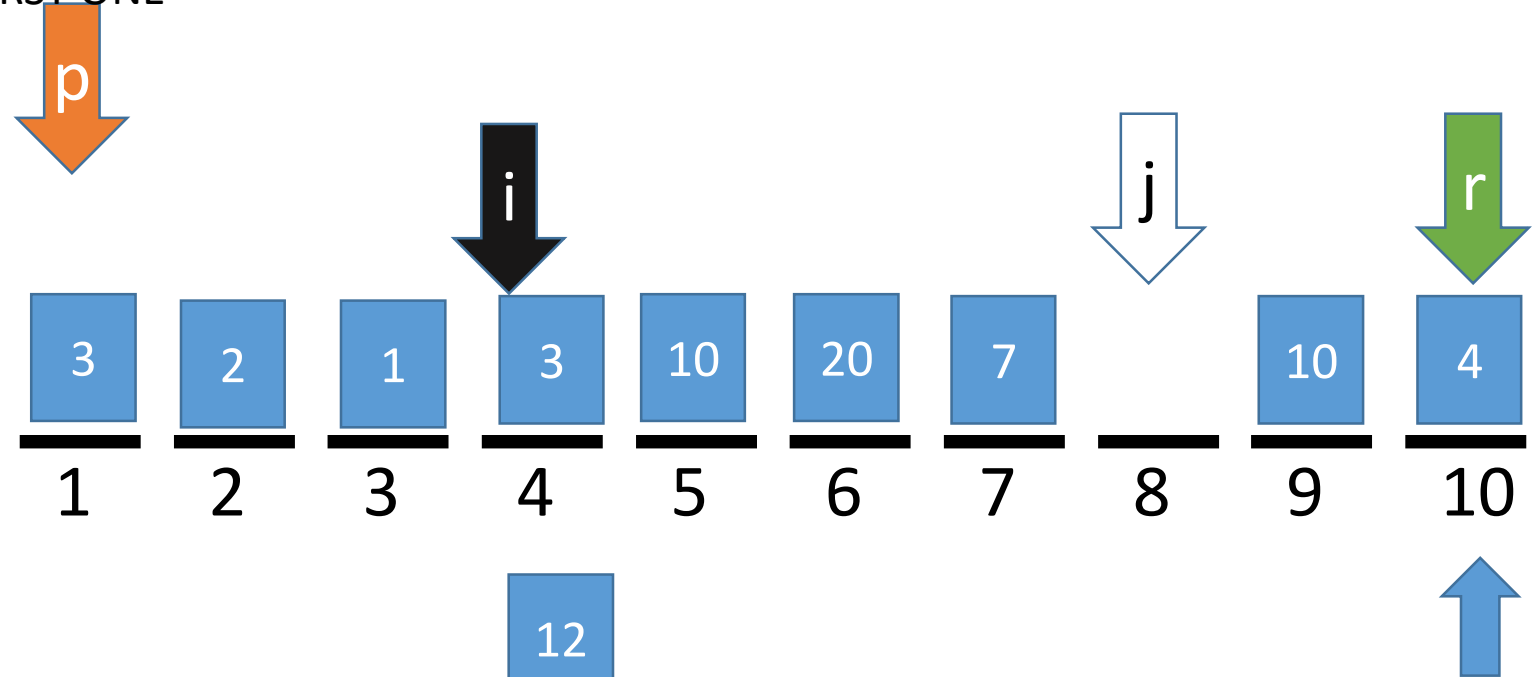
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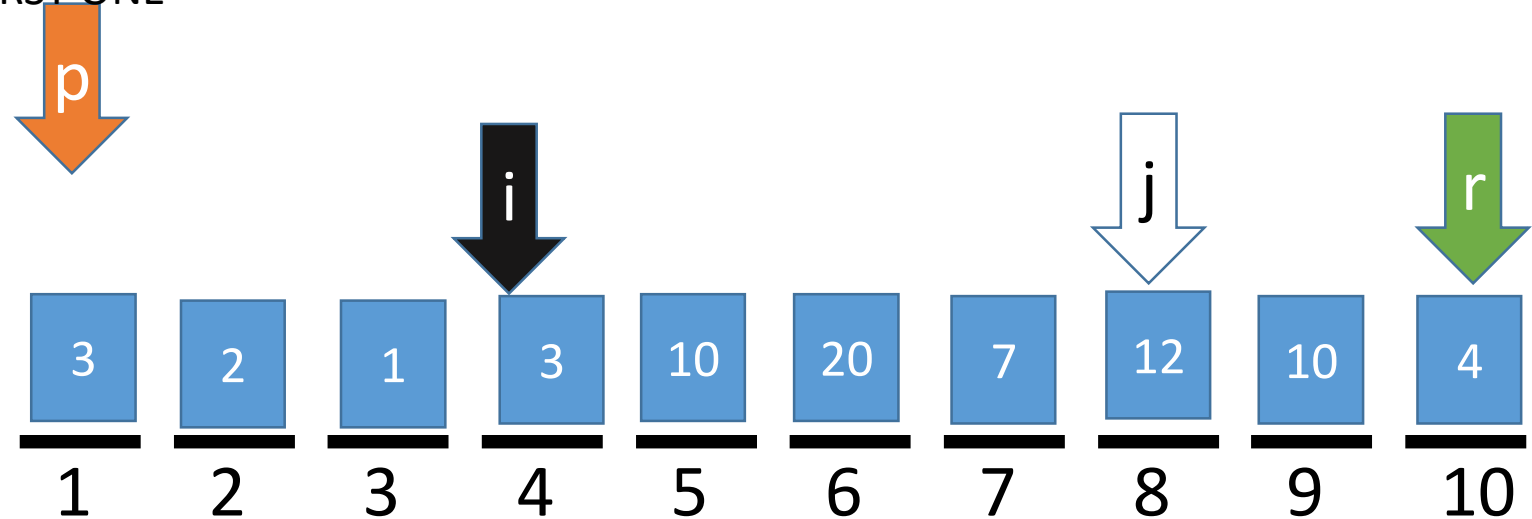
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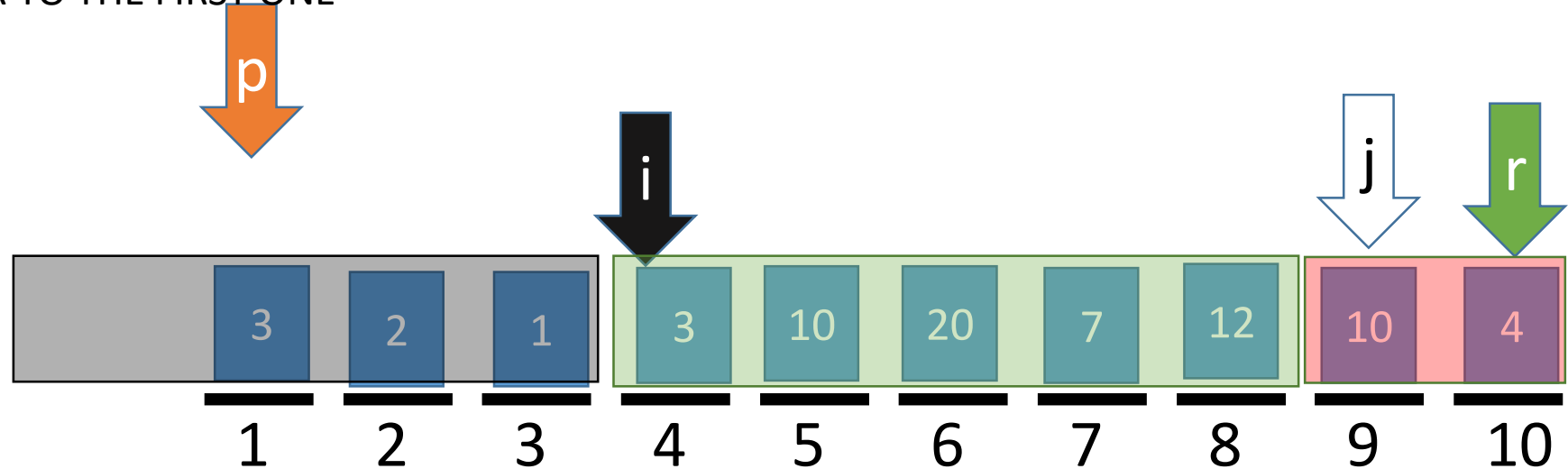
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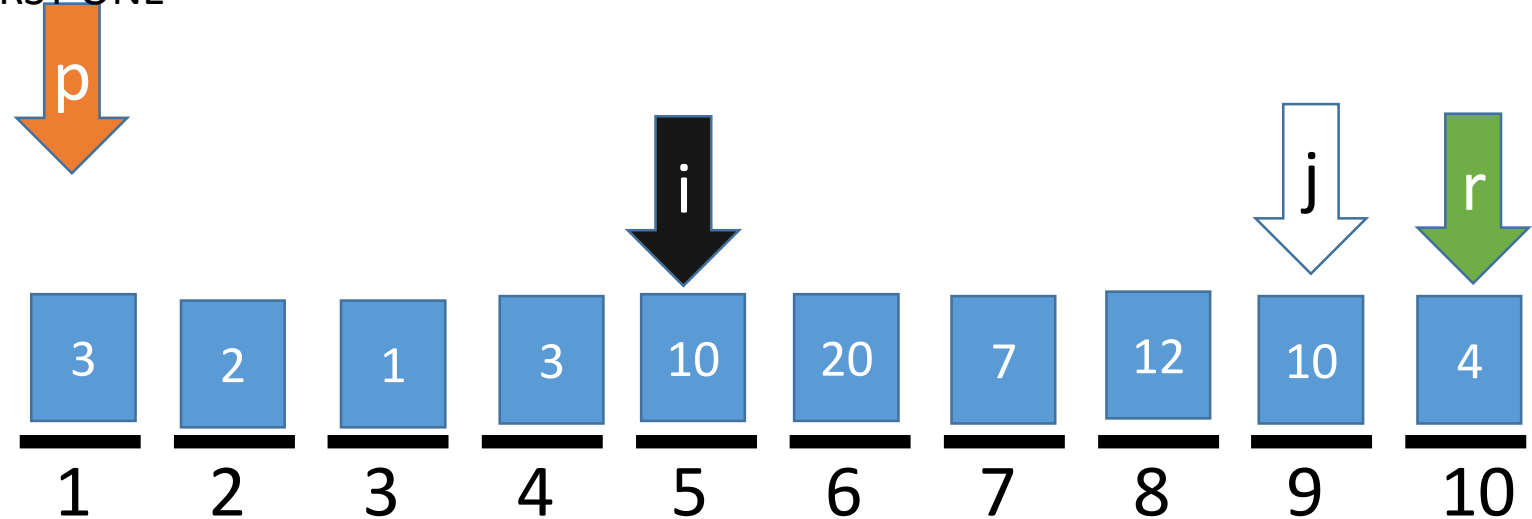
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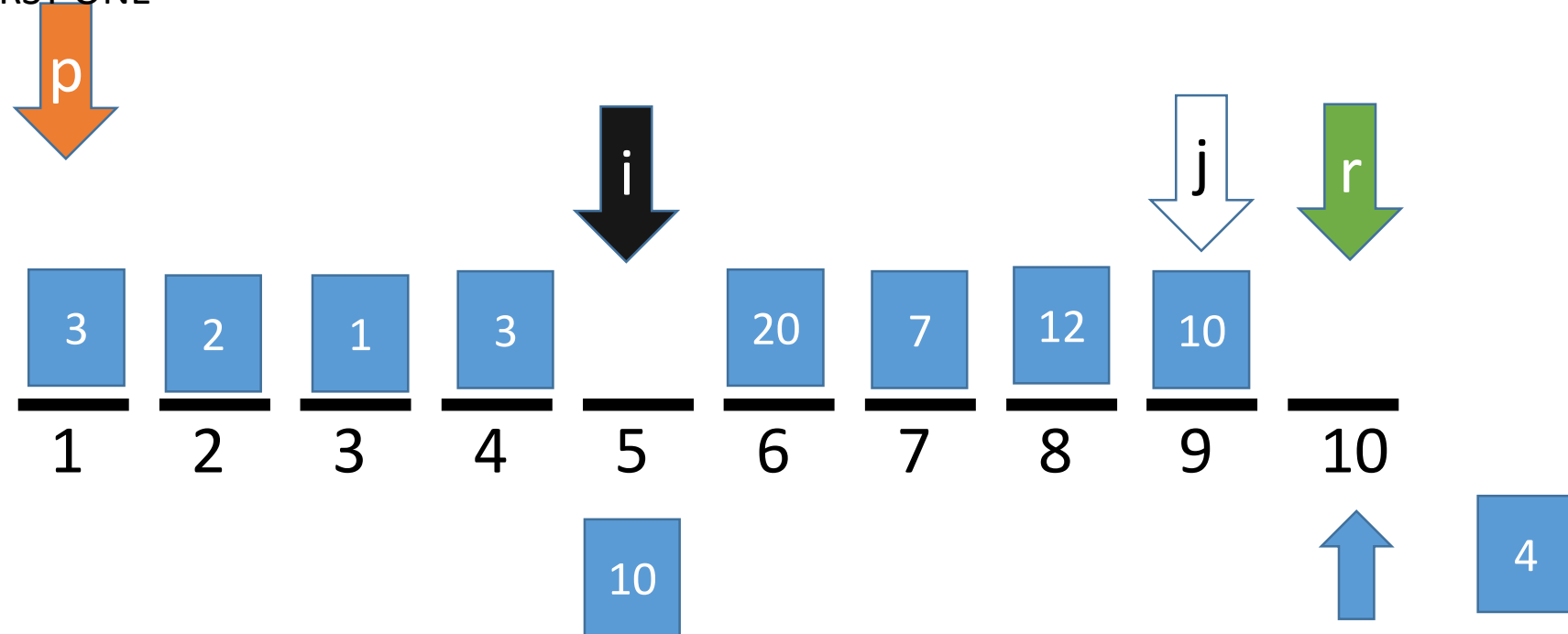
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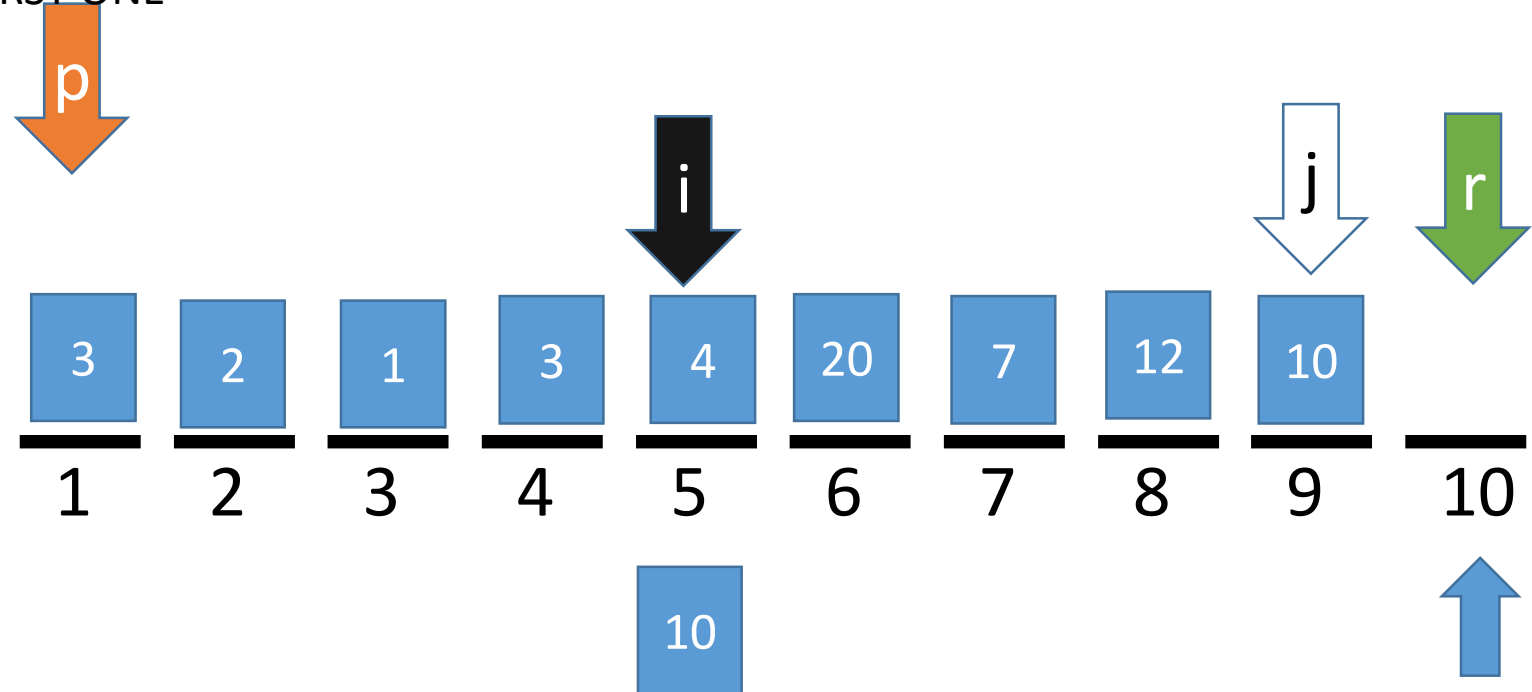


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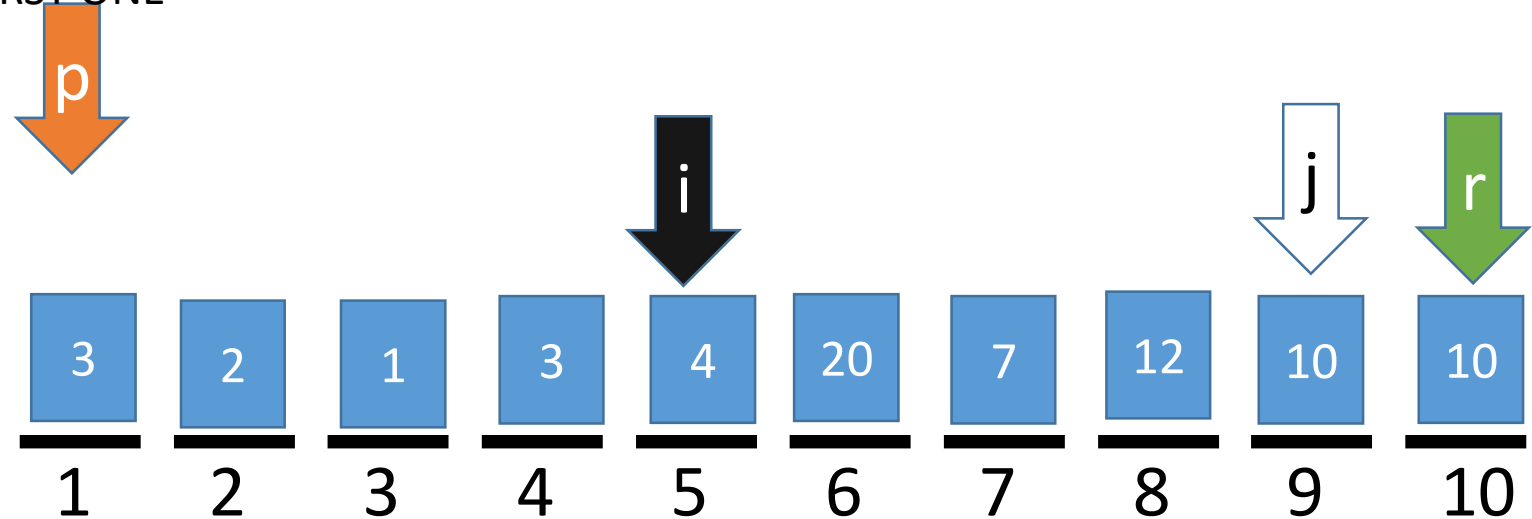
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Pivot
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Pseudocode for Partition

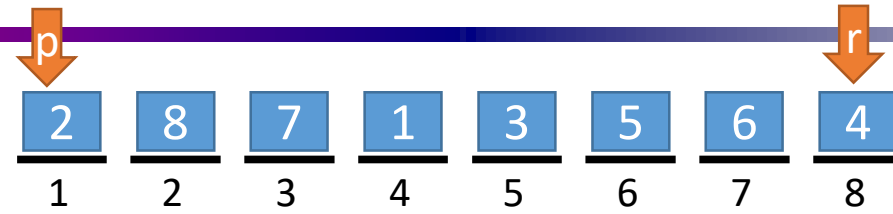
- For the first case:

PARTITION(A, p, r)

```
1   $x = A[r]$ 
2   $i = p - 1$ 
3  for  $j = p$  to  $r - 1$ 
4      if  $A[j] \leq x$ 
5           $i = i + 1$ 
6          exchange  $A[i]$  with  $A[j]$ 
7  exchange  $A[i + 1]$  with  $A[r]$ 
8  return  $i + 1$ 
```

Back to Quick Sort

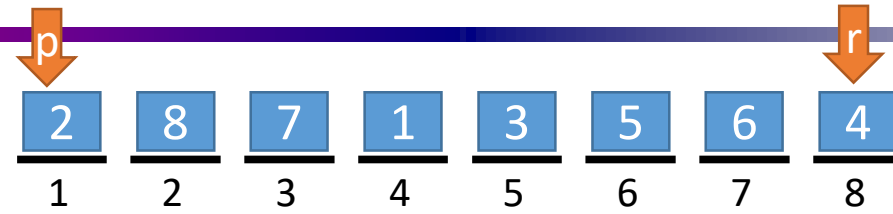
```
QUICKSORT( $A, p, r$ )  
→ if  $p < r$   
  2    $q = \text{PARTITION}(A, p, r)$   
  3   QUICKSORT( $A, p, q - 1$ )  
  4   QUICKSORT( $A, q + 1, r$ )
```



Back to Quick Sort

QUICKSORT(A, p, r)

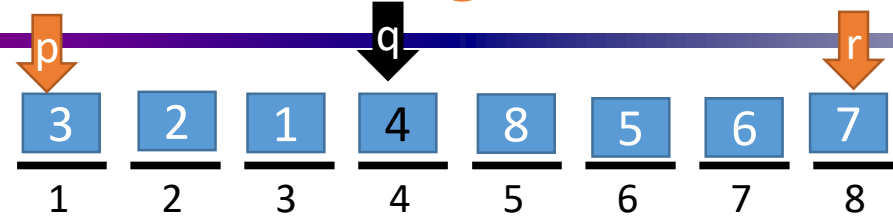
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QUICKSORT(A, p, r)

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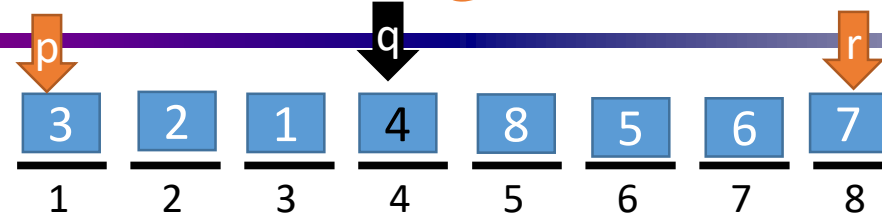


Back to Quick Sort

QUICKSORT(A, p, r)

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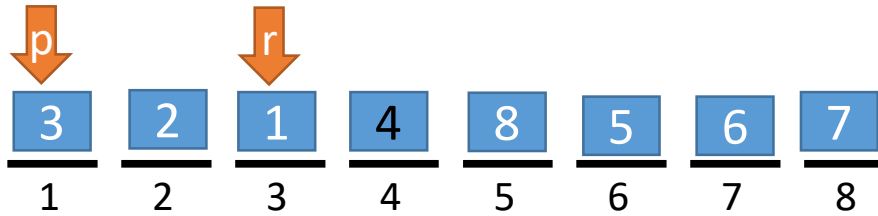
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QUICKSORT(A, p, r)

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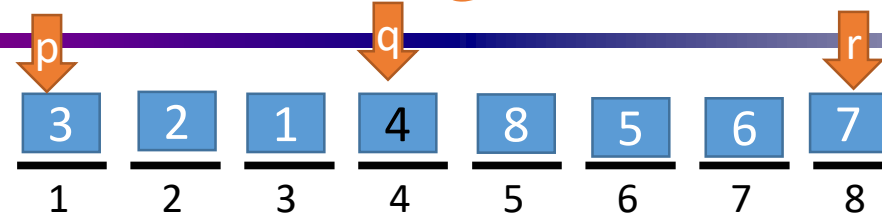


Back to Quick Sort

QUICKSORT(A, p, r)

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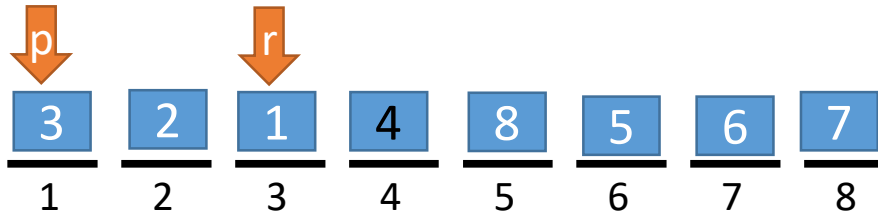
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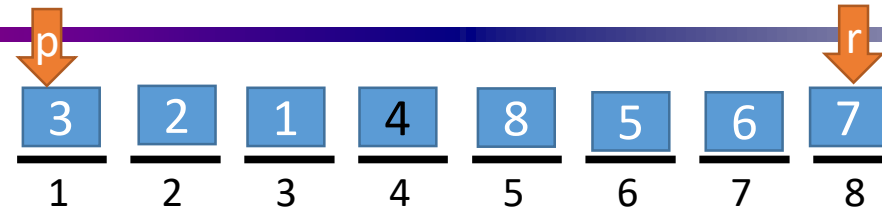


Back to Quick Sort

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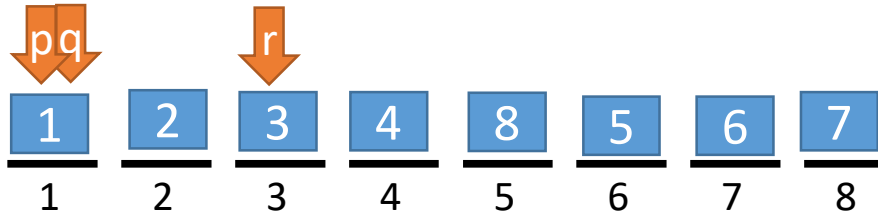
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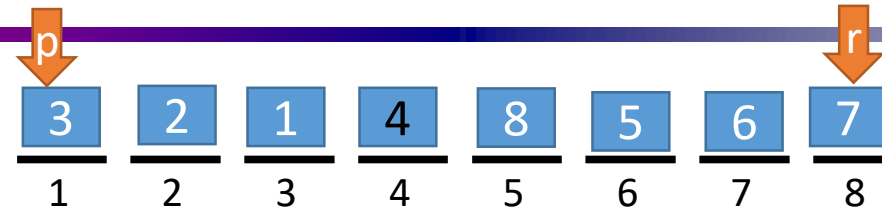


Back to Quick Sort

QUICKSORT(A, p, r)

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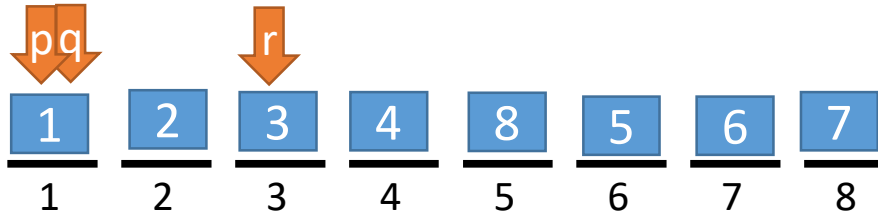
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QUICKSORT(A, p, r)

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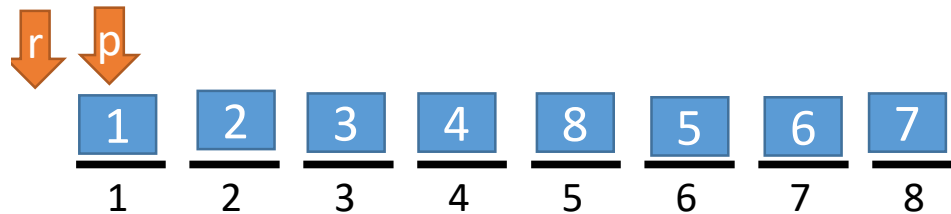


NO ELEMENT BEFORE PARTITION

QUICKSORT(A, p, r)

```

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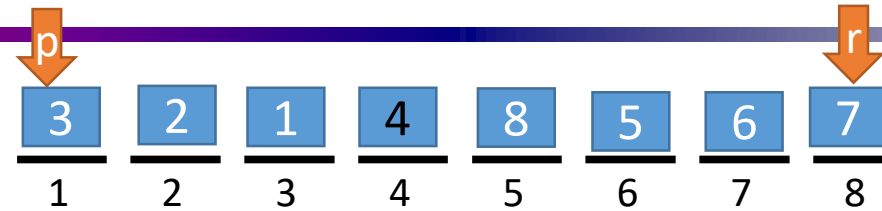


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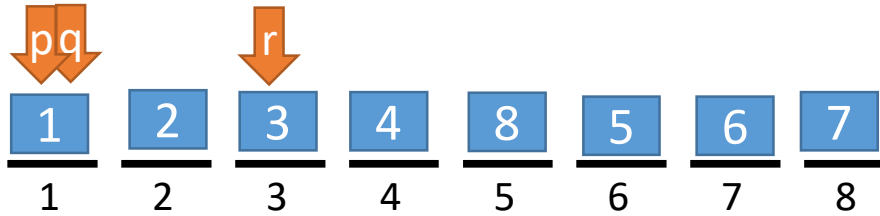
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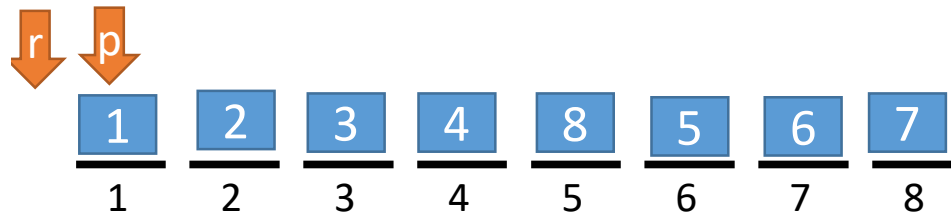
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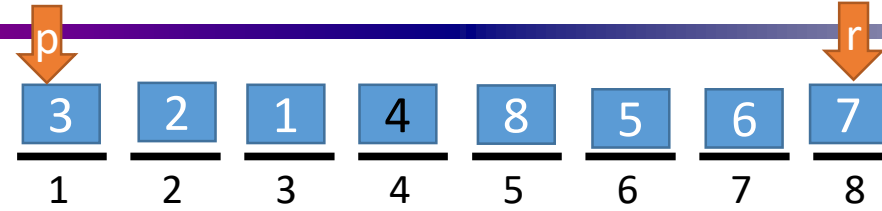


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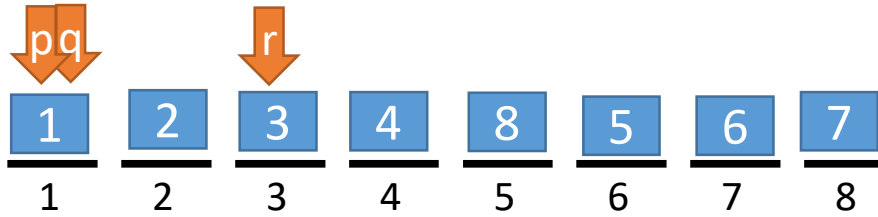
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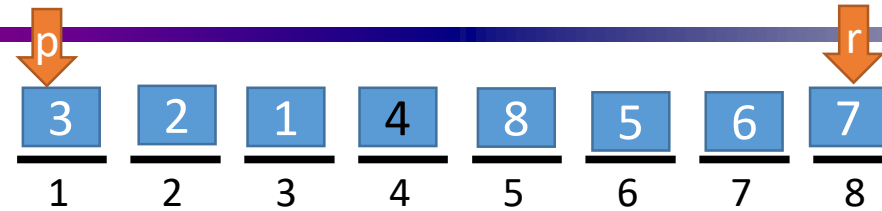


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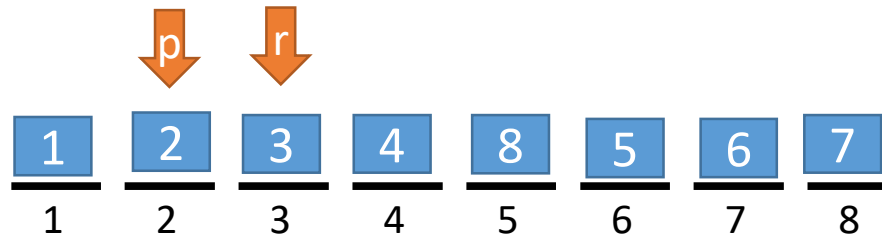
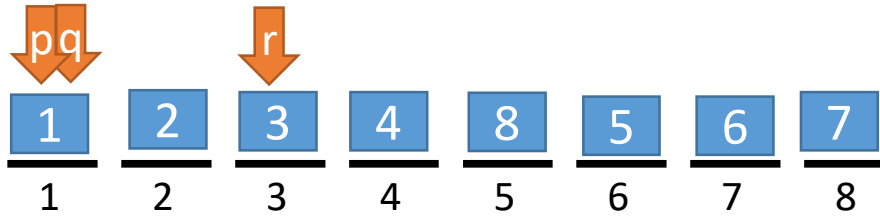
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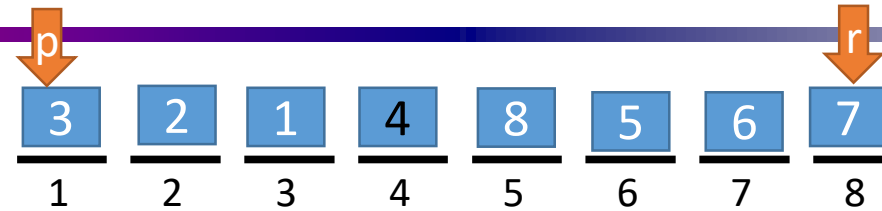


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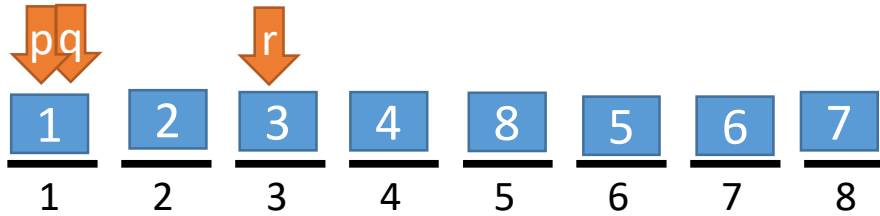
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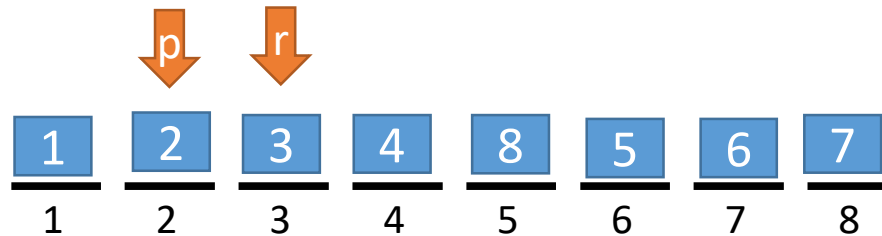
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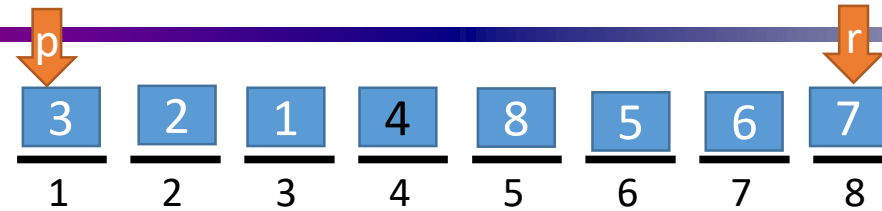


Back to Quick Sort

QUICKSORT(A, p, r)

```

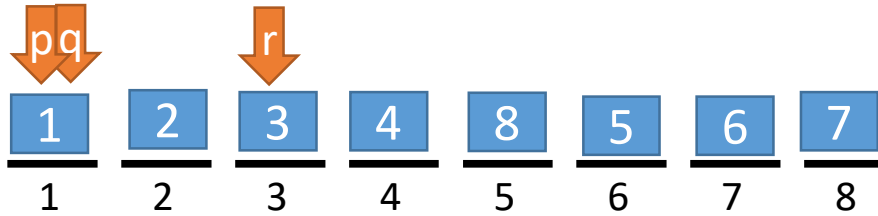
1  if  $p < r$ 
2     $q = \text{PARTITION}(A, p, r)$ 
3    → QUICKSORT( $A, p, q - 1$ )
4    → QUICKSORT( $A, q + 1, r$ )
    
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QUICKSORT(A, p, r)

```

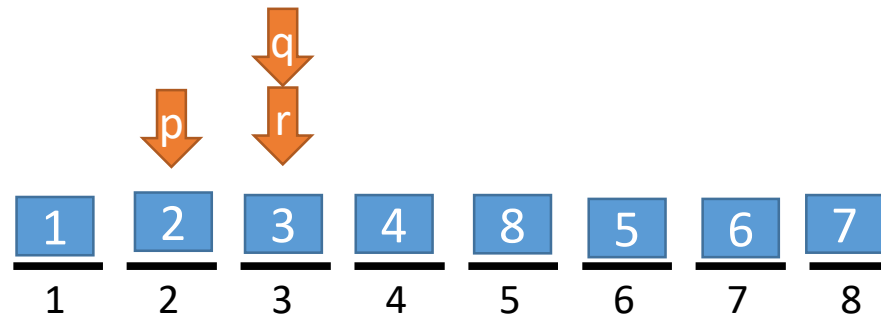
1  if  $p < r$ 
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QUICKSORT(A, p, r)

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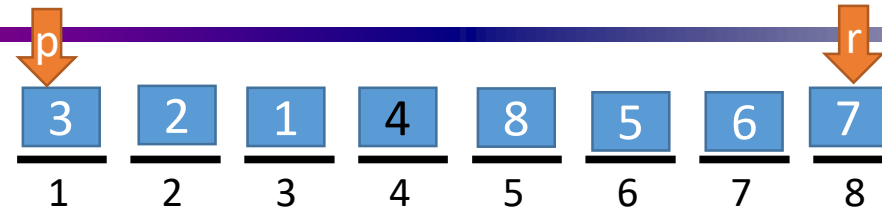


Back to Quick Sort

QUICKSORT(A, p, r)

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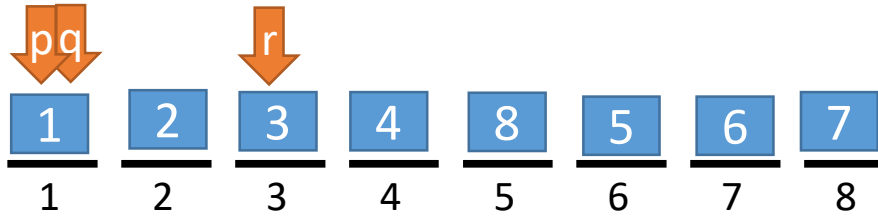
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QUICKSORT(A, p, r)

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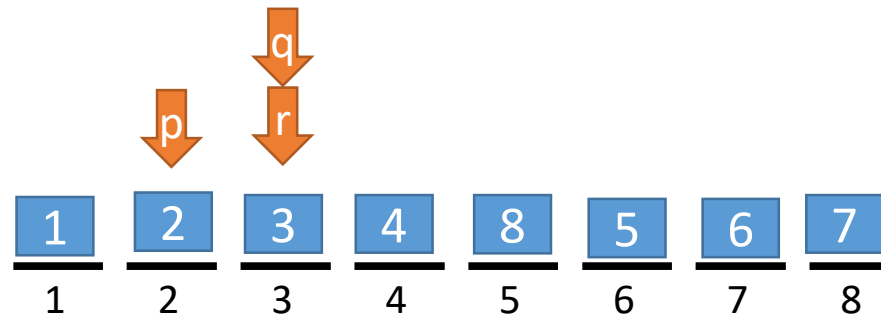
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QUICKSORT(A, p, r)

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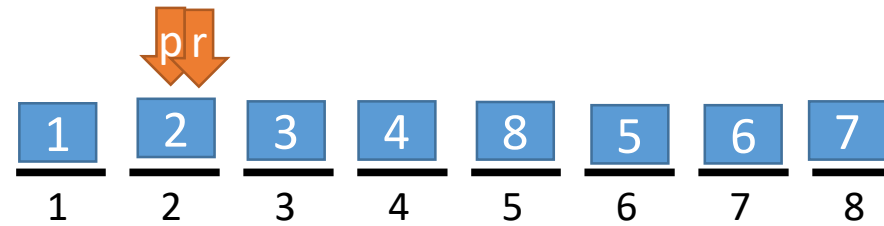


ONE ELEMENT BEFORE PARTITION

QUICKSORT(A, p, r)

```

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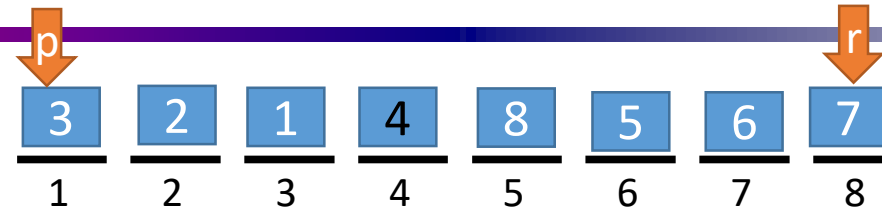


Back to Quick Sort

QUICKSORT(A, p, r)

```

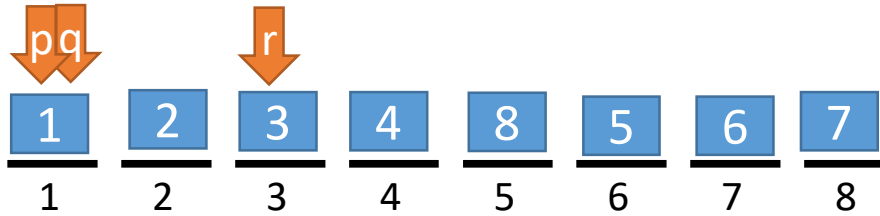
1  if  $p < r$ 
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QUICKSORT(A, p, r)

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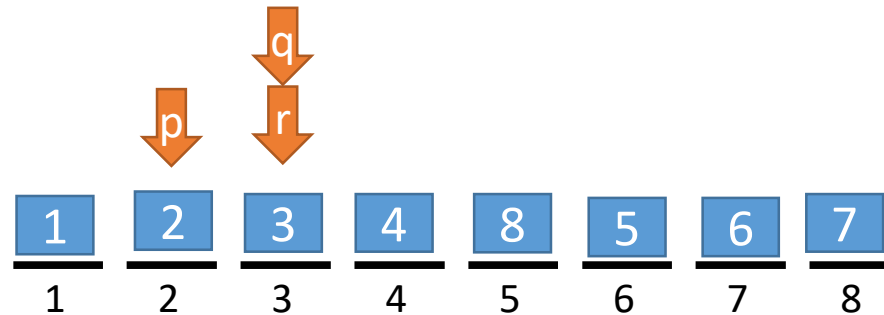
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QUICKSORT(A, p, r)

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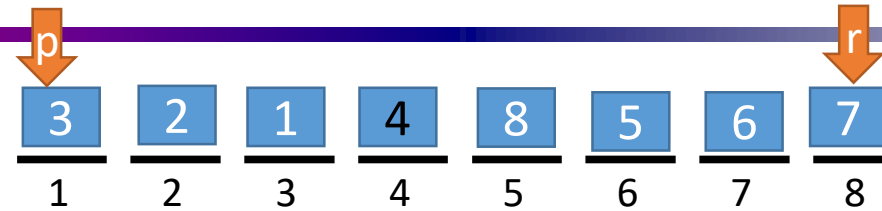


Back to Quick Sort

QUICKSORT(A, p, r)

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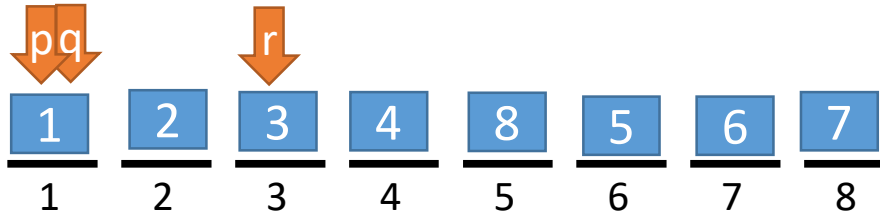
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QUICKSORT(A, p, r)

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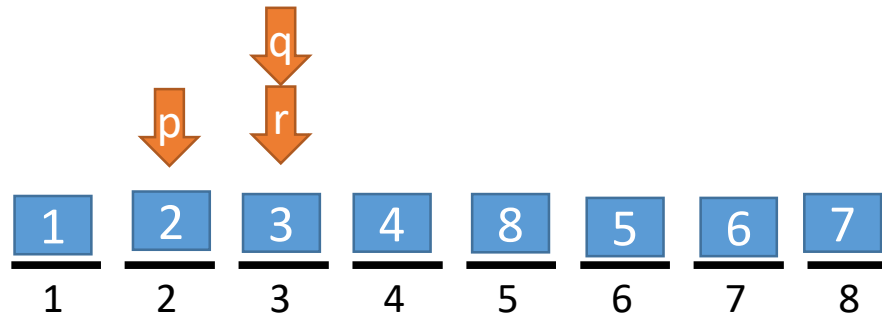
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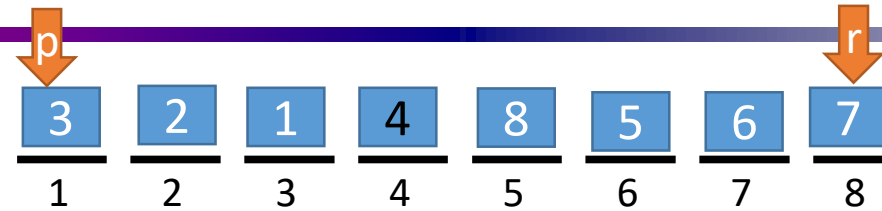


Back to Quick Sort

QUICKSORT(A, p, r)

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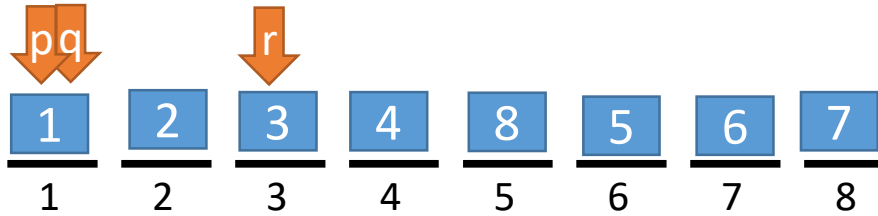
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QUICKSORT(A, p, r)

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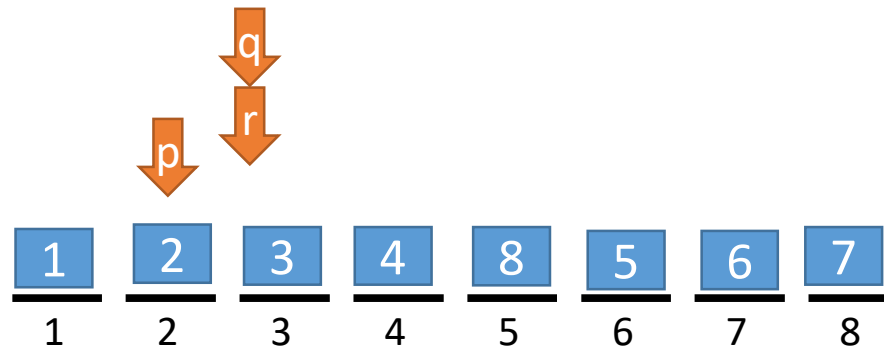
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QUICKSORT(A, p, r)

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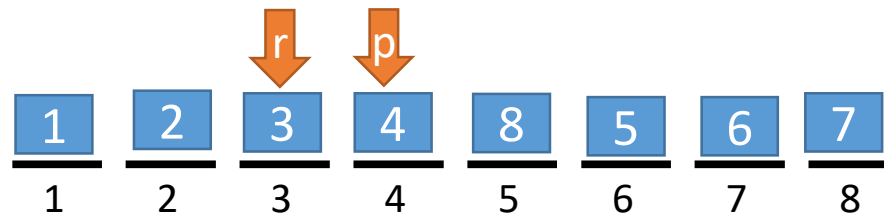
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→ QUICKSORT(A, p, r)

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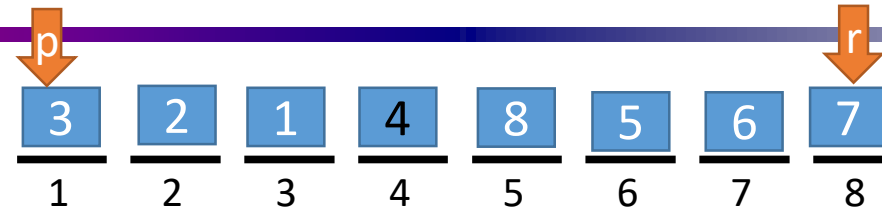


Back to Quick Sort

QUICKSORT(A, p, r)

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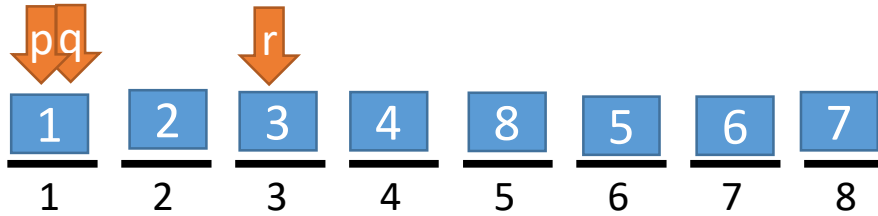
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QUICKSORT(A, p, r)

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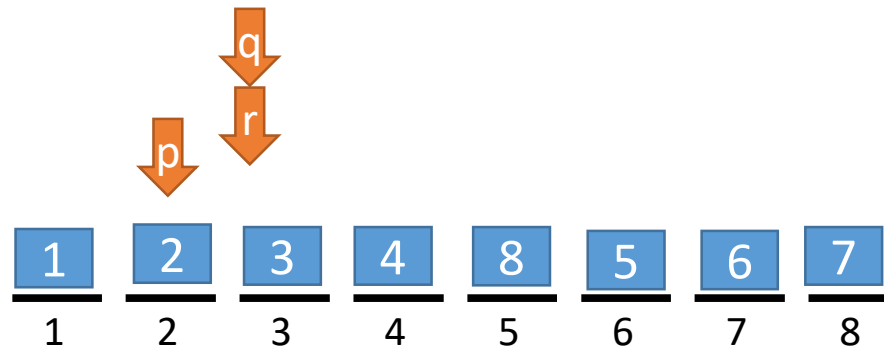
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QUICKSORT(A, p, r)

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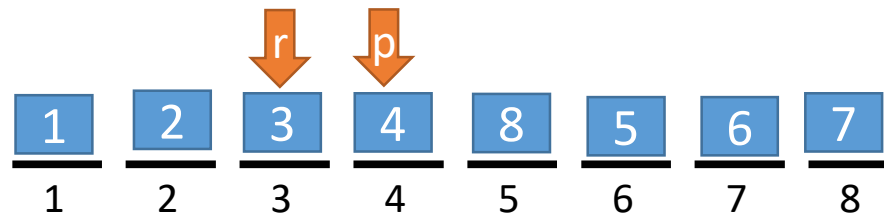
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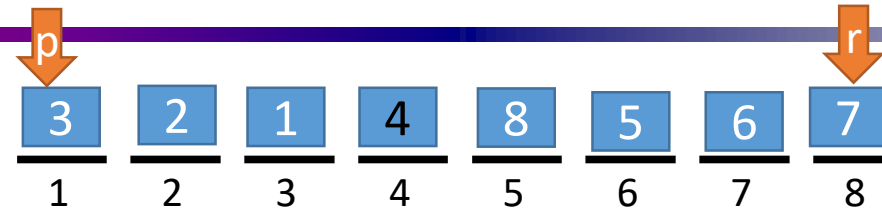


Back to Quick Sort

QUICKSORT(A, p, r)

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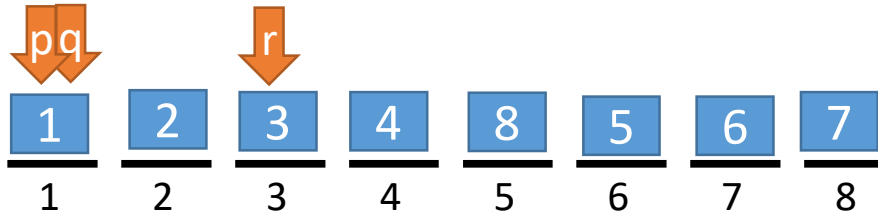
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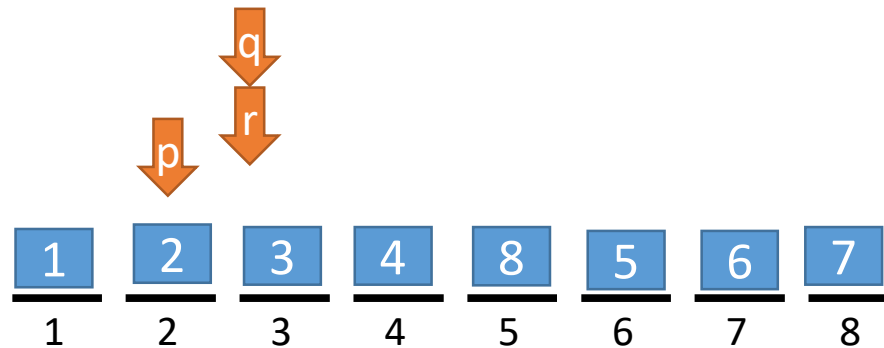
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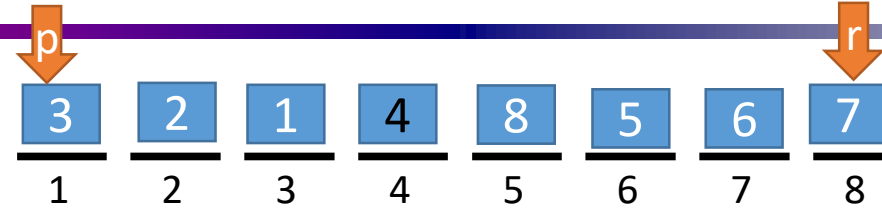


Back to Quick Sort

QUICKSORT(A, p, r)

```

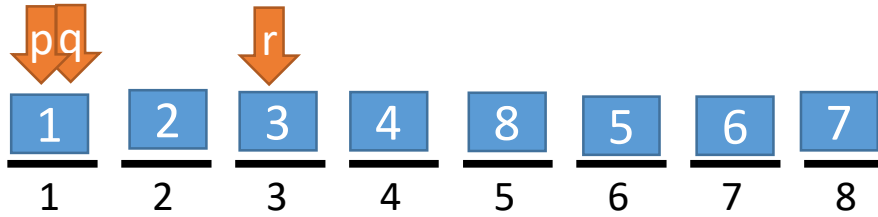
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QUICKSORT(A, p, r)

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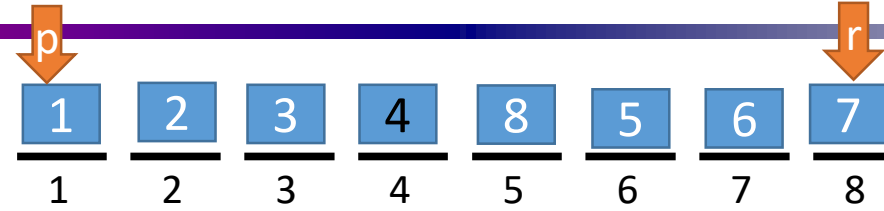
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Back to Quick Sort

QUICKSORT(A, p, r)

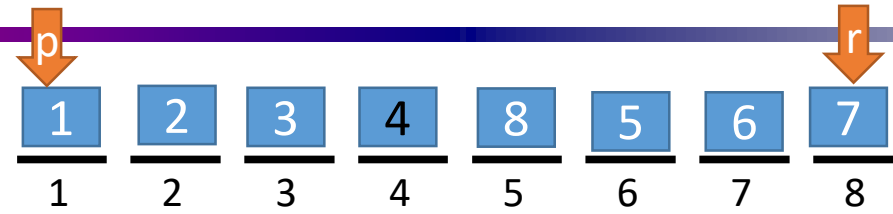
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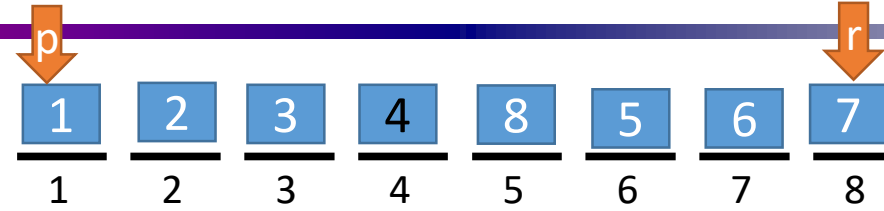


Back to Quick Sort

QUICKSORT(A, p, r)

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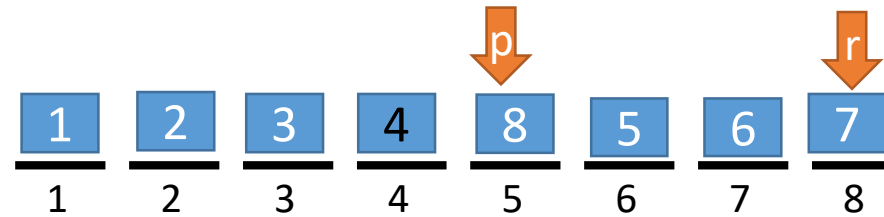
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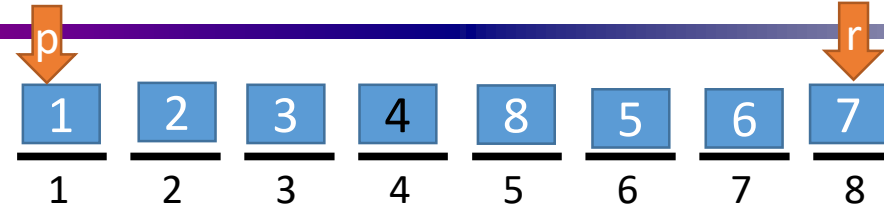
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Back to Quick Sort

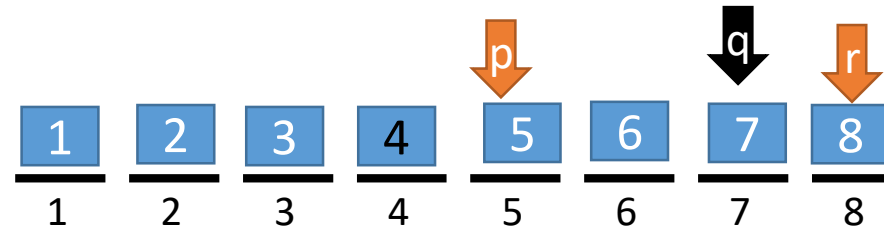
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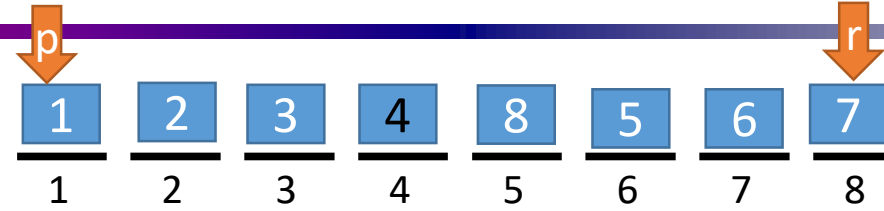
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Back to Quick Sort

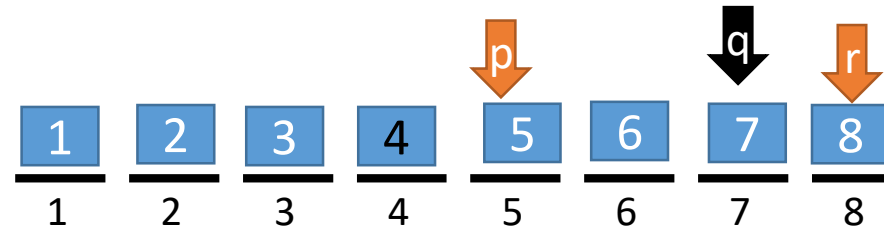
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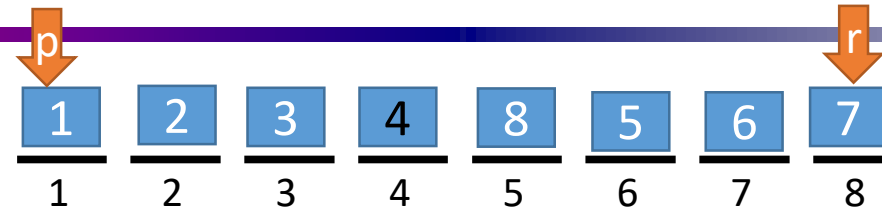


Back to Quick Sort

QUICKSORT(A, p, r)

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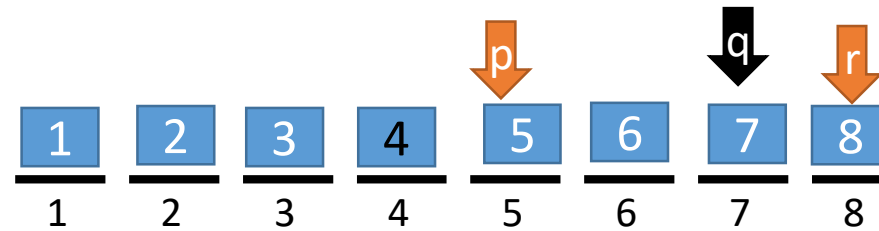
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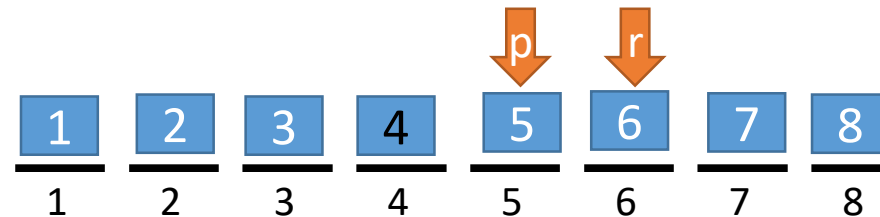
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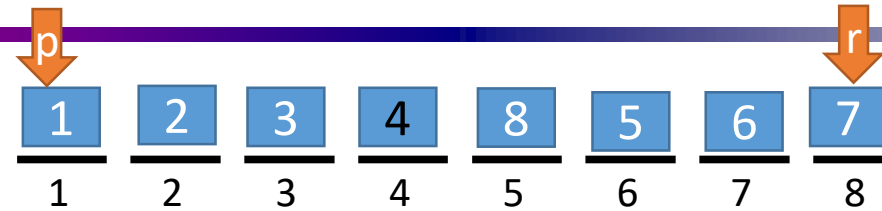


Back to Quick Sort

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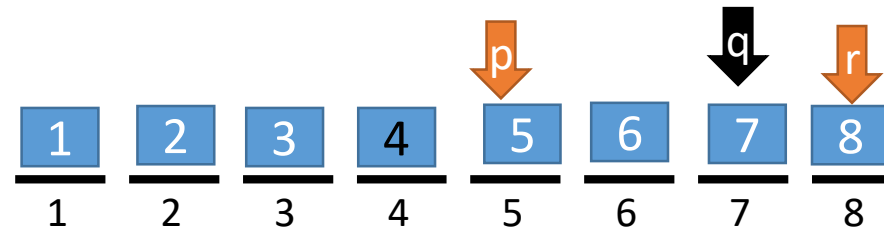
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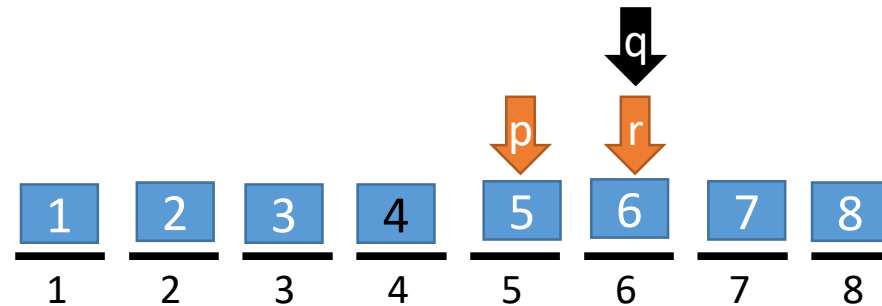
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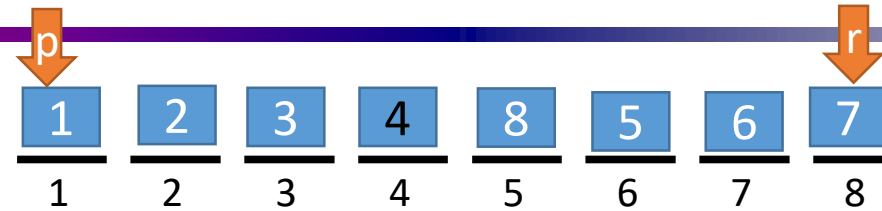


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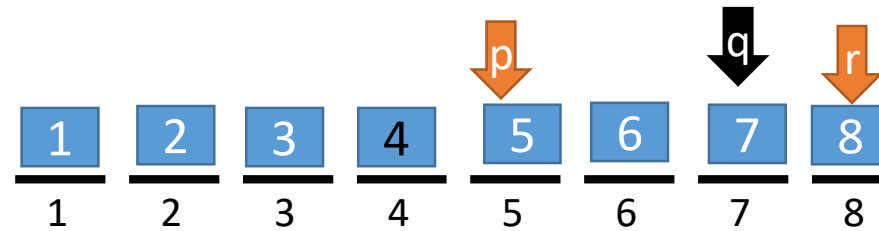
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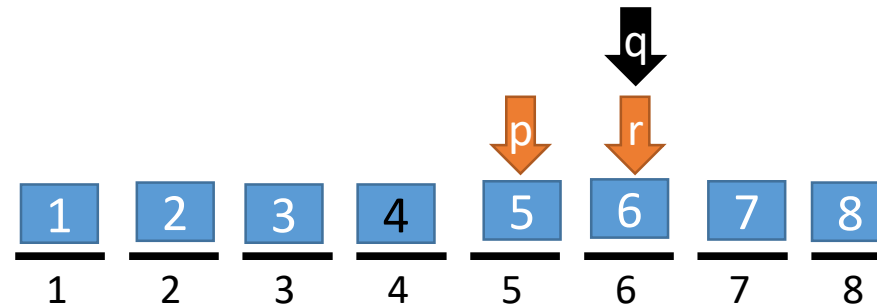
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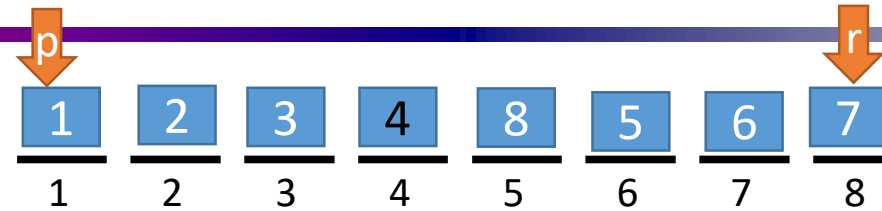


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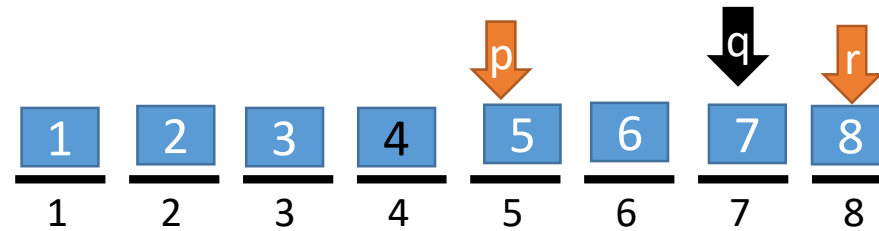
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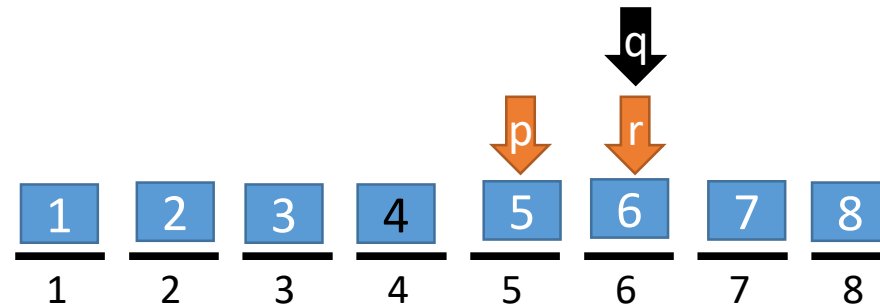
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```



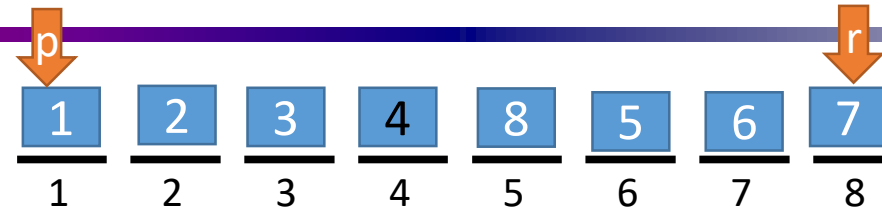
ONLY ONE ELEMENT BEFORE PARTITION

Back to Quick Sort

QUICKSORT(A, p, r)

```

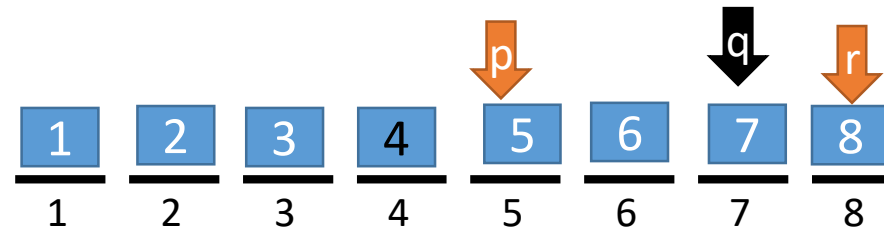
1  if  $p < r$ 
2     $q = \text{PARTITION}(A, p, r)$ 
3    QUICKSORT( $A, p, q - 1$ )
4  → QUICKSORT( $A, q + 1, r$ )
    
```



QUICKSORT(A, p, r)

```

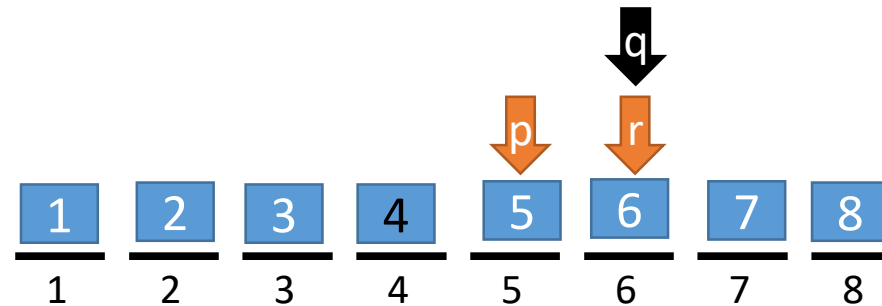
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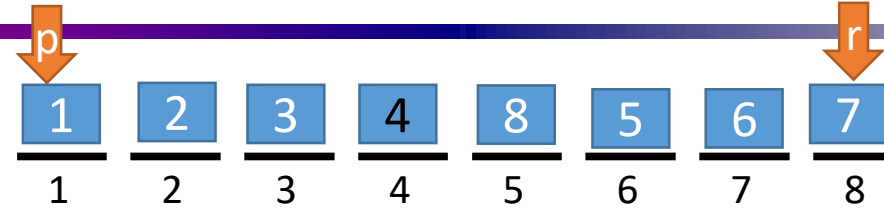


NO ELEMENT AFTER PARTITION

Back to Quick Sort

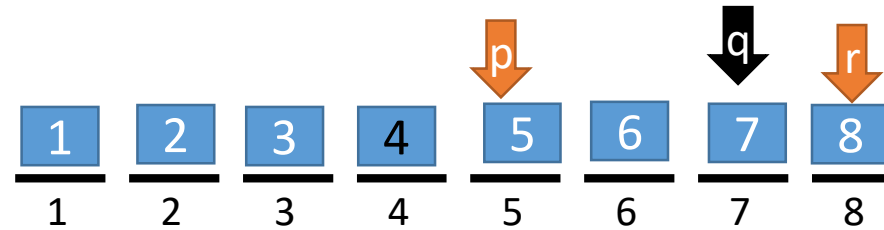
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QUICKSORT(A, p, r)

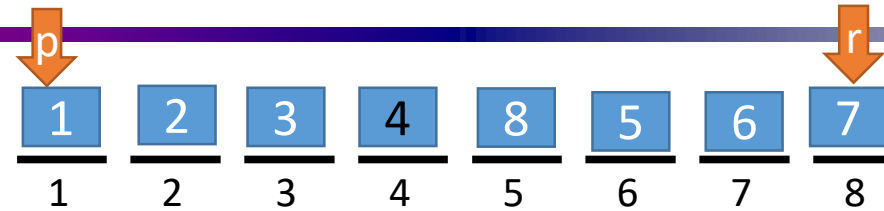
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Back to Quick Sort

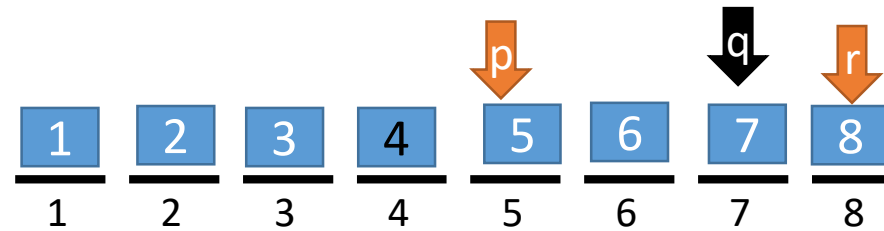
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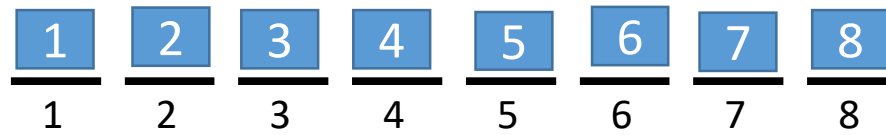
QUICKSORT(A, p, r)

```
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3    QUICKSORT( $A, p, q - 1$ )
4  → QUICKSORT( $A, q + 1, r$ )
```



ONE ELEMENT AFTER PARTITION

Back to Quick Sort



1	2	3	4	5	6	7	8
1	2	3	4	5	6	7	8

AND THE FOLLOWING END RESULT WILL BE OBTAINED

Time Complexity Analysis

- Best Case – $O(n \log n)$
- Worst Case – $O(n^2)$
- Average Case – $O(n \log n)$



Thank You